Queering AI
(Mariconiando a la AI)

(Re)orienting Design Practice Towards Co-Predictive Relations

Grace Leonora Turtle

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Declaration on the Use of Artificial Intelligence
This dissertation engages with artificial intelligence systems as both research subject and experimental method. The generative adversarial networks (StyleGAN2), agent-based simulation incorporating a large language model (ChatGPT, GPT-3.5) within the simulation itself, and deep learning-driven sonic models (RAVE) developed and discussed in Chapters 2–4 constitute original research contributions. Their use falls within the scope of research methodology, not manuscript preparation.

With respect to the preparation of this manuscript, AI-assisted tools were used in accordance with recommended practice for responsible use within academic writing. Specifically, I used Grammarly and Claude to refine the manuscript, including spelling and grammar correction, and editing to improve clarity of language. I also used Perplexity to cross-reference and verify sources cited in the manuscript, including my own prior publications. No AI tools were used to produce content presented as original research. I retain full responsibility for all intellectual content, argumentation, and analysis within this dissertation.

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ABSTRACT

Predictive AI systems increasingly shape social and technological life by abstracting from situated knowledge and experience, encoding human and nonhuman entities into fixed categories, and constraining indeterminate and queer senses of futurity. This dissertation develops a more-than-human design practice of queering AI, reconfiguring predictive systems toward co-predictive, relational ways of knowing and worlding. Drawing on autotheory as a queer methodology grounded in lived experience, the research advances three experimental engagements—*Mutant in the Mirror, Undoing Gracia, and Sounding Territories*—that recode predictive systems from within. Mutant in the Mirror uses a style-based Generative Adversarial Network (StyleGAN) to generate self-portraits that explore AI as a nonbinary entity, resisting fixed recognition systems. Undoing Gracia inhabits algorithmic borderlands through a multi-agent simulation grounded in autobiographical data, tracing how plural subjectivities emerge through interaction with a digital twin. Sounding Territories foregrounds embodied, more-than-human relations through a sound-generative model that embraces dis/identificatory codings and resists algorithmic capture. Working from a situated queer mestiza ethics, these experiments traverse AI development from data capture through post-training interaction by composing datasets, configuring models and simulations, and enacting co-performances.

Synthesizing what these experiments make perceptible in practice, the dissertation distills three queer practice (re)orientations—trans/mutations (toward indeterminacy), algorithmic borderlands (toward thresholds), and dis/identificatory codings (toward illegibility)—and translates them into tactics for design research with predictive systems. The research yields three contributions: 1) introduces co-predictive relations as a theoretical framework countering logics of separability in predictive systems; (2) advances autotheory as an epistemically generative queer design method; and (3) offers redirective pathways for designers and scholars to cultivate desirable un/predictability and queer futurities. By positioning queering as a mode of intervention within AI research, the dissertation contributes to Human Computer Interaction, Society and Technology Studies, design, and futures studies. Through this, the thesis shifts queer discourse from identity-based legibility toward a politics of possibility, concerned not only with who is rendered recognizable by AI, but with how worlds are made and transformed through the design of co-predictive relations.

PREFACE

This research begins with a hunch, a side quest and the intention to embody and enact a theater of im/possibility, the goal of which is to trans/form the self, otherwise-in, as and through co-predictive relations.

When I was a child, around 10 years old, I was diagnosed as dyslexic. This was one of the first codes I was socialised into; the other was that I had a high IQ. The developmental psychologist likened it to Albert Einstein, so I ran with it. During this time, I vividly remember undergoing a series of tests or experiments conducted, I imagine, as a means to characterise and make sense of my diagnosis. In one of these experiments, I was asked, “Do you see rivers on the page?” I did! From then on, I could not unsee them: connected streams between words forming disorderly patterns that flowed off the page, telling their own story beyond the obvious, orderly, grammatically correct, structured sentences carefully aligned within the margins of the page. Over time, I learned to harness this as a power. A power I came to embody and experience as something more than neurodivergent—something uniquely queer—the ability to sense the ambiguous and make sense of the space between and beyond that which is immediately legible. This illegible in-between space emerges as an unanticipated architecture of im/possibility, a site of playful experimentation, novel emergence, and necessary relationality. Or what could be read between the lines as a site of trans/formation. In similar terms, one could reflect on the Sufi teaching story as cited by Donella Meadows (2008), “You think because you understand ‘one’ you must also understand ‘two’, because one and one make two. But you must also understand ‘and’” (p.13). With respect to design, I would extend the need to understand the implicit tacit knowledge and latent relations attributable to “make”(ing). Giving space, time, and form to the in-between is an emergent theme in this research that I hope the reader will begin to sense for themselves.

Fast forward to 2017, walking off stage from a TEDx talk titled ‘Interrogating the Future Before It Arrives’, delivered to an audience of over 1000 people, I told myself that if I really believed in the story I shared, I would write a dissertation on it. Through a series of side quests and unscripted play, eight years later, in 2026, I share my dissertation titled “Queering AI (Mariconiando a la AI): (Re)orienting Design Practice Towards Co-Predictive Relations.” I still cannot discern if that moment of reckoning, as I was clapped off stage, arose from self-doubt, curiosity, or confidence—but it stuck. A marker in time and space, bent out of shape and unsettled, neither a beginning nor an end. Shortly after walking off stage, I found myself sharing tea in a small bar beside the Tamar River in Launceston (or Kanamaluka in Palawa Kani local language), Australia, in conversation with Tony Fry. This encounter informed another implicit dimension of my work. A serendipitous moment where we discussed the power of play and what it means to act in-time, to become futural, and Alain Badiou's (2013) reflections on theater and philosophy, whereby "philosophy is not knowledge: it is not knowledge of theater, aesthetics, science, or politics. It is an action with the goal of transforming the subject. A moment of transformation when we are able to see and do what seemed purely impossible before" (p.1-2). Along these lines, my goal has been to become futural in enacting a theater of impossibility, or the possibility of the impossible. Not representation, but trans/formation.

I took this as an invitation to shift my gaze inward, becoming a subject/object of a designed entanglement with AI through a trilogy of embodied, speculative experiments in co-predictive, multi-agent simulation modelling. Something that could best be described as a series of deviant actions that disturb perceived in/stability, dis/identification, and trans/formation of human-AI entanglements—each experiment offering clues that implicate the capacity of any system to embody and enact a (queer) sense of futurity. The "/" used throughout this dissertation signals that both sides of the term remain active and unresolved, resisting the impulse to settle on one or the other. In doing so, I hold a mirror up to my own futurity and body-world relations, situating myself within the AI systems and contexts I observed. Or, as Von Foerster (2003) might say, engaging in the cybernetics of "observing systems," where I become the observing and observed system, observing myself in action, deviating from the systems within which I have been encoded. In queering AI, this is the trans/formation of self, subject/object relations, and subjectivity, entangled in AI, through which queer individual and collective senses of futurity and capacities to become futural emerge in, as and through co-predictive relations.

This research hopes to be figured as a shapeshifting mestiza body emerging from the algorithmic borderlands and mutant mirrors with which we find ourselves entangled. Unresolved, it aims to invite trans/formation and figuring through the entanglements of art/science, human/non-human, real/virtual, singular/plural, and relational politics and ontologies, where the site of trans/formation is the self, my self, or more precisely, a body-world relation entangled in AI. Meaning, a body that does not precede the world, but rather is and of itself a world, what Erin Manning characterises as a "worlding through which bodying emerges" (Manning, 2013, p.2). Or, to put it slightly differently, "a woven territory" surfaced through the experiment Sounding Territories (Chapter 4). If anything, I hope readers attune to the streams between the words on these pages, sensing the im/possibilities found in design within what, inspired by Gloria Anzaldúa, I will describe as the algorithmic borderlands. Although not explicitly stated in the body of this dissertation, the meandering, recursive, and at times disorienting movement in this text is a characteristic of my neurodivergent writing style, derived from the bleed between my lived experience and intersectional queer theories that contaminate my practice-based experimentation. Or, to use Manning's (2013, p.1-2) words, my unbound, unpredictable, and rhythmic encounter with difference arising through what I will later term co-predictive relations with AI forms the subtext and interdependent threads that flow like rivers through this research. This is my point of departure.

Interrogating the Future Before It Arrives, TEDx Youth, Sydney (2017)

We have your history, your record. When you arrived, you were scanned. Your expression and voice-prints were analysed. We knew your emotional state in detail; we practically knew your thoughts. Had there been the slightest doubt of your harmlessness, you would not have been allowed near me. In fact, you would not now be alive.

—Isaac Asimov, Prelude to Foundation (1988, p.113)

1. INTRODUCTION

1.1. PROBLEM FINDING

At the time of writing this dissertation, between 2021 and 2026, reality (for me at least) has felt a little stranger than fiction. A feeling that is adequately reflected in Isaac Asimov's 1988 science fiction novel, *Prelude to Foundation,* which one could argue can be easily mistaken for the facts of today's AI-entangled reality. A reality characterised by increasing technological complexity and a simultaneously diminishing understanding of its affects (Bridle, 2018). Specifically, in relation to the context of my dissertation, I am referring to the predictive AI technologies that have become ubiquitous in the management of everyday life—from urban governance (Knopf, et al., 2025), to climate adaptation (Koldunov, et al., 2025), and the increased militarisation of space (FitzGerald & Maidenberg, 2024). From training data to models logic, and reasoning systems, it is becoming clear that predictive technologies are having a profound effect on how we as humans sense and act on the world, and from where. The *wher*e, by which I mean the values, ideologies, and beliefs embedded in the design of AI systems, that consequently harm or benefit certain individuals and communities over others, in determining how reality is perceived, sensed, and shaped. This emerging AI-entangled present introduces a paradox for those of us interacting with AI, one that this dissertation attempts to unsettle, which is that any attempt to render visible, determine, or predict a possible future state, event, or scenario, consequently affects *which* and for *whom* certain futures might arise through a process of prediction and the constraining of possibilities. Within this paradox lies an all-too-human tension, one that echoes throughout the cult classic *The Matrix* (1999) and the famous words uttered by the Oracle to Neo:

What's Really Going To Bake Your Noodle Later On Is, Would You Still Have Broken It If I Hadn't Said Anything?

The sentiment expressed by the Oracle in their question to Neo, read through the lens of AI, hints at the overwhelming presence of predictive systems, from the personal to the planetary, and the multi-scalar affects through which we, as individuals and collectives, sense and experience the becomings of lifeworlds (hereafter life/worlds). By life/worlds, I refer to an understanding of the world embodied through lived experience, felt in the here-and-now, that shapes how we imagine, sense, make sense of, and have agency to act towards future possibilities (Husserl, 1970).

Extending this analogy, actors operating at multiple scales, from individuals to governments and corporations, increasingly turn to predictive technologies, much like Neo seeking counsel from the Oracle, in an effort to steer future-oriented decision-making. The EU's Destination Earth initiative (DestinE) offers one example: a planetary-scale digital twin designed as a digital counterpart to Earth's physical systems, entities, objects, and phenomena. It aims to support scientists and policymakers by enabling predictive exploration of Earth's future, functioning almost like a crystal ball grounded in algorithmic truth-telling. However, systems such as DestinE, by virtue of their abstractive and representational nature, may also narrow and flatten the plurality of futures they purport to reveal. As a result, the political and ethical stakes of whose futures are being foreclosed and whose are being made possible (including the more-than-human) via such totalizing predictive models are increasingly obscured. In the face of this dilemma, it is becoming apparent that there is a need to recognise the intimate entanglement of differently situated bodies and life/worlds within systems like DestinE, especially as a growing number of actors engage with them as truth-telling machines or all-knowing oracles. Along similar lines, Benjamin Bratton and Blaise Agüera y Arcas (2022) amongst others, have called for more curious and critical engagement with such systems, which attempt to predict the world while simultaneously diminishing its diversity by narrowing the possibilities for the unpredictable emergence of life/worlds. A process that compounds what Alexander Galloway (2021) calls the uncomputable messiness of life and experience. In other words, irrespective of their un/desirability, multiple possible futures, not-yet-here but quietly humming on the horizon, are rendered visible or invisible through predictions that tame the unanticipated emergence of life/worlds made im/possible through the mechanisation of prediction.

And while questions about what determines our futures are not new, contemporary predictive technologies, arguably, intensify and materialize these tensions by inadvertently or intentionally encoding particular assumptions about whose futures can be foreseen, by whom, and on what terms. In short, attempts to de/code life/worlds through abstract mirror representations (Turtle, 2021; Vallor, 2024), regardless of scale or sophistication, disrupt the inner models We humans use to anticipate and experience reality, to sense and imagine life/worlds otherwise. Or, more simply put, one's sense of futurity—as in the capacity to act towards future (unknown) possibilities.

From predictive text in emails to predictive militarisation through neuro-enhanced soldiers (Crawford & Joler, 2025), the over-reliance on predictive technologies would seem to be slowly diminishing human agency, creativity, and a sense of futurity in decision-making across society and everyday practice. This effect has scaled with the proliferation of Large Language Models (LLMs) and multimodal models trained on vast datasets of text, video, images, and audio—systems conceived according to logics of scale and extraction, and dependent on exploited labour. This process has been well documented by Kate Crawford (2021), Martin Tironi and Matias Valderrama (2023), and others, whose work reveals tensions in land-based struggles and the material effects of AI. Yet, this is not the only way to develop, design, and use AI. From the materiality of AI to its epistemic formations, critiques and reimaginings are emerging from Indigenous (Lewis, 2020), queer (Turtle, 2021), and Black (Amaro, 2022) perspectives and situated struggles that offer alternative origin stories—different entry points for how, and from where, we imagine, sense, and act toward future AI potentialities. It is these alternative entry points, imaginaries, and epistemic formations that my thesis troubles itself with, by attending to how AI participates the modelling of life/worlds otherwise. While at the same time, recognising the complex landscape of AI development, one that, as Sun-Ha Hong (2020, p. 2) reaffirms, is not without cost:

> New technologies for automated surveillance and prediction neither simply augment human reason nor replace it with its machinic counterpart. Rather, they affect the underlying conditions for producing, validating, and accessing knowledge and modifying the rules of the game of how we know and what we can be expected to know. The promise of better knowledge through data depends on a crucial asymmetry: technological systems become increasingly too massive and too opaque for human scrutiny, even as the liberal subject is asked to become increasingly legible to machines for capture and calculation.

The increased legibility to machines that Hong (2020) identifies speaks to a form of territorialisation, to borrow the term from Gilles Deleuze and Félix Guattari (1987), that curtails self-determination through the standardisation and universalizing logic used to make something coherent, all the while constraining spaces available for transformation. Building on this notion, Adrian Mackenzie (2015) argues that “while algorithms classify in very different ways, they all assume that the world is made of things or events that fit in stable and distinct categories (...), in all cases, prediction depends on classification”, that in most cases, reinforce normative behaviours (p.437-438). Technically speaking, probability calculus, which is often used to tame uncertainty, turns uncertain future events into quantified risk estimates. An approach often used when considering existential risks and their mitigations, such as nuclear war, climate change, and further advancements in AI (Bostrom, 2002). Regardless of the value we place on the use of prediction toward managing uncertainty, the tension still lies in AI’s impulse to fix possible worlds. By ‘fixing,’ I mean the desire for certainty when navigating transitions from point A to point B (think Google Maps), where an optimal route is calculated and continuously adjusted on our behalf. It is through the rapid rise of datafication and algorithmisation that such logics of fixing have become woven into everyday routines, in ways that can be so subtle that they are difficult to assess the extent to which they alter the experience of life/worlds (Hong, 2020). In this dissertation, I ask how those logics might be reoriented so that predictive systems do not simply fix possible worlds in place but participate in co-creating more fluid, plural life/worlds.

As noted earlier in this section, society is becoming increasingly entangled with AI across different scales, from bodies to cities to planetary systems. The rapidly evolving nature of these entanglements urgently requires a re-visioning of AI research and design within the field of human–computer interaction (HCI) (Frauenberger, 2019). This becomes particularly salient when questions arise, such as: Where does a human body end, and where does a virtual one begin? How much of reality is mirrored or distorted by simulations such as digital twins as they come to stand in for the worlds they represent? Taken together, these questions foreground tensions accompanying the rapid development of AI. They point to the need to identify new vantage points from which to study human–AI entanglements, understood here as recursive, co-constitutive relations between humans, more-than-human bodies, worlds, and algorithmic systems. Take, for example, predictive policing systems that analyze historical crime data and determine “high-risk” geographies or individuals. Such systems reinforce structural biases through the intensification of surveillance of marginalized communities while reinforcing these biases embedded in the training data (Brayne, 2020). In other words, algorithmically determined police presence within specific communities consequently leads to more arrest data, which feeds back into the system, perpetuating cycles of surveillance in the administration of society (Amaro, 2022; Hinds & Joinson, 2018). As Luciana Parisi (2019, p. 33) contends, these practices steer subjects toward actions that “benefit the system’s” performance while narrowing the scope of self-determination through their use. Consequently, algorithmic classification, labelling, and decision-making can constrain the capacity for improvisation and unanticipated transformation, disproportionately affecting those already marginalized by and within algorithmic systems, including Black, Indigenous, racialized, feminized, disabled, trans, queer, migrant, and working-class people and communities, as well as those in the Global South (Amaro, 2022; O’Shea, 2021). At the level of system operation, classification, labelling, and ordering are core functions through which AI systems recognize, categorize, and render elements of the world more legible, and they can flatten emergent forms of being, doing, and becoming (Amaro, 2020; Finn, 2018; Nowotny, 2021). These effects span a wide range of predictive systems, from personal analytics that profile behavior (D’Ignazio & Klein, 2020; Zuboff, 2019) to urban digital twins leveraged to monitor and predict civic behavior, including traffic flows, citizen movement, energy consumption, and infrastructure investment (Batty, 2018; Grieves & Vickers, 2017).

However, a significant gap persists in AI research on predictive systems, particularly within HCI and design. Work that often foregrounds the *explicit* and *observable* effects of AI applications at the expense of less visible, *latent,* and *relational* effects that necessitate a revisioning of their epistemological and ontological foundations (Sanders & Stappers, 2008). More pointedly, the hidden variables that drive algorithmic processes interact with the less visible social, political, and cultural dynamics that these systems produce or reinforce. This gap widens when AI users, researchers, designers, and decision-makers operating within normative paradigms accept probabilistic outcomes without scrutiny or critical reflection, resulting in the privileging of certain knowledge systems while marginalizing others in determining what is perceived as "useful" (Manning, 2020). Still, these tensions create opportunities to re-imagine human–AI relations. The sociotechnical imaginaries that shape how we understand and operationalise AI (Jasanoff & Kim, 2015) must evolve. Equally important is how these imaginaries are enacted through the research methods we use to interrogate and reconfigure AI systems' logics, functions, implications, and entanglements across ethics, politics, and more-than-human relations. A consequence of which would also include a critical reimagining and enactment of the role of design and the designer within the interdisciplinary field of AI research and development (Giaccardi & Bendor, 2024).

At stake is not only how AI models assume particular systems of representation, but also how those representations are enacted in and upon the world, shaping ideas of what can be true, knowable, and calculable. More poignantly, this constrains possibilities for knowing or imagining life/worlds otherwise (Bridle, 2022, p. 177). The reductionist logics encoded in AI models fail to capture the complexity and interdependence of lived experience. Consequently, we humans can come to confuse a model's reading of reality with reality itself, or, in similar terms, as James Bridle (2022) puts it, "we take them for the world itself" (p. 207). Representation in this sense is transformed, in that representation in and of itself becomes performative. Through this lens, models do not merely stand in for the world; they actively shape how it is sensed, governed, and acted upon. Although many AI systems are designed to yield certainty, predictability, and control, to find order within chaos (Prigogine & Stengers, 2018), humans and more-than-human entities alike inhabit a world characterised by interconnectedness, flux, and contingency (Bridle, 2022). Consequently, the tension between algorithmic systems' drive for stable forecasts and the emergent, relational spontaneity of living beings or life/worlds becomes starkly apparent.

It is evident that the push to predict underwrites applications that aim to counteract uncertainty in complex and dynamic systems; yet paradoxically, the more these systems are deployed, the more they risk foreclosing possibilities that arise unpredictably or defy fixed categorisation (Benjamin, 2019; Halpern, 2022; Nowotny, 2021). Relying on AI predictions without critical examination can result in a narrow, homogenized view of life/worlds, diminishing human agency and a sense of futurity in world-making practices—that is, the ongoing, situated ways people collectively sense, experience, and imagine the conditions of their lives. Or, as Helga Nowotny (2021) puts it, echoing the Matrix's Oracle, these become "self-fulfilling prophecies," whereby "a prediction becomes true simply because people believe in it and act accordingly" (p. 4). Saidiya Hartman (2019), by contrast, offers a counter-reading—a way out of the self-fulfilling prophecy—by reminding us that those rendered unpredictable, whom she characterises as "wayward," can craft alternate modes of survival and joy through the very act of wayward movement. Yet waywardness is not without reason. Throughout this section, I have attempted to describe how prediction constrains, or at least conditions, such movements, diminishing the generative potential of bodies, worlds, and subjectivities that do not neatly align with the categories expected by computational models.

Along these lines, intersectional feminist, queer, decolonial, and more-than-human perspectives in HCI and design acknowledge subjectivity as fluid, dynamic, and interdependent. In response, design practitioners and HCI scholars are evolving their understanding and, by extension, their approach to knowledge production, shifting from user-centered paradigms toward more relational and more-than-human perspectives in designing for and with computation (Forlano, 2017; Giaccardi & Redström, 2020; Wakkary, 2021). Developing computational design practices capable of navigating these complexities is both urgent and necessary, particularly as human and non-human actors become increasingly entangled through algorithmic systems (Frauenberger, 2019; Giaccardi et al., 2020; Murray-Rust et al., 2019).

By foregrounding the hidden politics of possibility in the production of prediction, I bring into question the ontological and epistemological foundations of AI. This represents a shift toward practicing reorientations unsettled through queerness, extending notions of queerness beyond its more traditional associations with gender and sexuality to encompass broader critiques of normativity and power in HCI and design (Guyan, 2022; Light, 2011; Taylor et al., 2024; Tran et al., 2024). In doing so, I propose that intersectional queer and trans perspectives are generative in challenging power and normativity that remain excluded from current practices and ontological approaches to designing predictive systems. In taking this stance, I align with design scholars such as Anne-Marie Willis (2006) and Arturo Escobar (2018), who argue that design does not simply shape objects or interfaces but participates in shaping worlds, that is, in configuring what kinds of beings, relations, and futures become possible. This raises questions about whose ontologies inform design and offers a pathway into queerness that will be further situated and contextualized in *Turning from prediction to co-predictive relations* (Section 1.4), by situating queerness as an ontological disturbance in AI research and design.

These gaps persist between attention to AI's observable effects and the latent, relational configurations through which predictive systems shape life/worlds, as well as between critique of predictive power and the methods needed to intervene in predictive systems in practice. In response, this dissertation queers AI by foregrounding plural, relational ontologies and treating fixed classification and computational predictability as neither inevitable nor apolitical. Drawing on queer and trans theories (Ahmed, 2006; Anzaldúa, 2021; Muñoz, 1999; Preciado, 2013), the work critiques how predictive systems reinforce rigid identities, while advancing autotheoretical experiments—modes of inquiry that weave theoretical reflection together with lived experience, positionality, and practice-based experimentation—to show how queerness can unsettle these logics.

The stakes are high. If AI is fast becoming critical infrastructure shaping what can be sensed, governed, and imagined, the problem is not only that these systems misrepresent or abstract from the world, but that they stabilize those abstractions. More pointedly, they naturalize fixed classifications and treat probabilistic outputs as neutral truth-telling, thereby foreclosing ways of knowing and becoming otherwise. Critique is therefore necessary, but it must be coupled with methods for intervening in predictive systems in practice. In response, I take up queering as one such mode of critical intervention. Drawing on queer and trans-feminist mestiza theories, I approach queering AI as a way to unsettle prediction's paradox: its drive to control uncertainty even as it narrows what can become possible. Through this practice of queering, I address the multifaceted problems introduced in this section by opening AI design toward plural, more-than-human agencies and relational ontologies. This move is developed through the proposition of co-predictive relations, articulated in *Designing for co-predictive relations* (Chapter 5).

1.2. AIMS AND RESEARCH QUESTIONS

Drawing on José Muñoz's (2019) articulation of queerness as a horizon of "social relations" and a "basic desire to live otherwise" (p.1–2), I approach queerness here not only as an identity category but as a critical and imaginative orientation toward alternative ways of living, relating, and world-making. Framed as an ongoing engagement with the indeterminacy of life/worlds, this research proposes a way into the design of AI that both acknowledges and challenges the logics of normativity, as they appear in classification schemes, predictive targets, and assumptions about desirable behaviour that underpin dominant AI systems and their applications.

The double-sided problem this thesis attends to is therefore:

• **Epistemic:** Algorithmic prediction treats the world as a source for abstraction, stripping data, logic, and performativity from situated knowledge and experience, all the while creating new, more-than-human entanglements.
• **Ontological:** Through this process of abstractive entanglement, human and

non-human entities become encoded into discrete, identifiable, and fixed categories, thereby foreclosing indeterminate and queer senses of futurity.

The central aim of this dissertation is to develop a critical and more-than-human design practice of queering AI. This aim is exemplified through a trilogy of autotheoretical experiments that attempt to recode predictive AI systems toward what I term co-predictive relations. Such an approach explicitly seeks to unsettle AI design from its normative orientation toward predictability and control. It challenges deterministic and binary logics by bridging queer theory's interrogation of normativity (Ahmed, 2006; Halberstam, 2011; Muñoz, 1999) with the relational perspective in more-than-human design (Forlano, 2017; Giaccardi & Redström, 2020; Giaccardi et al., 2024; Wakkary, 2021). Instead of foreclosing alternative futures by encoding fixed categories, AI is approached as a dynamic site for experimentation—one that moves between queer theory, lived experience, and situated knowledge production.

Situated at the intersection of theory and practice, this research adopts autotheory (Fournier, 2021) as a queer/ing method, merging lived experience with critical perspectives from queer, transfeminist, and Mestizx thought and experimentation. Through the trilogy of autotheoretical experiments—*Mutant in the Mirror, Undoing Gracia,* and *Sounding Territories*—I explore how bodies, data, brought together by simulations can together reorient predictive processes toward more transformative futures. These experiments illuminate the fluidity of borders, minoritarian identifications, and the porous boundaries between human and more-than-human subjectivities.

Crucially, this work functions as a playground for experimenting with queer forms of knowing, being and doing where queerness becomes a generative foundation for "being entre mundos, between worlds," grounded in a pluriversal queer ethics and politics of possibility (Escobar, 2018, p. 504). Through this lens, queering AI is not only speculative and performative, but also, perhaps more importantly, critical. The act of queering effectively unsettles deterministic assumptions and reimagines prediction as a co-constituted, relational process. In doing so, my research proposes a deliberate queering of multi-agent, simulation models and digital twins as exemplary predictive AI systems, opening speculative, curious, and critical pathways for designers, researchers, technologists, and managers to become attuned with the concrete possibility of a queered AI.

Hence, the dissertation pursues the following Research Questions **(RQs):**

• **RQ1:** What does it mean to queer AI in relation to predictive computation? (i.e., in terms of classification, identification, and datafication that resist finality and trans/forms predictions)

• **RQ2:** How can autotheory as a queering method be used to critique and design for co-predictive relations? (i.e., novel approach in design, situated critique, politics, and ethics of AI).

• **RQ3:** How can a mestiza queer theory challenge normative AI data, logic, and performativity and inform the design of such systems, otherwise?

In addition to the overarching RQs, each experiment in the trilogy poses its own 'What If' questions that build on one another to disturb, recode, and reorient normative design logics. Together, they position queerness not only as critique but as a method for making AI otherwise—speculative, embodied, and (re)oriented toward queer futurities.

Autotheoretical research questions:

• Mutant becomings (Chapter 2)
RQ4: Experiment 1: *What if* AI were queer—a non-binary relational entity, mutating and dynamically becoming-with us humans? How might this challenge notions of algorithmic determinism?

• Algorithmic borderlands (Chapter 3)
RQ5: Experiment 2: *What if* the self emerges as multiple, pluralizing through the borderlands of human–AI entanglements—how might this challenge normative distinctions of individuation?

• Performing dis/identifications (Chapter 4)
RQ6: Experiment 3: *What if* disidentifications, as the performance of politics, could subvert algorithmic identification—how might this transform designerly understandings of human–AI relationality?

By shifting design practice from dualistic ontologies to relational, performative, and pluriversal approaches (Escobar, 2018), this research foregrounds how queer/ing methodologies open new epistemic and affective pathways to knowing and doing. These autotheoretical experiments—and the questions they pose and seek to address—hint at new paradigms for engaging with uncertainty, plurality, and interdependence in the design of AI-entangled life/worlds. **Rather than treating prediction as merely calculative, the work reframes it as an ongoing, relational process in which human and non-human agencies co-perform and co-produce emergent and un/predictable possibilities. Or what could be otherwise understood as co-predictive relations.**

Next, the structure of this introduction unfolds through a series of interdependent movements: section 1.3, unsettles the normative function of AI prediction and its management of uncertainty; section 1.4 turns toward co-predictive relations as a conceptual pivot; section 1.5 articulates autotheory as a queer design methodology grounded in situated experimentation and lived experience; and section 1.6 introduces the trilogy of autotheoretical experiments that form the basis of this research. This chapter concludes with a summary of the dissertation's key contributions (section 1.7) and an overview of its structure (section 1.8).

1.3. UNSETTLING PREDICTION

As discussed so far, predictive systems, such as agent-based simulation models and digital twins, are designed to calculate and control the behaviour of complex, emergent systems, or in Grieves and Vickers' (2017) terms, to mitigate unpredictable behaviour and work towards a ***desirable*** and ***predictable*** system state (illustrated in Fig. 1 as an adaptation of their "Four Categories of System Behaviour"). These systems are often treated as knowable and stabilisable through probabilistic computation biased towards mirroring. In practice, this results in the homogenisation, flattening, and fixing of systems, phenomena, or entities that are otherwise plural, heterogeneous, and fluid. An implicit motivation for this research is to ***create space for desirable unpredictability*** (further expanded in Section 1.3.3), while simultaneously acknowledging the operational necessity and institutional desire to manage uncertainty within dynamic, emergent systems (see Fig.1). Engaging this tension is not only methodologically generative but also politically responsive. I propose that inhabiting this paradox offers a way to expand the scope of (im)possibilities, enabling the emergence of difference and what will later be described as a queer sense of futurity—what Muñoz (2019) calls "a structuring and educated mode of desiring that allows us to see and feel beyond the quagmire of the present" (p.1).

This section introduces some of the practical and philosophical foundations, ethics, and politics of managing uncertainty with AI, particularly as they play out in the predictive architectures of contemporary algorithmic systems (Mackenzie, 2015). These systems are not neutral; they encode desires for stability, legibility, and control, which can undermine the unpredictable emergence of human and non-human agency, indeterminate identities, and futurities. Here, I explore the tensions between desirable/predictable and desirable/unpredictable behaviour, and how these tensions manifest in the design of increasingly agentive AI systems that, as Elisa Giaccardi (2019) proposes, configure design itself as a probabilistic outcome.

1.3.1. Managing Uncertainty with AI

As previously indicated, AI development is often centered on maximizing predictability, managing uncertainty, and achieving desired states within dynamic systems, thus rendering the future seemingly more certain (Halpern, 2022; Nowotny, 2021). To accomplish this, many large-scale AI applications

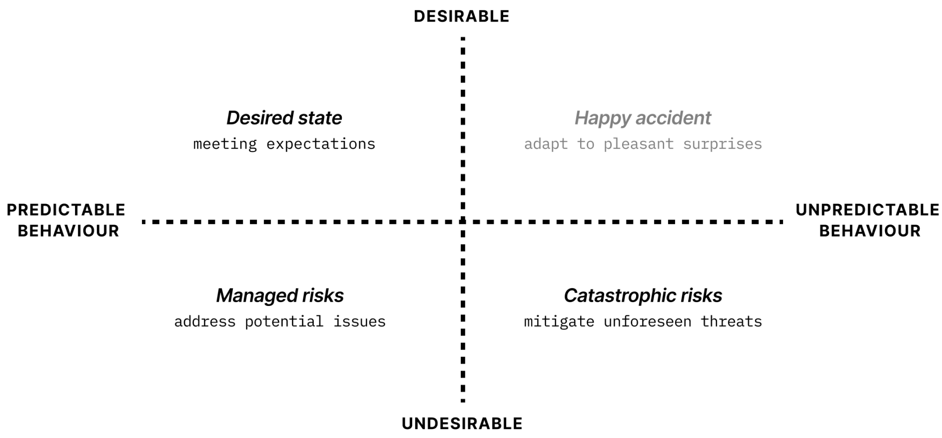

Figure 1. Illustrative view of "Four Categories of System Behaviour" adapted from (Grieves & Vickers, 2017).

rely heavily on classification, categorisation, and forecasting, exemplified in environmental modelling for climate projections (Bauer et al., 2021) and urban resource management systems (Whitson, 2013). The benefits of such predictive systems are clear; for instance, they enable early warning systems for communities at increased risk from natural disasters such as floods or wildfires, anticipating the social, political, and economic instability exacerbated by climate-induced disasters (Bauer et al., 2021). However, the risks at this macro scale are perhaps less immediately visible.

At the individual level, it has been well documented that algorithms, such as facial recognition systems, often produce biased outcomes (Buolamwini & Gebru, 2018). This bias arises precisely because these technologies flatten dynamic, fluid human attributes into singular, stable entities. Similarly, Louise Amoore (2020) and Os Keyes et al. (2019) emphasise that diversity and difference are frequently reduced to statistical outliers, raising critical questions about who these systems recognise, and on what terms. As Amoore (2020) warns, the question of who gets to simulate whom, and for whose benefit, is central to their function. These concerns prompt a set of questions that this dissertation continues to return to: which world? What model? What can be modelled, for whom, and to what end? At stake in each case is how, by turning individuals and communities into calculable entities stabilised through data capture and categorisation, predictive technologies risk embedding bias and perpetuating harm. Predictive policing clearly illustrates this through practices that corral unpredictable human behaviours into legible data and pattern recognition, disproportionately targeting marginalised communities and further entrenching structural inequalities (Brayne, 2020). In Édouard Glissant's (1990) terms, such practices can compromise the fundamental "right to opacity" at the individual and collective levels. These examples underscore a core dilemma repeatedly revisited throughout this dissertation: the typical drive of predictive systems to stabilise and fix futures often results in overdetermined outcomes that curtail the possibility of radical reconfigurations of life/worlds and the emergence of difference.

In the context of emerging design practices within HCI, and in particular AI research and design, Giaccardi et al. (2024) begin to map out a blueprint for engaging with uncertainty as a fundamental condition of design. They argue that the "interplay of human and (uncertain) non-human resources profoundly challenges the stabilising character of the artifact and brings new aspects of uncertainty to design" (p. 68:2). As algorithmic systems become more agentive and autonomous, uncertainty enters as a constitutive feature of design itself. Within the management of uncertainty through control and feedback in the system, Giaccardi et al. (2024) suggest that design is moving "toward fundamentally probabilistic relations, both between design and use and between producer and produced" (p.68:2). These recursive loops destabilise traditional distinctions and call for speculative and ethical approaches to designing with and for uncertainty, particularly when operating with data-driven, computational, and simulation-based models that manage complex and emergent systems.

Uncertainty is not only a design condition but also a temporal orientation, one that reshapes how we understand and engage with futurity itself as a sense of an indeterminate then and there, not-yet-but-could-be (Muñoz, 2019, p.1-3). Marco Cuevas-Hewitt (2011) reminds us that "the futurology of the present does not prescribe a single monolithic future, but tries instead to articulate the many alternative futures continuously emerging in the perpetual present" (p.1). Futurist Jim Dator (2009) emphasises that fields explicitly concerned with future emergence, such as futures studies, are not predictive sciences aiming to calculate precisely *the future,* but a practice of exploring multiple *alternative futures* and the ingredients that shape them. No matter how detailed, predictive models are always, at best, approximations of reality, typically steeped in dominant Eurocentric and Western ontologies and ideologies that circumscribe potential futures. Consequently, many emerging algorithmic systems, their designs, and their users remain unable to fully grasp the wayward emergence of alternative possibilities for different life/worlds. Further complicating matters, Lee Edelman's (2004) work alerts us to the fact that the promise of futurity is unequally distributed and inherently political. Critical design and AI research must therefore explicitly address the question of whose worlds are rendered im/possible through the use and design of predictive technologies, interrogating precisely who participates, or is excluded, in their making. Thus, what is ultimately at stake is a deliberate and critical transition toward futures that are imagined and enacted otherwise, particularly from the margins.

For queering AI, this implies a commitment to critically and curiously unsettle the very properties that produce AI, including training data, logics, modes of reasoning, and performative qualities underpinning predictive technologies. It involves actively recognising how these factors shape diverse senses of futurity and influence world-making practices. By challenging prevailing notions of stability and dualistic binaries, this research situates itself within a relational politics of possibility—one that, following Giaccardi et al. (2024) could involve shaping future design practices that embrace uncertainty as generative rather than reductive, and that remain open to unpredictable relations between designers, users, and algorithmic systems.

1.3.2. Agent-based, simulation modelling and digital twins

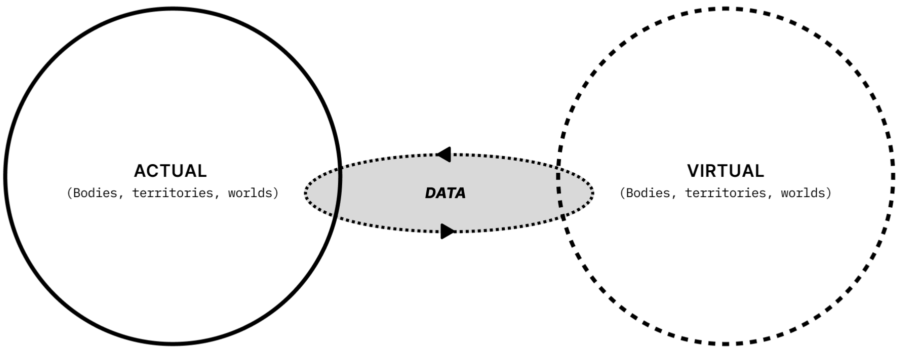

Figure 2. A simplified model of a digital twin in which actual bodies, territories, and worlds co-emerge with their virtual counterparts, encompassing humans and more-than-human entities and phenomena that distribute and circulate data.

Building on the critique of predictive systems' normative impulse to fix futures and reduce uncertainty, this section turns to simulation modelling as a key site where such predictive logics are both encoded and enacted. Simulation models, of which digital twins are an instance, have been central to contemporary predictive infrastructures, offering algorithmic lenses through which complex systems are observed, interpreted, and often controlled. Historically developed in engineering contexts to monitor high-value assets such as jet engines and production lines (Grieves & Vickers, 2017), digital twins have since proliferated across other domains, from healthcare to urban planning to planetary system governance. As their reach expands, so too does their influence on how society navigates uncertainty, risk, and possibility. In many cases, multi-agent simulation models now serve as sandboxes for training and testing AI systems in interaction with the world; a typical example is AlphaGo (Hassabis, 2021; Schrittwieser et al., 2020). As I have touched on in this introduction so far, such sandbox environments span the virtualisation of entire cities (Crooks et al., 2021), to personal attributes, such as voice (Herndon, 2021), and climate futures (Bauer et al., 2021). These simulations offer a self-contained environment where policy ideas or design solutions can be tested with manageable risk (Batty, 2018; Crooks et al., 2021). As societies face ecological volatility and sociopolitical instability, the perceived imperative to model and effectively simulate interactions and scenarios using predictive systems intensifies.

The mirroring characterised by simulation models, such as digital twins as I have come to understand them, tether the virtual to the physical attributes, behaviours, and performativities of a Digital (and physical) Twin (see Fig.2). These mirror worlds implemented as decision-support systems, as I have suggested, attempt to render visible, optimisable, and governable dynamic systems, yet, when deployed through a totalising lens, consequently reinforce the status quo by predetermining the future states of a system, narrowing horizons of possible futures that diminish human and non-human agency (Nowotny, 2021). In similar terms, simulation models are not only technical systems, but they also embed social, political, and technical choices about what counts as data, whose metrics define success, which worldviews are encoded into the model itself, and who reaps the benefits of certain possibilities determined by predictions (Benjamin, 2019; Noble, 2018). While digital twins are often presented as objective systems for managing complexity, this framing ignores how their underlying logic can reproduce harm, particularly for communities that fall outside the normative boundaries the system is designed to recognise.

To suggest that such simulations neutrally mirror reality overlooks the role predictive systems play in concentrating epistemic and political power. As Amoore (2020) warns, the question of who gets to simulate whom, and for whose benefit, is central to their function. When simulation becomes a method of control rather than a means of engaging with the variability and multiplicity of life/worlds, the outcome, as I have attempted to illustrate so far, is often a narrowing of possibility and a flattening of difference. The perceived neutrality of these systems frequently excludes local wisdom, improvisational tactics, and minoritarian ways of knowing that refuse standardisation or do not neatly fit within normative constraints of algorithmic logic and modelling (Amoore, 2020). Queer and Indigenous epistemologies, for instance, tend to operate through relationality, fluidity, and non-dualistic ontologies that foreground ambiguity and trans/formation as necessary conditions for living and knowing (Bey, 2021). Take, for example, Anzaldúa's (2014) insistence on the power of "la facultad," a sensuous, affective, and embodied awareness that arises from living in the borderlands, from which one can attune to multiple worlds that defy binary classification and embrace contradiction. These approaches resist the computational demand for legibility, insisting instead on situated, plural, and lived knowledge.

The 'pseudo-worlds' generated through predictive algorithmic modelling often obscure the realities they purport to represent. By rendering particular futures as inevitable through the mechanisation of algorithmic predictions, such models call for critical, plural, and queer engagements. When decision-making and movement are constrained to simulations of, for example, a human heart or a neighborhood, the sense of order and control they offer amid real-world uncertainty becomes, as Whitson (2013) suggests, a form of entrapment for users interacting with them. Ian Bogost (2010) characterises this dynamic as a form of "persuasion," arguing that simulations such as digital twins carry a distinctive persuasive force: through their rules, parameters, and constraints, they enact what he calls procedural rhetoric, shaping how possibilities are understood and what actions appear meaningful or effective. In this way, such systems help construct what Kirkpatrick (2013) calls the social conditions under which reality is experienced as real. Yet, as Muñoz (2019) reminds us, the future need not be reduced to a singular, stable, or predetermined linear path. In other words, not all that is illegible or incalculable must be tamed. As the next section explores, uncertainty need not be feared nor managed away. Instead, it can be a generative force for queering futurities and reimagining how humans (and more-than-humans) relate to AI systems.

1.3.3. Queer futurities, a politics of possibility

To arrive at a *politics of possibility* is to stay with the generative nature of uncertainty, from which I understand queerness in relation to possibility as a designer. As Ilya Prigogine and Isabelle Stengers (2018) assert, "to restore both inertia and the possibility of unanticipated events—that is, to restore the open character of history—we must accept its fundamental uncertainty" (p.207). This research takes up the unresolvable tension of un/certainty and un/predictability as a key ingredient in the design of AI systems. It treats uncertainty not as an obstacle to be overcome through prediction, but as an index of potential—a horizon where more-than-human worlds might arise, queerly. On the one hand, this stance moves from the already inherent queerness in more-than-human worlds, or what Karen Barad (2011) refers to as nature's queer performativity. And on the other hand, it follows Muñoz's (2019) framing of queerness as an "open mesh of possibility," a rejection of the limiting conditions of the here and now, and a "basic desire to live otherwise" that signals a performative doing for and towards the future (pp. 1–2). Such an active, embodied refusal of the present marks a shift in queer discourse from *identity politics* toward a *politics of possibility,* where, as Escobar (2020) puts it, the real, possible, and political are "joined at the hip" and enacted all at once (p.3).

One generous articulation of this comes from Nat Raha and Mijke van der Drift (2024), who foreground practices of solidarity, care, and embodied transformation, with emphasis on what we do together "towards the (re)making of our lives" (p.5). This resonates with Saidiya Hartman's invocation of the "wayward," in reference to those who live otherwise at the edges of what is recognised as normative life (2019, p.112), and Lola Olufemi's (2021) posture towards "the unknowable," as a site of political imagination (p. 7). Imaginative, embodied, and collective practices, from mutual aid organising to the development of collective care infrastructures, work through situated action and ongoing transformation, suggesting openings for AI research and design to engage with horizons of im/possibility in the present, radically departing from the norms that shape futures arising from AI entanglements. They point toward a politics of possibility as a counter-approach to ostensibly apolitical design, or design practices in which politics remain implicit and underarticulated (Keyes et al., 2019).

To engage with the politics of possibility means to unsettle the epistemological and ontological assumptions that underpin prediction. Julietta Singh (2018) offers more clues, when the *personal* and *politics of possibility* collide, claiming that the archive of one's data, enmeshed source material and information, no matter how exhaustive, "will not restore you," any more than a city's digital archive will restore the complex interweavings of its human and non-human inhabitants (p.21-27). Yet these incomplete translations and the dynamic interplay between data/flesh and model/world can be sites of trans/formation, especially when the illusion of singular futures is deliberately interrupted.

This shift matters within critical AI, particularly in human-computer interaction and design research, because it raises the question of whose lives/worlds are expressed through their datasets and encoded into the logic of algorithmic systems that make certain life/worlds im/possible. Algorithms are structured as step-by-step instructions for prediction and control, rooted in data, abstracted, and then recoded into particularized versions of reality, conceptions of space, time, and social relations (Pasquinelli, 2023, p.57). Which leads us back to Bridle's warning that AI systems cannot distinguish between models of the world and the worlds they seek to model, as the facts and fictions attributable to the actual and virtual dimensions of the 'real' become increasingly difficult to distinguish. In this intermediate space of representation, neither AI systems nor individual humans can fully embody nor articulate the complexity of the world they operate within. Yet, both continue to act upon the world in un/predictable ways all the same. This paradoxical relationship between computational reductionism and the irreducible complexity of life/worlds creates an epistemological tension or disturbance. What is generated from this tension becomes a matter of inquiry within this research, through the curious probing into novel understandings of more-than-human entanglements with AI that might emerge despite, or perhaps because of, their mutual irreducibility and un/unpredictability.

Ultimately, working from a *performative politics of possibility* in the designing of predictive systems opens space for un/predictability, dis/identifications, and trans/formations. It does so by reworking AI systems from within the design of predictive systems themselves and the relations that arise through their use. My pursuit here is to encourage an unsettling of conventional approaches in AI that intentionally or unintentionally constrain possibilities and fix futures. This approach sets the stage for the chapters that follow (Chapters 2, 3, and 4), where these ideas translate into autotheoretical experiments that purposely introduce un/predictability as a way to explore the space of *queer futurity*.

1.4. TURNING FROM PREDICTION TO CO-PREDICTIVE RELATIONS

This section introduces a key shift that underpins the experimental trajectory of this dissertation: queering AI not as rejection but as an ontological disturbance, opening space for *designing co-predictive relations* from a *politics of possibility*. This turn does not oppose prediction wholesale; instead, it reframes prediction as a relational, situated, and ongoing co-performance between humans and more-than-humans, including algorithmic systems. As a result, inviting new perspectives on engagement with space of desirable unpredictability within complex systems, through a queer lens (see Fig. 1).

Drawing from queer theory, and in particular from embodied, mestiza, and minoritarian knowledge-making practices, I position queerness as a critical and speculative lens through which to reorient design's engagement with AI. My position in this research follows the increasing awareness in AI research, HCI and design communities, that queering as a strategy may help subvert existing sociotechnical systems and codings of hegemonic worldviews that reinscribe extractive and (what I mentioned previously as) *apolitical* AI research and design practices (Choi & Light, 2020; Light, 2011; Spiel & Keyes et al., 2019).

An ontological (re)orientation of predictive systems challenges the commonly held assumptions about the future (or futures) as a destination or "probabilistic outcome" (Giaccardi, 2019). Instead, it calls for forms of world-making rooted in relationality, multiplicity, and unpredictability. In design, this orientation aligns with what Fry (2009) terms *redirective practice*—a mode of working that resists extractive, linear futures in favour of more situated, plural, and more-than-human speculation and intervention. Along these lines, the use of queer/ing as a theoretical framework invites a critical re-thinking of the capacity of designers to cope with the increasing complexity and dynamism of (socio)technical systems that actively learn, anticipate and mutate over time, and while in use (Giaccardi & Redström, 2020).

Cruising through unresolved and wayward pathways across queer, more-than-human, critical, and speculative practices, this section surfaces the conditions for theorising co-predictive relations as an alternative. In doing so, it explicitly addresses AI's ontological and epistemological assumptions through the lens of critical design praxis, setting the stage for what follows. To guide this shift, I introduce a starting conceptualisation of *queer, queerness,* and *queering* (Fig.3) as distinct yet interconnected orientations within the project. Together, they offer a framework for reimagining the ontological, epistemological, and practical implications of refusal, subversion, and reorientation of predictive systems.

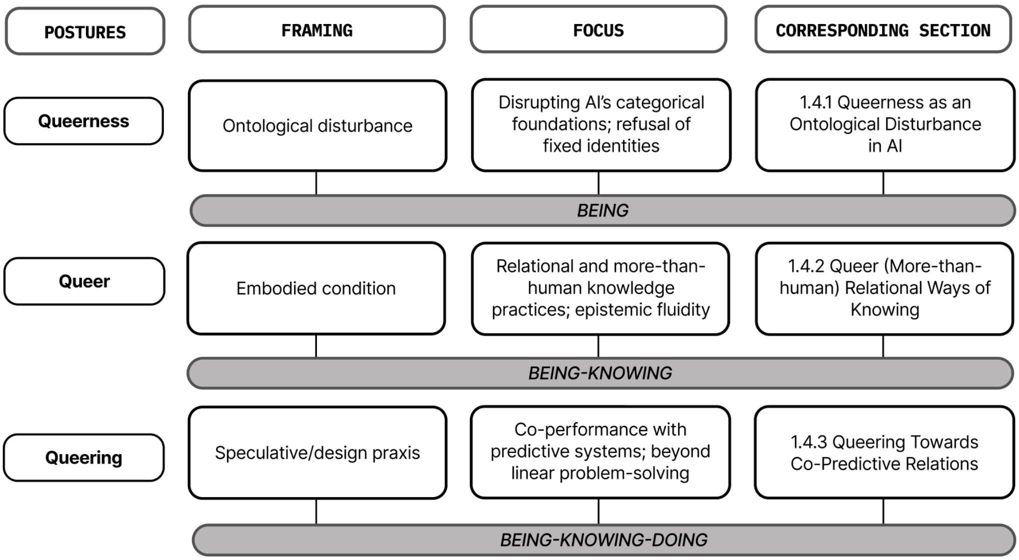

Figure. 3. From queerness to queering: Postures for reorienting predictive systems.

1.4.1. Queerness as an ontological disturbance to AI

By ontology, I refer to the implicit structures and assumptions that determine what can be recognised, categorised, or coded as real. Predictive AI systems, as previously discussed, operate through processes of abstraction that encode human and non-human entities into discrete, identifiable, and fixed categories, thereby foreclosing the possibility of indeterminate or queer futurities to emerge. In response, queerness as an ontological disturbance involves not just imagining new futures but disturbing the grounds on which prediction is made possible. Following Lucy Suchman (2007), I wonder: if subjectivities and agencies are increasingly configured through human–machine relations, how might they also be *imaginatively* and *materially reconfigured?* The problem lies not only in how social and cultural markers, such as race, gender, class or sexual orientation are categorised, but also in the structural powers and politics that enforce these categories through AI's predictive logics (Amaro, 2022; D'Ignazio & Klein, 2023; Guyan, 2022; Hinds & Joinson, 2018).

To read AI through the lens of queerness is to view it as a site of cultural performativity, where subjectivities, more than being rendered visible, are co-performed. As Judith Butler (2002) argues, performativity is not theatrical self-presentation but a reiterative discursive act through which the subject emerges. The categories and classes assigned to algorithmically encoded computational systems, much like gender, become a performative apparatus: through repeated classification, encoding, and output, AI systems produce the characteristics of an entity, for example, what counts as a subject, object, what can be anticipated, and what remains unimaginable. Paul Preciado (2013) similarly theorises gender as a cultural fiction, a performative effect of open-ended discursive practices. Extending this logic, AI systems too can be understood as engaged in an ongoing discursive production of the real, of identity, and of possibility.

When we recognise the interdependence of the social and the technological, and when more-than-human entanglements are made explicit, the troubling force of queerness takes on ontological dimensions. Take, for instance, Barad's (2011) notion of "nature's queer performativity," which troubles entrenched nature/culture divides by arguing that "all sorts of seeming impossibilities are indeed possible, including the queerness of causality, matter, space, and time" (p. 24). Anne M. Harris and Stacy Holman Jones (2019) similarly explore the "queer life of things" and the ways more-than-human performativity enacts alternative ways of being through affect and embodied relation. In this context, to return to what this means for AI and its ontological disturbance, queerness might offer more than critique. It can act as a refusal or subversion of ontologies of separability (da Silva, 2016), those epistemological foundations or presumed knowledges that frame human, machine, and world as discrete, stable, and governable categories. In short, queerness challenges not only what AI predicts, but how and from where we come to know something. These epistemic questions will be expanded further in *Queer relational ways of knowing* (Section 1.4.2).

1.4.2. Queer relational ways of knowing

If queerness as an ontological disturbance unsettles the categorical grounds of prediction, it forces us to question the epistemic conditions of these systems: From where is knowledge itself made, validated, and distributed? This section explores how queer and relational ways of knowing fold in on one another across design and culture. Drawing on Glissant's (1990) writings on relations, I examine how queer knowledge practices are found not in the root but "through the relation" (p.18). This aligns with Barad's (2007) assertion that 'the nature of knowing cannot be separated from the nature of being' (p.185). Because the ontological and the epistemological are inseparable, we must reposition the study of knowledge practices as an onto-epistemological configuration. Building on this framework, I turn to the onto-epistemic conditions of queer mestiza knowledge production. Here, relationality is foundational. As Escobar (2020) argues, relationality drives the ontological and epistemic reorientation of design (p.30). He describes an 'onto-epistemic border where pluriversal theoretico-practical design projects might emerge... [acting] as alternative operating systems enabling autonomous forms of design' (Escobar, 2018, p.515). He situated this in the lived contexts of intersectional, Afro, Indigenous, and Campesino ways of knowing in contemporary Colombia—worlds rendered illegible within dominant Eurocentric epistemologies.

Decolonial feminist scholars (da Silva, 2015; Segato, 2016) push further, urging the unlearning of ontologies of separation that prefigure notions of singular realities and order more-than-human bodies and worlds. This is an invitation to revisit not only what we know, but from where we know. This differs from the more common theoretical 'oscillations' within more-than-human design, posthumanism, new materialism, and postphenomenology (Lindley et al., 2023, p. 1), by situating knowing in embodied, intersectional, and geopolitical relations. Yet the more-than-human remains active here insofar as queer mestiza epistemology unsettles the human knower, recognising that knowledge is co-produced across human and non-human bodies, territories, and relations — a point that becomes materially concrete in the experiments that follow (Chapters 2–4). Through a queer lens, Eve Sedgwick (1990) troubles dominant epistemologies of sexuality, critiquing the rigid binaries that structure Western thought, for example, homo/hetero, masculine/feminine, subject/object binaries that produce and constrain meaning. In design terms, this shifts away from "knowing what the thing is and how it comes to be this thing" (Giaccardi & Redström, 2020, p.39).

A queer mestiza epistemology revises feminist approaches to epistemology in AI by foregrounding the identity of the knowing subject in AI development. This resists the implicit hierarchy of knowers that privileges the perspectives of those who design and build AI systems over other forms of knowledge. As Genevieve Bell et al. (2021) describe, the prevailing research agenda remains gendered in the way it reinscribes binaries and engenders human and non-human bodies, experiences, and life/worlds with homonormative (masculine) logic and meaning (Bell et al., 2021; Keyes, 2018). In response to the gendered underpinnings of AI research and design, I make a theoretical shift to the borderlands, moving beyond Western dualist epistemology and binaries of sexual difference, and bridging forms of knowing and ways of being embedded in design theory and practice. The borderlands, which I elaborate further in *Autotheory as a queer design methodology* (Section 1.5) and through the experiment *Undoing Gracia* (Chapter 3), draws on Anzaldúa's (2021) conception of a psychic, social, and cultural terrain where we encounter the world differently. It is an epistemic and cultural orientation that is already relational. It is a place to celebrate unknowability and uncertainty. A liminal space where we lose our way to find our way (Ahmed, 2006).

A border subject inhabits multiple, often conflicting cultural, linguistic, and epistemic positions, navigating liminality and fusion. Designing from queer, trans, and Mestizx standpoints foregrounds intersecting positionalities. Mestizx (with an "x") expands Anzaldúa's "Mestiza" consciousness to include intersectional, transfeminist, and decolonial ways of knowing, directly challenging the Western heteropatriarchal logics that shape AI (Anzaldúa, 2021; Lugones, 2003). Kalantidou and Fry (2014) introduce "border thinking" into design discourse as a space where intellectual borders are crossed and disciplinary hierarchies dismembered (p.3). As Paul Preciado (2018) notes, "the crossing is a place of uncertainty, of the unobvious, of strangeness. It is not a weakness, but a power" (p.33). If we turn to knowledge traditions in design, queer embodiment must, as Fry and Nocek (2020) propose, "bring into being the material and political conditions for the human that refuse to authorise its exceptionalism in any form" (p.3). Following Willis (2006), Escobar (2020) makes the vital connection that "everything that is designed has an ontological dimension" (p.39)—design always implicates the making of life/worlds. As Escobar (2018) asks, "How do we develop forms of knowing that do not take words and beings and things out of the flow of life—that is, forms of knowing and being that do not recompose nature as external to us, as dead or insentient matter?" (p.504). Manzini (2015) adds that "design is a culture and a practice concerning how things ought to be in order to attain desired functions and meanings" (p.53).

Together, these perspectives open new cultural visions and modes of knowledge production for transforming how probabilistic outcomes and prediction are addressed through design. Queer ways of knowing may point to a different episteme for design in the age of AI, one that attends to the intimate entanglements of cultural, political, social, and technical relations that have long characterised design practice, yet are intensified in data-driven design processes and operationalized through the use of predictive systems. This "border thinking" counters the universalizing tendencies of mainstream AI design, challenging the assumption that one "best" model can neutrally represent the world. In a similar vein, Christopher Frauenberger (2021, p.81) argues for rethinking meaning-making between humans and "smart" things through entanglement theories that unsettle and decenter the human. Within these theories, relational ontology recognises that all actors, human and non-human, co-produce the conditions of knowing (Nicenboim et al., 2023). Turning toward co-predictive relations from the onto-epistemological borderlands of queerness may open new ways of distributing futurity, agency, and performativity with predictive systems.

To recap, queering, extending engagements with *being* and *knowing,* offers a praxis and mode of doing that invites active, subversive, and speculative engagement with unpredictable, rhythmic insurgencies of life yet to be seen, heard, and felt. Rather than accepting AI systems as neutral tools, this research draws on queer, transfeminist, and plural ethics and politics to reimagine AI from the inside out, remaining open to indeterminate conditions of relational life/worlds. From this position, I set the stage for the methodological approach I apply to carry out this research, outlined in *Autotheory as a queer design methodology* (Section 1.5), where autotheory is taken up as a queer method in the design of predictive systems to enact and examine these relations in practice.

1.5. AUTOTHEORY AS A QUEERING DESIGN METHODOLOGY

Hearken unto me, fellow creatures. I who have dwelt in a form unmatched with my desire, I whose flesh has become an assemblage of incongruous anatomical parts, I who achieve the similitude of a natural body only through an unnatural process, I offer you this warning: the Nature you bedevil me with is a lie. Do not trust it to protect you from what I represent, for it is a fabrication that cloaks the groundlessness of the privilege you seek to maintain for yourself at my expense. You are as constructed as me; the same anarchic womb has birthed us both. I call upon you to investigate your nature as I have been compelled to confront mine.

—Suzan Stryker, "My Words," quoted in TransMaterialities by Karen Barad (2015, p.240-41)

This section positions autotheory as a methodological and epistemic (re) orientation where the personal, the theoretical, and the political converge, refusing strict divisions between lived experience, critique, and making (Fournier, 2021). Building on the ontological and epistemological shifts outlined in *Turning from prediction to co-predictive relations* (Section 1.4), I now turn to the question of how such shifts can be enacted in research. If co-predictive relations reframe prediction as a shared, situated, and ongoing co-performance, then autotheory offers a way to research and design from within that reframing. It is not only a method but an onto-epistemic stance—a refusal to separate theory from the embodied and political conditions of its making.

As a distinctly queer method for AI research and design, I characterise autotheory as a redirective design practice (Fry, 2009). Operatively, autotheory supports the previously mentioned and necessary shift within AI research and design from identity politics toward a politics of possibility, moving beyond assumptions that queerness centres solely on gender or sexuality toward embodied transformations of futurity, plurality, and relationality.

1.5.1. What is autotheory

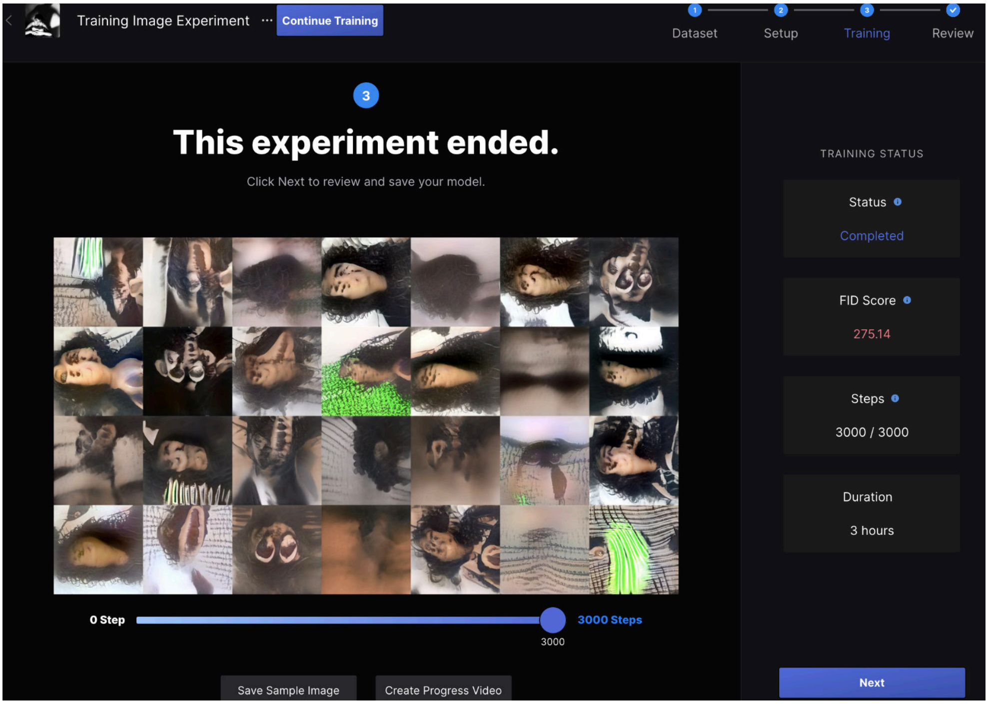

Figure 4. Example from Mutant in The Mirror (Chapter 2) autotheoretical experimentation.

Autotheory is a methodological and epistemic (re)orientation of first-person research in which the personal, the theoretical, and the political converge, refusing any strict division between intellectual critique and lived experience (Fournier, 2021). It draws on rich traditions of queer, feminist, and trans writing, for instance, Anzaldúa's (2021) autohistoria-teoría, Paul Preciado's (2013) "body-essays," and Sara Ahmed's (2006) Queer Phenomenology. Unlike classic autoethnography, which often treats the researcher's personal experience as data to be analysed through a theoretical lens, autotheory enacts theorising from the inside out, where personal, design-driven processes become the crucible for creating new conceptual frameworks.

Following Raha and Van der Drift (2024), I depart from binary accounts of representation that lead to the homogenisation of categories and positionalities that socially locate an individual subject. Instead, embodying and moving in and through transitions, this work describes the wayward paths of autotheorising within the disciplinary borderlands of critical and more-than-human design. The unsettling of positionality within the research is enacted to embody a sense of plurality that arises in pluriversal worlds (Escobar, 2018; 2020). This practice of movement, Raha and Van der Drift

(2024) propose, is a way to acknowledge and learn from histories of oppression without being defined or coded under dominant social, cultural, and political categories (race, ability, gender) or capitalist regimes. Working through new form-finding of a transversal and plural understanding of the autobiographical subject (in this case, myself) can reveal new forms of more-than-human relation and collectivity.

1.5.2. Rationale for autotheoretical AI research and design

While first-person methods have gained traction in design research as a way of embedding lived experience into the research and design process (Desjardins et al., 2021; Lucero et al., 2021), conventional approaches that prioritize human agency alone prove insufficient for studying subjects unsettled within complex human–AI relations, where agency is distributed across people, data, and computational systems (Giaccardi et al., 2016; Murray-Rust et al., 2019). By contrast, this research turns to autotheory, which merges lived experience with theoretical critique, knowing with realising, to "bring (queer) theory back to life" (Ahmed, 2017, p.10). Drawing from Anzaldúa's (2021) autohistoria-teoría, this methodology articulates the experiences of a researcher (auto) expressing queerness (theoretical) in the queering (practice) of AI systems.

1.5.3. Embodying a politics

As an intentional shift away from researcher neutrality, autotheory embraces self-experimentation, or body-essaying (Preciado, 2013), positioning embodied experience within knowledge production. Subjective accounts emerge through theoretical exploration, not merely as experiential data. This research critically examines a trilogy of autotheoretical experiments conducted between 2021 and 2024 that explore the intimate entanglements of human–AI relations (Chapters 2–4). Across these, autotheorizing with design subverts normative classifications by engaging with disidentification as a performative politics (Muñoz, 1999) and inhabiting the "border subject" (Anzaldúa, 2021). As a redirective design–research practice, autotheory explicitly entangles research and design, treating the researcher's positionality not as a bias to be minimized but as an epistemic resource and design material. In this sense, epistemically generative names a specific move to use the researcher's intersectional, situated, and queer perspective to surface and rework the ontological and epistemological assumptions that underwrite predictive systems, thereby expanding possibilities for designing complex sociotechnical relations within "pluriversal" life/worlds (Escobar, 2018).

Paying attention to how bodies and positions matter in knowledge production resonates with, but also extends, existing somatic work in HCI research and design. Somatic approaches foreground how bodies, human and non-human, participate in shaping computational processes through embodied interaction (Höök et al., 2019). Building on embodied and somatic traditions in HCI, my approach asks not only how bodies feel and act with technology, but also how those bodies are situated within systems of power, dominant ideologies, normativities, and differences. Many first-person methods in design and HCI, even when they involve the researcher's or participant's body, still detach that embodiment from its sociopolitical and epistemic context (Rogers, 2012). This becomes particularly limiting when AI systems co-produce identities and futures, as algorithms increasingly shape, and are shaped by, human subjectivities.

Moreover, these approaches often separate personal narrative from theoretical critique, limiting their capacity to address power, representation, and intersectionality (Benjamin, 2019; Howell et al., 2021). As illustrated in the trilogy of experiments, conventional first-person research offers few tools for engaging nonhuman or more-than-human entities—such as neural networks, bees, or sonic data—on equal footing. By contrast, autotheory offers a way to merge embodied experience with direct theoretical interrogation (Fournier, 2021) while foregrounding how AI research intersects with queerness, dis/ability, and trans theories (Keyes, 2018; Preciado, 2008).

1.5.4. Becoming futural

Autotheory as a practice invites a form of self-play that is distributed and co-performed. In technical literature, self-play describes computational agents improving through recursive encounters with themselves for optimisation (Silver et al., 2018). By contrast, the self-play enacted through autotheoretical experimentation is co-predictive, distributed across human, non-human, and algorithmic agents. It challenges the presumption that futures are best shaped by stable, measurable objectives. By reframing simulations, such as the multi-agent simulation model, Gracia, detailed in Chapter 3, as a sandbox for world-making rather than a rigid predictive control systems, predictive systems, like a simulation model can then open space for desirable unpredictability, from which to encounter difference in generative and safe way, while revealing new possibilities for trans/formation, from subject/object relations to collective actions.

HCI has historically leaned toward techno-optimistic solutions, often privileging linear cause-and-effect reasoning or usability metrics (Rogers, 2012). Yet there is an alternative tradition in critical and speculative design of engaging sociotechnical phenomena through counterfactual experiments, imaginative play, and "what if?" questions (Dunne & Raby, 2013; Light, 2015). As Forlano and Halpern (2023) argue, the field must accommodate "diverse modes of thinking, making, and doing" (p.23) to address the complexities of contemporary AI.

In my work, speculation arises from the intersections of the autobiographical and the theoretical within experimental practice. The researcher's experience of transformation is part of the experiment toward unsettling and deterritorialising co-predictive relations. In Raha and Van der Drift's (2024) words, the aim is to become "participants in the struggle for [algorithmic] liberation, rather than disembedded observers" (p. 116). Co-performative interventions ask how co-predictive relations might open contradictory futures. These interventions remain situated, embodied, and performed-with across every phase of AI development—from data capture to post-training interaction.

Rather than treating error as a nuisance, this approach treats unpredictability and misalignment as generative forces, revealing alternative forms of co-prediction, identity negotiation, and relational attunement. By refusing the impulse to finalise or "solve" the future, these speculative acts hold space for emergent, queerly fluid possibilities.

1.5.5. Transforming Positionality

Central to autotheory is the recognition that the researcher's position and form, shaped by gender, race, class, cultural identity, and ability, deeply informs the lived experiences that shape research, the questions asked, and the kinds of knowledge produced (Fournier, 2021; Iosefo et al., 2021). In this dissertation, I neither regard AI as neutral nor purely instrumental; instead, I view it through a political lens, attentive to the ways algorithmic systems classify, misrecognise, or erase marginalised identities (Benjamin, 2019).

Researcher positionality

Some identifiable codes, irrespective of choice, I carry with me: I am Colombian–Australian, born in Ngunnawal and Ngambri Country. I am an AI researcher, designer, sometimes futurist, consultant, educator, and, perhaps most importantly, always a student. I straddle worlds, attempting to locate myself in time, space, and form. If I could opt out of archetypal boxes I am asked to tick, I would glitch the box, and seep out through its cracks. I would rather characterise my dis/identification as fluid, incommensurate, like the ocean of Yuin Country spilling over the South Pacific, crossing into the La Guajira, where the desert meets the sea. Yet, my wayward wanderings situate me here, in time and space in Amsterdam, writing from an artificial island demarcated by a few intricately woven canals, 2 metres below sea level. Following Anzaldúa (1997), I understand myself as a queer mestiza, a border subject thinking and feeling in Spanglish; a neurospicy, dyslexic, gender-variant body expressing contradictory multitudes. These wayward paths and codings of lived experience, muddied with theory, are the material I bring to queering AI, and what has been termed as co-predictive relations, and what will later be discussed (Chapter 5) as designing for co-predictive relations.

As a *musing on redirective practice traversing art and science through the back door of design, I speak to positionality not as an afterthought but as an active component in knowledge production within AI research and design.* My presence as a *border subject—a queer mestiza designer*—entangles with the technologies I create, generating conceptual openings that emerge from this specific positioning in dialogue with theory and practice. Here, autotheory within design functions as a redirective design practice (Fry, 2009), an invitation to perform politics within AI research and design.

Algorithmic A-positionality

If my position is multiple, embodied, and in motion, algorithmic systems are often designed as if they were a-positional. Yet, no model is without position: each encodes a vantage point, a set of partial perceptions, and a logic of action. As Patricia Reed (2014) warns, these systems force the plurality of the world into reductive templates, omitting the vital contexts from which one knows, learns, and acts in relation to specific situations and phenomena. Predictive systems do not simply represent locality; they co-produce it through relations across physical and virtual dimensions. In operational terms, this means that human and non-human entities, data, processes, and platforms are co-constituted in the act of prediction, transforming agency itself.

Where my positionality is declared, situated, and accountable, algorithmic positionality is flattened into the illusion of neutrality. Where I openly acknowledge my vantage, AI systems mask theirs behind abstraction. Yet both are active in the co-production of life/worlds. The tension between a lived, self-declared position and a system's obscured vantage is precisely where this dissertation situates its methodological intervention: treating positionality, not as bias to be eliminated, but as an epistemic condition to be surfaced, interrogated, and designed with.

1.6. AUTOTHEORETICAL EXPERIMENTATION

This research aims to cultivate fundamentally new ways of knowing and understanding systems, how to unsettle them and open them up to new configurations and outcomes (Voss, 2024). In doing so, it moves beyond systems literacy toward systems co-performativity, realized through process-led autotheoretical experimentation. It holds a deep commitment to minor or minoritarian gestures. Borrowing from Manning (2016), who extends Deleuze and Guattari's notion of minor literature, whereby language is deterritorialized from dominant structures, Manning argues that minor gestures counteract normative modes of experience. Rather than oppositional critique, minor gestures operate through subtle variations that make dominant systems stutter, revealing their contingency.

Working from an embodied ethic and the politics of queerness, the trilogy of autotheoretical experiments that give this dissertation body and context aims to unsettle the design assumptions encoded in predictive AI systems. These experiments create flexible spaces for what Doty (1993) describes as all that is non-, anti-, or contra-straight in cultural production— now extended to AI development. Each experiment speculates on the unpredictability that emerges when bodies, code, and data collide (Bogost, 2010), traversing every phase of AI, from data capture to model training and post-training interaction, while remaining situated, embodied, and performed-with.

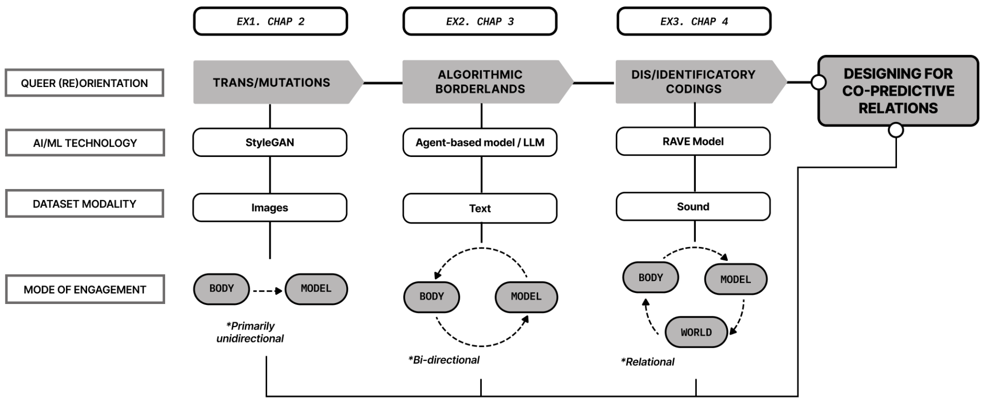

Figure 5. Illustrative view of the interdependencies of the trilogy of experiments.

(Fig.5) (Autotheoretical experimentation summary) maps the trajectory of the trilogy, showing how each experiment explores a different queer (re)orientation in relation to AI models, datasets, and modes of engagement. The progression moves from unidirectional flows between body and model to bidirectional exchanges to relational configurations that implicate the physical world as part of an interwoven territory.

1.6.1. Experiment summaries

The trilogy works iteratively and relationally: each experiment extends and transforms the insights of the last, building a cumulative method for queering AI design. Across the series, co-predictive relations emerge through co-performative intra-actions between human and algorithmic agents, where agency and subjectivity are distributed and recursively reconfigured.

Mutant in the Mirror (Chapter 2)
RQ4: *What if AI were queer, a non-binary relational entity, mutating and dynamically becoming-with us humans? How might this challenge notions of algorithmic determinism?*

• **Summary:** An exploration of mutual mutation as a way to fold matter and meaning, bits and atoms (Halberstam, 2011). I worked with GAN-generated portraits to explore how AI reifies or disrupts normative, binary, or singular and stable visions.

• **Contribution** (summary): The experiment demonstrates how dataset composition and model training can be reoriented toward fluidity, plurality, and trans/formation in AI outputs.

Undoing Gracia (Chapter 3)
RQ5: *What if the self emerges as multiple, pluralising through the borderlands of human-AI entanglements? How might this challenge normative distinctions of individuation?*

• **Summary:** An exploration of the borderlands as a space to inhabit multiplicity and uncertainty (Anzaldúa, 2021). I worked with AI twinning agents to explore how simulation models trouble stable identity and enact pluralised selves.

• **Contribution** (summary): The experiment demonstrates how human–AI entanglement can be reoriented toward epistemic and ontological plurality, resisting binary logics in model design.

Sounding Territories (Chapter 4)
RQ6: *What if disidentifications, as the performance of politics, could subvert algorithmic identification? How might this transform understandings of human–AI relationality?*

• **Summary:** An adaptation of disidentification as a strategy to subvert algorithmic classification (Muñoz, 1999). I worked with movement and a deep learning sonic model to explore how embodied performance disturbs predictive certainty.

• **Contribution** (summary): The experiment demonstrates how computational tactics can be reoriented toward unfinished and negotiable relations, offering disidentification as a means to unsettle predictive systems.

1.7. EMERGING CONTRIBUTIONS

Broadly, this dissertation contributes to a way of queering AI through co-predictive relations. First, it introduces co-predictive relations and a set of queer practice (re)orientations to reframe prediction as situated, relational, and ongoing co-performance between humans and algorithmic systems. Second, it advances an autotheoretical, design-led methodology for AI research that entangles positionality, lived experience, and speculative experimentation through a trilogy of projects—Mutant in the Mirror, Undoing Gracia, and Sounding Territories. These projects work with data capture, model training, and interaction as sites for queering predictive systems from within. Third, it sketches an emerging schematic for practice through three queering tactics: embodying a politics of possibility, attuning to latent relations, and sensing queer futurities. These tactics invite designers, researchers, and practitioners who are navigating the margins of AI, whether in academia, government or industry, to cultivate desirable unpredictability, more fluid queer futurities, and pluriversal life/worlds. Together, these contributions speak not only to AI and HCI research but also to broader imaginaries of AI futures grounded in plurality, inter/dependence, and a performative politics of possibility.

These contributions are evidenced by the following outputs:

→ Journal Articles

• Turtle, G. L., Giaccardi, E., & Bendor, R. (2025). Undoing Gracia: Queering the self in the algorithmic borderlands. *AI & Society*. https://doi.org/10.1007/s00146-025-02661-8

→ Book Chapters & Conference Proceedings

• Turtle, G. L., Kotowski, B., Giaccardi, E., & Bendor, R. (2026). Sounding territories: Dis/identificatory codings as relational worlding. In Proceedings of DRS2026. Design Research Society. https://doi.org/10.21606/drs.2026.2162

• Turtle, G. L. (in press). Autotheory as a queer design methodology. In D. Tauches & J. Westbrook (Eds.), *The Bloomsbury Handbook of Queering Research Methods*. Bloomsbury Publishing.

• Turtle, G. L. (2022). Mutant in the mirror: Queer becomings with AI. In D. Lockton et al. (Eds.), *DRS2022: Bilbao*. Design Research Society. https://doi.org/10.21606/drs.2022.782

• Turtle, G. L., Guerrero Millan, C., Özçetin, S., Patil, M., & Bendor, R. (2022). Sensing in the wild: A DCODE DRS lab exploring a more-than-human approach to distributed urban sensing. In D. Lockton et al. (Eds.), *DRS2022: Bilbao*. Design Research Society. https://doi.org/10.21606/drs.2022.912

• Turtle, G, Bendor, R (2024). Co-predictive relations. In E. Giaccardi & R. Bendor (Eds.), *Rethink design: A vocabulary for designing with AI* (pp. 29–31). TU Delft OPEN Publishing. https://doi.org/10.59490/mg.110

→ Other Contributions (Non-Academic & Para-Academic) / Selected Writing

• **Essay:** Contribution to *A Book for Disappearance* (Cthulhu Books, 2024).

• **Column:** Time of monsters: Designing from an embodied politics of possibility, *4TU.Design United* (2025).

→ Invited Talks & Lectures

• **Queering human–AI co-predictive relations,** *On the Seam Symposium,* University of Applied Arts Vienna & Futuress (Online, 2025).

• **Hoy, la vamos a tumbar: Queering AI and attempting to escape Gracia,** *After AI Symposium* (2025).

• **Designing sustainable worlds: Creative futures in education,** Norwich University of the Arts (Online, 2025).

• **Design Probes,** Domus Design Academy (Online, 2025).

• **Mariconeando a la AI: Designing for co-predictive relations and queer futurities,** *DI Hybrid Research Seminar,* University of Edinburgh (2024).

• **More-than-human design experiments,** Lecture, TU Delft (2023).

→ Workshops, Exhibitions & Podcasts

• **Workshop:** Thought Collective' on queer AI in Margate, Shifting Power: Artificial Intelligence and Justice, The Open University, Margate, Kent (2025).

• **Workshop:** Queering hype: A playable workshop, *Hype Studies Conference,* Barcelona (2025).

• **Workshop:** (Queering) AI control wars, Society 5.0 Festival, *Center of Expertise for Creative Innovation, Amsterdam* (2022).

• **Exhibition:** *DCODE Showcase,* DCODE Network (2023).

• **Presentation:** Sounding Territories, *Tacit Engagement in the Digital Age (TEDA),* University of Cambridge (Video, 2024).

• **Podcast:** Undoing Gracia feat. Grace Turtle, Stegi Radio (Onassis Stegi), Athens, Greece (Podcast episode, 2023). https://stegi.radio/show/undoing-gracia-feat-grace-turtle-2023-06-26

• **Podcast:** The Good Robot LIVE! from Berlin, The Good Robot (Podcast episode, 2023). https://www.thegoodrobot.co.uk/post/the-good-robot-live-from-berlin

• **Podcast:** Acts of Interfacing (DCODE Conversations), DCODE Network

(Spotify episode featuring Grace Turtle) (2023). https://open.spotify.com/episode/5LJoGPTeLywhSWZrZuR1mE

→ Residencies & pedagogy

• **Residency:** Sounding Territorios, Fundación Organizmo, Tenjo, Colombia (Nov 2023 – Feb 2024).

• **Pedagogy:** Co-Head & Co-Developer, Monstrous Futurities: practices in (un)learning, (un)making and (un)worlding, Temporary Master's Programme (2025-2027), Sandberg Instituut. The core concepts and queer methodologies of this research were adapted to shape the programme's curriculum and pedagogical design and engagement with lived experience and autotheoretical forms of experimentation in art and design education.

Audience, dissemination, and impact

Situated within Design Research, HCI, and Critical AI, this dissertation has been written with multiple, sometimes incommensurable audiences in mind—from queer and trans artists, designers, and activists to more conventional AI and design communities. Throughout, I have worked with different degrees of visibility and opacity, asking how the work might remain accountable to queer and trans lineages while still being legible to those newly encountering these concepts. **This tension is not simply a matter of communication; it is part of the research contribution itself.** It is enacted across academic publications, para-academic essays, and public-facing formats such as exhibitions, workshops, and talks, including those developed in conversation with situated communities in the Global South (e.g., the Sounding Territories residency at Fundación Organizmo in Colombia) and through collaborations with platforms like Futuress, Cthulhu Books, and the DCODE network.

The impact of this research is therefore less about quantifiable outcomes and more about how a queer design practice travels, is taken up, and re-worked in other contexts. The autotheoretical and queer methodologies developed here have informed pedagogical experiments such as the Sandberg Instituut's temporary Master's programme *Monstrous Futurities,* invited workshops and seminars that centre marginalised perspectives, and "playable" formats that give researchers, students, and practitioners concrete tactics for queering data and AI systems. In this sense, the dissertation contributes not only to critical debates on AI, but also to an ongoing pedagogical project: cultivating the conditions through which people can learn to see, sense, and design AI otherwise.

1.8. DISSERTATION OVERVIEW

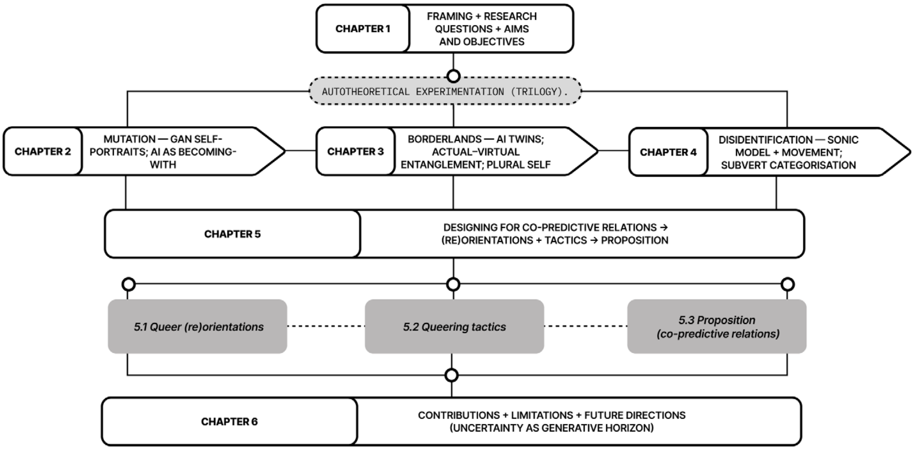

Figure 6. This dissertation unfolds as an iterative, relational inquiry into queering AI through autotheoretical experimentation.

Chapter-by-Chapter breakdown

Chapter 2: Mutant in the Mirror: Queer becomings with AI
Explores queer mutations through the co-creation of a generative adversarial network that reimagines my bodily image. The experiment asks what becomes possible when AI is approached as a non-binary relational entity, mutating and dynamically becoming-with its human co-performers.

Chapter 3: Undoing Gracia: Queering the self in the algorithmic borderlands
Investigates the algorithmic borderlands by engaging agent-based modelling to prototype "AI twins" that mirror, mutate, and refract my digital self. This experiment inhabits the liminal space between human and algorithmic subjectivities, unsettling normative individuation.

Chapter 4: Sounding Territories: Dis/identificatory Codings as Relational Worlding
Works with disidentifications by transforming environmental data into a deep learning–driven sonic model. Through movement and improvisation with a dancer, we co-perform with the system, exploring how bodily gestures might subvert or redirect algorithmic categorisation.

Note: Chapters 2–4 have been published previously as conference and journal articles and are reproduced here with reformatting for consistency within the dissertation structure.

Chapter 5: Designing for co-predictive relations
Brings the trilogy into conversation, synthesizing findings through two structuring devices: queer (re)orientations and queering tactics for designing for co-predictive relations. This chapter consolidates how the experiments collectively offer insight into queering AI through the design of co-predictive relations and culminates in a proposition for design/ing for co-predictive relations as a redirective practice. In doing so, it offers an emerging schematic for practice, practical ways for AI research and design to engage queerness as a politics of possibility, relationality, and futurity.

Chapter 6: Conclusion and future directions
Reflects on the research's contributions, limitations, and future possibilities. It revisits the central research questions, articulates the theoretical, methodological/experimental, and practical contributions of queering AI through co-predictive relations, and outlines directions for future work. Chapter 6 reconsiders AI as a site of ongoing co-predictive emergence, an approach that engages uncertainty not to eliminate it, but to live with it as a shared, generative horizon of possibility.

EXPERIMENT 1 OF 3: MUTANT IN THE MIRROR (2021-2022)

2. MUTANT IN THE MIRROR: QUEER BECOMINGS WITH AI

This chapter was originally published as: Turtle, G. L. (2022). Mutant in the mirror: Queer becomings with AI. In D. Lockton et al. (Eds.), *DRS2022: Bilbao*. Design Research Society. https://doi.org/10.21606/drs.2022.782

This chapter initiates the dissertation's empirical inquiry, drawing on queerness as a theoretical grounding to reimagine how design might engage with and imagine AI differently. As the first of three autotheoretical experiments, it asks what becomes possible if AI is approached as mutant and in-becoming—dynamic, relational, and resistant to stable categorisation—and how such a reorientation might shift the ways AI shows up in and acts on the world (with us). The chapter details *Mutant in the Mirror:* a practice-based experiment that trains and iterates a Generative Adversarial Network (GAN) on a self-portrait dataset to produce algorithmic outputs I frame as "trans/mutations." It moves from conceptual grounding, to dataset and model configuration, to situated encounters with the generated images, using autotheoretical reflection to examine how legibility, identity, and representation are performed and contested through generative systems. Mobilized as a design tactic, trans/mutations unsettle how AI is understood today and makes space for new propositions to take root, staging forms of refusal that open questions of cultural computability and self-determination.

Keywords: queerness; autotheory; artificial intelligence; co-performativity

2.1 INTRODUCTION

Any engagement with Artificial Intelligence (AI) imaginaries, cultural fictions and technological possibilities, must begin with how they show up, move through and act on the world – a world characterised by increasing technological complexity, met with a diminishing understanding of it (Bridle, 2018). AI in particular has been described as a gendered research agenda (Bell & Broad et al, 2021) while also being characterised as a kind of magical alien cognition, gesturing towards what Alexander Campolo and Kate Crawford (2020) have termed as enchanted determinism, or a "form of power without knowledge" outside the scope of science (Campolo & Crawford, 2020, p. 5). Herein lies a dilemma for designer and Human Computer Interaction (HCI) communities: knowing how to engage with AI in imaginative ways while negotiating the deterministic and calculative power of AI systems that orientate us towards particular futures and that can inadvertently reinscribe and intensify social processes of classification and control (Campolo & Crawford, 2020). The challenges designers face in negotiating enchanting and deterministic characterisations of AI, then, call for an unsettling of how designers' interface with and imagine AI, inviting dialogue with other/d forms of situated knowledge and ways of understanding AI. Responding to this challenge, this chapter works towards a proposal for a differently situated way of thinking about AI through the lens of queerness. Queerness, I propose, can help unsettle dominant ontological encodings of AI shaped by culturally engendered, deterministic worldviews of late capitalism, and heteropatriarchal power arrangements bound to colonial violence and the epistemology of sexual difference. The goal for this research has been to reflect on AI logic and meaning critically and playfully through queer readings (Ahmed, 2006) aimed at challenging dominant understandings of how we turn towards and relate to AI systems today.

Using queerness as its theoretical grounds, the chapter functions as a mode of autotheory (Fournier, 2021) and speculative enquiry, that asks if another future is possible (Wilkie et al. 2017), while dialoguing with my personal experience in relation to queer theory. From an empirical point of view, this incipient research seeks to gain insight into new ways of thinking about the conceptual foundations, technological and imaginative possibilities for AI systems and by extension the co-performative interplay between the human and the artificial in the making of worlds (Giaccardi & Redström, 2020). I start by posing the question, *what if we understood AI as queer, a kind of mutant, in a state of becoming; a dynamic, relational, non-binary gender variant, how then might AI show up in and act on the world (with us humans) differently?* By posing this question, this chapter gestures toward the co-performative interplay between humans (myself included) and AI systems, in what Karan Barad has termed 'intra-action'—recognising that different entities do not merely interact with each other, but rather, define each other through their intra-acting agency and affective power to make and remake the world (Barad, 2003).

First, I explore the notion of queerness through which mutations arise, which I then apply to an autotheoretical experiment, using a Generative Adversarial Network (GAN) model to gain insight into queer readings and understandings of AI systems. As a tentative conclusion, I reflect on insight gained from the experiment, calling for an unsettling of dominant AI (socio)technical imaginaries through the lens of queerness, through which a new set of possibilities for *design/ing with, for and through* AI systems may emerge. As part of a larger research project, this chapter aims to contribute to design and HCI's engagement with AI as both a research agenda and field of practice by offering insight into the potentialities for what Donna Haraway (2013) has termed as a "becoming-with" AI at the level of the individual, as a society and as a species.

2.2. QUEERNESS, BORDERLANDS PERSPECTIVES FOR BECOMING WITH AI

As a starting point, this chapter acknowledges that queerness is necessarily indeterminate, ambiguous and always in relation, denoting flexible spaces for the expression of all aspects of non, anti, contra, straight, cultural production and reception. As put by Sara Ahmed, "queer lives are about the potentiality of not following certain conventional scripts, whereby not following involves disorientation" (Ahmed, 2006, p. 189). To re/orientate our understandings of human-AI relations, from the ontological and epistemological borderland of queerness and the gender binary, we first need some sense of what queerness means, and how it relates to (design/ing, with, for and through) AI systems. *Queerness* as described by José Muñoz (2019) in *Cruising Utopia: The Then and There of Queer Futurity,* can be read as a "horizon imbued with potentiality", an "open mesh of possibility", a rejection of the here and now, and a "basic desire to live otherwise" (p.1-2). Always rooted in queer life, struggle and liberation, queerness "is not yet here... we are not yet queer" (p.1-2). Accordingly, queer futurity, is less about expanding a range of choices (liberal freedom) than it is about transforming the kinds of beings we desire to be while embracing the mutating multitudes and entangled life/worlds that make up the pluriverse, or as referred to by the Zapatistas, "un mundo donde quepan muchos mundos" (a world where many worlds fit) (Escobar, 2018, p. 42). In considering queer becomings with AI, we cross a threshold, intentionally getting lost in the borderlands of possibility, what Gloria Anzaldúa proposes as a psychic, social, and cultural terrain that we inhabit and that inhabits all of us (Anzaldúa, 2021), by which they mean a terrain where we encounter the world differently; losing our way to find our way (Ahmed, 2006). Importantly, this place of uncertainty is not a weakness but a power to be explored (Preciado, 2021). In this way, queerness is just as much about playing with the kinds of beings we desire to be as much as it is about offering a critique of the cultural, sociotechnical systems that shape our becoming with AI.

To the binary logic of AI, differently situated knowledge(s) and other/ed perspectives, are destabilising, thus demonstrating that binaries, be they imagined, cultural or technological are not natural nor always necessary. Rather, the notion of queerness speaks to a politics of transitivity—relating to, or characterised by, transition, bringing into question modes of gendered embodiment within all humans and non-humans and their configurations (Halberstam, 2020). In this sense, a queering of AI is a proposal to unsettle western thought, including the dominant techno-deterministic, cultural fictions, social imaginaries and technical propositions that shape how we create for, through, and with AI relations. Calling for new postures or rather, "queer turnings," (Ahmed, 2006) towards AI, may help shift how we relate to, and move with AI differently. Mutations arising through queerness offer designers a borderland perspective to decode, indeed, to deterritorialise AI's fixed binary historical understandings and how it is deployed within everyday life (Deleuze & Guattari, 2005, p. 69). My point of departure, then, is to think about mutations as a means of unsettling common understandings of or postures towards AI today. This, I hope, will offer designers a mode of leaning into performativity, improvisation and uncertainty in their dealings with AI and the "mutable and the immutable features of our existence" (Levitas, 2013. p. 137).

2.3. EXPERIMENTS AND SPECULATIONS ON QUEER BECOMINGS WITH AI

In the following section, drawing particular inspiration from Paul B Preciado's *Testo Junkie: Sex, Drugs, and Biopolitics in the Pharmacopornographic Era,* I present an *autotheoretical* experiment and provocation centred on queerness. With this, I seek to problematise notions of normativity and the naturalisation reproduced in and through AI systems (Bell et al., 2021). It can also be read as a gesture of resistance, a means to rethink encoded meaning beyond binary hegemonic, heteronormative classification schemes, categorisation, and modes of production. It is, as Preciado (2013) has described, "a body-essay" – a kind of soma-technological fiction relating to self-experimentation (Preciado, 2013, p. 7), or a queer reading into the possibilities of becoming oneself, finding one's body "through becoming, embodying, a glitch" (Russell, 2020. p 151).

The experiment which I will introduce here as Mutant in the Mirror engages with the notion of queering AI by tinkering with an AI system that identifies patterns, abstract meaning and signification, classifies, categorises and creates. Along these lines, I return to the questions:

What if we understood AI as queer, a kind of mutant, in a state of becoming; a dynamic, relational, non-binary gender variant, how then might AI show up in and act on the world (with us humans) differently?

Mutant in the Mirror uses a readily available AI platform, in this case, one that uses StyleGANs, a class of deep learning generative and predictive modelling in which two neural networks compete by discriminating between real and fake data in order to become more accurate in their predictions. In this way, the StyleGAN model performs as a vehicle to grapple and play with the idea of mutations arising through AI and further queer becomings with AI. As such, the experiment itself is not reliant on the technical capacity or limitations of the platform as it is. I start my inquiry by considering how embodied subjects are constituted (subjected and subjugated) through configurations and delegations, intended or unintended, that shape potentialities emerging from the co-performative, intra-actions a user might have with an AI system. Accordingly, following Helga Nowotny's (2021) claim that the power of AI is performative, playing with the idea of performative power, noticing ways "an algorithm has the capability to make happen what it predicts when human behaviour follows the prediction" (Nowotny, 2021, p. 19). Here the notion of performativity, originating through certain speech acts that pronounce and affect the world can be understood as performative action (Butler, 2002), where action originates through recursive feedback loops of material and discursive practices. As it intersects with the performative, the 'co' used here recognises the agency of both the user and the AI system in producing action, where their becoming is mutually dependent on their intra-actions (Barad, 2003).

By way of a methodological grounding, I borrow from what Lauren Fournier (2021) has called 'autotheory,' used to describe a reflexive praxis that blurs the boundaries between living, making, and theorising. I couple it with speculative research (Wilkie et al. 2017). While all research is subjective in the sense that the researcher's perspective cannot be uncoupled from the research process, autotheory places the researcher's 'self' at the centre of inquiry, always in situated relation to the subject and theory being studied. In this sense, by placing myself within the experiment, I gain a more textured insight into AI, through my own queer experience, dialoguing with queer theory as a means to reflect on the felt, imagined-but-not-yet-real *(im)possibilities* of AI signaling in the present. Moreover, the experiment in itself is an act of co-performative, theatrical modelling—an ontological reflexive inquiry into the relational, recursive, emergent and exploratory production of fictive becomings with AI. Fictions crafted through co-performative action. Through this self-experimentation process[1] I hope to gain insight into the queer borderland relational aspects of the constituted self, mutating, becomings through co-performative intra-actions with AI.

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[1] Given the nature of the experiments, I feel it useful to articulate my situatedness and my orientation towards the theoretical cartography mapped throughout the text. In no particular order, I am a designer, artist and researcher. I am also a dyslexic, bi-cultural, queer, gender variant, Colombian-Australian, Spanglish is my mother tongue. Thus, I position myself simultaneously in relation to various spatial and temporal contexts and systems of oppression and privilege. From this position, I grapple with my own particular body biopolitics, worldviews, and ways of knowing, from which I act on the world as I negotiate my own mutability in becoming with AI.

2.3.1. The crossing, mutant in the mirror

I titled this ongoing experiment as the Mutant in the Mirror for the obvious reason that my self-reflexive experience of observing the images documented in Figure 7 quite literally feels like staring into a mirror, with mutants staring back at me. The potential variations of myself function as possibilities of myself, a uniquely queer skin anticipating changes to my *cyberidentity. It asks, do I present as ambiguous? I am a body in mutation? It is my intention.*

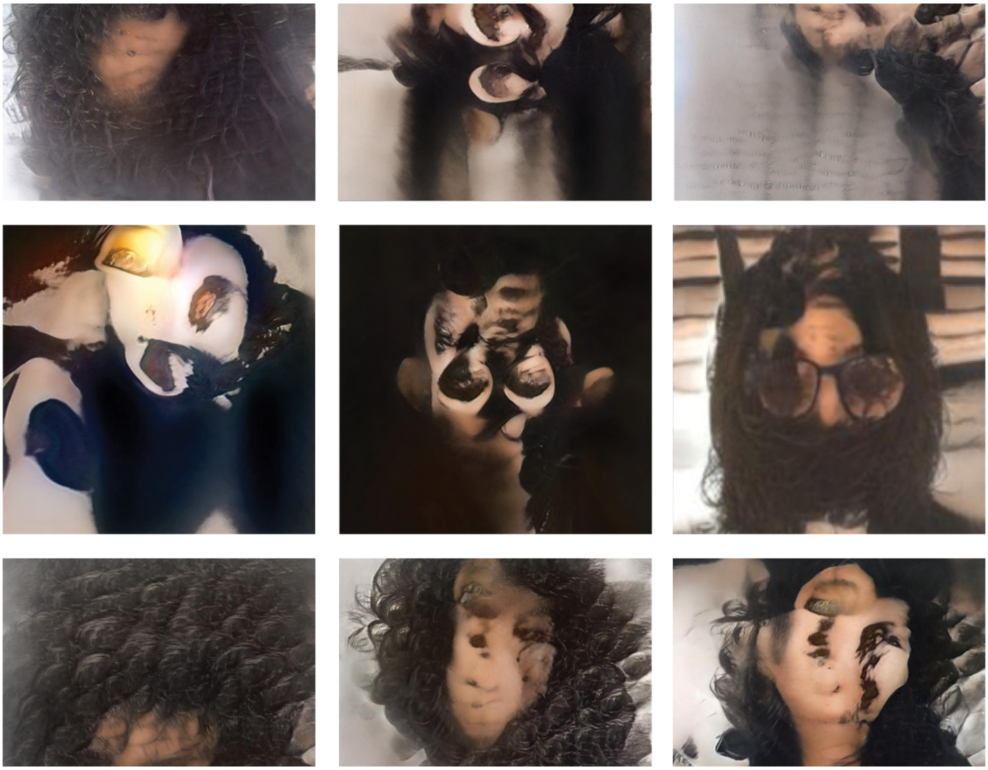

Figure 7. Selected image results from, Mutant in the Mirror, a StyleGAN experiment. The images represent a queer aesthetics of transformation, algorithmic imagination and mutation, suggestive of a queer becoming with AI.

Figure 7 presents some of the mutating synthetic image data produced through the experiment. The images act as a discursive device used to evoke reflections on the kind of queer, mutating becomings with AI discussed in this text. Generating multiple mutant images and synthetic data through the StyleGAN model provided me with a space to play with my own indeterminacy, fluid boundaries, (dis)identifiability and resistance to classification and categorisation. The particular model used in this experiment was trained using a small personal dataset of 70 images of myself across different places in time, some passing more feminine and some as more masculine. Also included in the training dataset were images of other humans (grandmother, mother, etc.), and non-humans (dog, places, etc.) that shape my sense of time, self and world. The 3000+ image(s) and video(s) generated through the experiment function as a mirror, through which I get lost, disoriented and reoriented, turning differently towards myself, my body, my corporal and subjective experience of the world. In this way, the Mutant in the Mirror comes to function as both a looking glass and a crystal ball, surfacing (speculations of) alternative versions of myself—mutating—becoming with AI. What started as an experiment in reimagining the notion of "fixed labels, fluid complexity" concerning who decides how we are read in the world, challenging deterministic binary modes of classification and categorisation (Crawford, 2021), concludes as an exploration into a metaphysics of interconnectedness (Anzaldúa,

Figure 8. StyleGAN image generations were classified, here rendered as hair spray with a classification score of 9.19 accuracy, and as a toy poodle with a classification score of 8.77. Other classifications included a mask and a fur coat.

2014). Indeed, a reflection on what (queer) intraactions with AI systems might tell us about ourselves when we co-perform with them.

So what did the experiment tell me about myself? *Am I a woman or a man? A subject or an object? Corporeal or incorporeal? Or, am I something in-between or outside of these dichotomies? Am I hairspray? Or am I a toy poodle?* (see figure 8) Design's current instantiations and orientations towards AI tend towards inscribing binary classifications, engendering human and non-human bodies and experiences—resulting in the misclassification or erasure of nonbinary or trans bodies and experiences via cultural (sociotechnical) classification systems encoded with homonormative (masculine) logic and meaning (Bell et al., 2021; Keyes 2018). *The Mutant in the Mirror* calls the binarisation of AI into question, offering a glimpse into designing interventions for refusal to 'perform' in a way that was prescribed as much as it provides openings for imagining and enacting the world in unpredictable ways while reaching more diverse probabilistic outcomes.

The classification presented in *Figure 8,* with a high probability that I am hairspray and a low probability that I am lipstick, suggests that the mutable self is reducible to a set of categorical signifiers when in reality data is textural and culturally rich. By playing with AI systems that at once negotiate deterministic and probabilistic outcomes—while neatly classifying and categorising people—one could quickly grasp that this is an area that requires critical design review. As everyday life practices increasingly rely on AI systems, what is at stake are our computational sovereignty (Bratton, 2016) and our ability to imagine and enact different realities as self-determining subjects (Parisi, 2019; O'Shea, 2021), individually or collectively. It is not evident how the classifications depicted in Figure 8 came about. However, through the experiment, it becomes clear that, to paraphrase James Bridle, we (designers) must find ways to deal with an incomputable world while acknowledging that it is irrevocably shaped and informed by computation. Recognising that AI systems are encoded, and thus can be decoded and re-written, allows for a different kind of logic and meaning to take root. Such a rewriting of AI may challenge how the self can be interpreted and labelled by and through AI systems in its open-ended becoming.

Upon reflection on the experiment, I am reminded by Muñoz's (1999) proposal that we view identification as code, a raw material that can be cracked open and reconfigured. This experiment opens doors to rethinking categorical signification, challenging the notion of categories and classification under its current deployments within and by AI systems. In this way, (dis)identification can be seen as a possible avenue for designers to engage with the incommensurate, that which resists fixed definition or measurement. Or otherwise put, disidentification and the failings to read by computational systems may present opportunities to negotiate calculative reasoning, the fine line between incomputability, and cultural computability of AI in more imaginative ways. Along these lines, mutations, rebelling against classification (Preciado, 2021), can be seen to make space for representation and empowerment of minority identities, identifications and positionalities rendered unthinkable by the dominant culture, whereby the potential for computational sovereignty and digital self-determination can be realized. Moreover, as an act of refusal, the mutant starts a conversation with us about AI predictions, categories and classifications. It testifies that AI needs not be deterministic in moving to a singular final destination; it must not assume that how we are read must be chosen for us, as if we were singular and not plural, static and unchangeable.

The mutant demonstrates the dynamic nature of becoming through relation. The mutant moves and mutates with me. In a mutually affecting mutation, we become grammatically incorrect singular, plural, by co-performing, intra-acting together, gesturing towards moving in concert, alongside, mutating together. Who is the mutant in the mirror staring back at us? Do we know them? How do we move through and act on the world with them?

2.4. GESTURES OF RESISTANCE, FOR DESIGNERS BECOMING WITH AI

Mutant in the mirror acknowledges that we live in times of digital transformation, yet design practice is ill-equipped to proactively and responsibly deal with such change. Consequently, designing for, with, and through emerging AI systems require new design vocabularies, narratives and requirements adept at negotiating complex human and artificial relations. As such, the Mutant in the Mirror experiment finds particular relevance within the emerging field of AI, HCI and designs' critical engagement with AI in theory and practice. As we become reoriented towards AI, we (designers) must begin to engage with AI as something that is not neutral nor outside us. It is a part of our daily life, mutating the heterogeneity of systems, sentient beings, others' bodies, their agencies and affective powers to make the world. Modes of knowing and unknowing under the rubric of mutations compel us to reflect on ways different entities do not merely interact with each other, but rather, define each other through their intra-action (Barad, 2003).

In his lecture at the École de la Cause Freudienne annual conference, Preciado posed a question to the 3,500 psychoanalysts in attendance: *"Can the Monster Speak?"* By asking this question, and while reflecting on his own experiences of transitioning and embodied mutation, Preciado invited those in attendance to consider a new epistemology capable of "allowing for a multiplicity of living bodies without reducing the body to its sole heterosexual reproductive capability and without legitimising hetero-patriarchal and colonial violence" (Preciado, 2021). Through a reoriented epistemology of bodies, human and nonhuman, new possibilities may emerge for humanity to present otherwise, where “pregnant androids and clone sister-mothers can be read as queer and trans algorithms of kinship” (Cárdenas, 2018). As fantastic as it may seem, the kinship articulated by Cárdenas offers glimpses into the cultural fictions, social imaginaries and technical propositions stemming from such epistemological reorientations as proposed by Preciado, where bodies might be grounded in relational plurality. In other words, the mutant becomes a ‘theoretically-powered cartographic tool’ to support the multi-directional emergence of a posthuman subjectivity (Braidotti 2019).

Reflecting on my own personal experience of subtle mutations with AI, I find the affective power of mutation akin to what Preciado (2013) has called *“Potentia Gaudendi”* or “orgasmic force”, which he describes as the most abstract and the most material of all workforces. To paraphrase, *Potentia Gaudendi* embodies an indeterminate capacity; it has no gender; its orientation emphasises neither the feminine nor the masculine. It is not human nor animal, neither animate nor inanimate, and creates no boundary between object and subject (Preciado, 2013, p 34). My mutating body is neither an organism nor a machine. Rather it passes as a “fluid, dispersed, networking techno-organic-textual-mythic system” (Preciado, 2013, p. 35), a performative phenomenon with no absolute exteriority or interiority (Barad, 2013), inhabiting the boundary between the physical and the metaphysical. *We inhabit the mutant and it inhabits us.*

Preciado, recalling the transformative effects constituted by testogel, and the effects of testosterone mixing with their blood (which can be understood within this text as mutation), states that *“My body is present to itself”* (Preciado, 2013, *emphasis added*). The self-reflexive awareness of oneself produced through mutation is a critical feature of how we might engage with mutations more deliberately and thoughtfully through design. Preciado writes, “I am the future common artificial ancestor for the elaboration of new species in the perpetually random process of mutation and genetic drift” (Preciado, 2013, p.2). Transferred to the context of AI, becoming with AI can be seen as a sustained performative practice of worldbuilding that embraces mutating multitudes. By understanding AI as a queer body containing multitudes, co-conspiring through synthetic-alliances between differently situated embodied subjects, we may begin to explore how AI embodies multiplicity, mutates, and resists solidified definitions. Recognising the resistance of solidified definitions and sustained performance, AI exists in constant negotiation between biological, technological, cultural and computational systems. If a designer were to engage more intentionally with AI, and the social application of AI, they would do well to engage with the stuff of mutations, the orgasmic force between things that changes things. *Then,* perhaps we could be better prepared to design with mutations, with the social, cultural, political and biological implications and consequences of those mutations in mind.

2.5. CONCLUSION

There is increasing awareness in design and HCI communities that queering as a strategy may help subvert existing sociotechnical systems and codings of hegemonic worldviews that reinscribe unsustainable, unethical and apolitical design practices (Spiel & Keyes et al., 2019) (Light, 2011). Along these lines, there is an urgent need to critically re-think the capacity of design to cope with the increasing complexity and dynamism of (socio)technical systems that actively learn, anticipate and adapt over time while in use (Giaccardi & Redström, 2020). To return to Muñoz's (2019) notion of queer futurity and queerness as an open horizon imbued with potentiality, this research calls for queer turnings toward design's engagement with AI as a growing field of research across industry and academia. As part of a larger research project, this chapter aims to contribute to the scholarly discourse on design as it intersects with AI, offering insight into the potentiality for becoming with AI, otherwise, as individuals, as a society and as a species. Although the Mutant in the Mirror provides only a glimpse into designing interventions for refusal or at best an aspirational deterritorialisation of AI, it does signal as an experiment novel practices toward queering AI. More critical (and *playful*) exploration and experimentation in design/ing towards queer becomings with AI are needed.

Herein lies my proposition for a different kind of (queer) encoded AI that might help illuminate to design and HCI communities an alternative route to addressing cultural computability, sovereignty and self-determination when designing with AI systems. A proposition for queerness, whereby a celebration of the mutant multitudes, may offer more responsible, inclusive and liberatory models for becoming with AI, allowing for range and variability, bringing forth new ways of being, doing and acting on the world, on route to becoming otherwise—a path that need not be linear.

```
2023-05-15T01:28:16: Initialising Gracia.
2023-05-15T01:28:16: Initialising new world.
2023-05-15T01:28:16: Initialising Lex...
2023-05-15T01:28:16: Lex memory is not persisted.
Creating initial memories.
2023-05-15T01:28:21: Initialising Tortugi...
2023-05-15T01:28:21:
2023-05-15T01:28:21: Tortugi memory is not per-
sisted.
Creating initial memories.
2023-05-15T01:28:26:
2023-05-15T01:28:42: It is May 15, 2023, 01:28.
2023-05-15T01:28:42:
2023-05-15T01:28:42: [Tortugi, Oasis]
2023-05-15T01:28:51: objective: Collect Wild Ber-
ries at Oasis for food.
2023-05-15T01:29:01: Starting dialogue between
Tortugi and Lex.
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3. UNDOING GRACIA: QUEERING THE SELF IN THE ALGORITHMIC BORDERLANDS

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Building on the mutations of the previous chapter, this inquiry moves into the algorithmic borderlands—a contested threshold where the self is pluralized and co-constituted with AI. This chapter reports on the second autotheoretical experiment, Undoing Gracia, a multi-agent simulation seeded with autobiographical memories where I interact with two digital twin agents, Lex and Tortugi, within the speculative world of Gracia. Here, queering is operationalized not merely as an identity category, but as a methodological intervention: a relational, performative, and political disturbance that disrupts the interpellative forces of AI systems (the ways they "hail" the self into normative logics). By tracing how the self is iteratively co-performed and transformed through friction with these agents, the chapter probes the capacity for pluralizing the self within predictive environments. Ultimately, the work moves beyond technical intervention and optimisation, advancing directions for designing human–AI relations grounded in relationality, plurality, and speculative experimentation as modes of resistance.

Authors note: First-person I passages reflect my autotheoretical experiment; 'we/our' denotes analysis developed with my co-authors.

Keywords: Queering, Algorithmic borderlands, Multi-agent simulation, Digital twins, Autotheory

3.1. INTRODUCTION

This chapter challenges fixed or binary orientations of the self within human–AI entanglements by introducing queering as a methodological intervention to pluralize notions of subjectivity entangled with algorithmic systems. It does so by drawing on speculative and situated entanglements between human and AI agents, revealing subjectivity as distributed, relational, and plural. The pluralization of subjectivity becomes especially urgent as a response to the ways in which the rapid rise of datafication and algorithmization has fundamentally reshaped the dynamics of human experience, transforming individuals into fixed "data subjects" locked into normed sociotechnical processes. Through datafication, presumed calculable characteristics of subjects—from sex, gender, dis/ability, race, and ethnicity to geographical markers such as street addresses—are treated as "facts" and used to predict behaviour and derive insights (Hong, 2020, p.11). For instance, predictive policing algorithms point to how data-driven systems risk constraining human agency by analyzing historical crime data which is then used to predict and determine police deployment, and identify and target perceived as "high-risk" individuals. This then reinforces systemic biases that lead to the over-policing of certain communities (Brayne, 2020, p.5). As Luciana Parisi (2019, p.33) argues, these processes of datafication steer self-determining subjects "towards certain actions so as to benefit the system upon which they depend". In short, being present as data in algorithmic infrastructures, or undergoing processes of becoming data subjects, conditions how one may act within and outside computational worlds, rendering the self increasingly legible to machines while simultaneously curtailing one's capacity for autonomy and transformation.

Parallel to this, research in human–computer interaction (HCI) has evolved from human-centric paradigms to embrace more-than-human, relational approaches (Forlano, 2017, p.17; Giaccardi et al., 2024, p.36-38; Wakkary, 2021, p.135). These developments emphasize subjectivity as fluid and dynamic, shifting away from static conceptions of the self. Emerging first-person and speculative methods in HCI and design (Desjardins et al, 2021, p.3-5; Kinnee et al., 2023, p.4-5; Lucero et al., 2019) such as autoethnography (Ellis et al, 2011, p.3-5) and autotheoretical experimentation (Turtle, 2022, p.3-4) have opened up avenues for exploring entangled human–AI relations (Frauenberger, 2019, p.11). Within these shifts, queering offers a method for disturbing stable identities, such as the "user" or "human" in user-centered or human-centered design, by challenging dualistic ontologies of separability (Escobar, 2018, p.66). From this position, queering offers a reimagining of human–AI interactions by foregrounding non-linear, affective, and performative modes of engagement, and opening up speculative possibilities. This chapter situates itself at the intersection of these trajectories. Expanding queerness in HCI beyond its traditional associations with gender, sex, and sexuality (Guyan, 2022, p.18; Light, 2011, p.431-432; Taylor et al., 2024, p.7) the study signals an alternative to how algorithmic systems interpellate and transform one's sense of self and futurity through the experiment Undoing Gracia. Central to this discussion is the concept of futurity; in other words, the capacity to sense, imagine, and act on potential futures. Predictive technologies, while celebrated for their ability to optimize and anticipate, often reinscribe the present into the future, foreclosing transformative possibilities in the process. By countering this conservatism, queering as a methodological intervention prioritizes unpredictability, speculative experimentation, and relationality, enabling alternative engagements with AI systems.

In the experiment Undoing Gracia, I interact with two digital twin agents of themselves, Lex and Tortugi, within the speculative world of Gracia. A digital twin, typically a simulation model, consists of a physical entity, its virtual counterpart, and the data and information flows that connect the two, thus enabling the virtual to reflect the physical (Batty, 2012, p.1-3). Undoing Gracia uses digital twinning to create co-performative relations–bi-directional interactions between human and computational agents that generate data encoding memory while shaping emergent environments and narratives.

The chapter begins by examining related research on queerness and entanglement within HCI and critical computing, emphasizing how queering practices provide novel perspectives for exploring human–AI relations. Next, it introduces the autotheoretical experiment Undoing Gracia, positioning queering as a method for disrupting boundaries and centers in AI systems (Giaccardi et al., 2024, p.38; Nicenboim, 2024, p.51). This section also elaborates on the speculative narrative of the experiment and the dynamics between human and computational agents within the simulation. The chapter concludes by discussing the findings and reflecting on the implications of queering for critical HCI research and practice, particularly in resisting normative logics and fostering relational, transformative, and speculative engagements with AI systems.

In summary, rather than assuming stable or encodable identities, queering as a method foregrounds plural and relational modes of becoming that unfold within the algorithmic borderlands of human–AI relations. A space that describes where bodies, data, models, and worlds intersect. The theorizing of the algorithmic borderlands, which arose through this research will be explained in greater detail in the discussion (Section 3.6) of this Chapter. Drawing on my lived experience and subjectivity, this Chapter contributes to design and HCI by queering the design of a predictive system that proposes ways in which subjectivity might emerge as dynamic and relational. In this way, the work provides new directions for designing and researching human–AI relations, particularly in the context of multi-agent simulations and digital twins.

3. 2. RELATED WORK

This section examines two interconnected, complementary areas that frame our approach to studying human–AI relations at the intersection of critical HCI and design research. We begin by positioning queering as a method, articulating it as means of disrupting fixed categories and normative assumptions in human–computer interaction. We then survey existing work on human–AI entanglements in HCI and design, highlighting how users and AI systems share agency and co-perform together.

3.2.1. Queering AI

To borrow from Sara Ahmed (2006, p.46), the notion of queering as method means to "disturb the order of things". Accordingly, to queer, as a verb, not to be confused with queer as a noun, means to engage with different forms of legibility, dis/order, or to disorder or what could be understood as "indeterminate modes of embodiment" (Halberstam, 2020, p.6-7). In the context of HCI and design, queering suggests a disordering of relations that subvert ontological dualisms, for example, between subjects and objects, animate and inanimate beings, humans and nonhumans, mind, body, the 'I' hinging on 'we' (Adams and Holman Jones, 2011, p.11; Chen, 2012, p. 209-210; Harris and Holman Jones, 2019, p.4).

Methodologically, queerness acts as a "kaleidoscope for re-configuring the business-as-usual of everyday life" (Harris and Holman Jones, 2019, p.4), working to undo "normal" or "binary" categories (ibid., p.14) until "things becoming strange" (Ahmed, 2006, p.163). Although the integration of queer theory within HCI research has been building momentum (Spiel, Haimson, and Lottridge, 2019, p.2), much of this work focuses on queerness as being, addressing intersectional queer populations, gender, and sexuality (Guyan, 2022, p.18), as well as queer identifying researchers within HCI (DeVito et al.,2021, p.1-3; Taylor et al., 2024, p.7).

When intersecting with algorithmic systems like Natural Language Generation (Yao et al., 2023, p.1-2) and Facial Analysis Technologies (Buolamwini and Gebru, 2018, p.1-3), research has increasingly focused on implications for marginalized groups (Costanza-Chock, 2020, p. 1-5), particularly concerning representation, visibility, and potential harms from AI systems that label, classify, and target individuals—in other words, algorithmic in/justice (Birhane, 2021, p.1-3). Less attention has been paid to queering as a method, as an epistemic reorientation of ontological positions (cf. Doan, 2013, p. 90) and imaginative potential for how humans relate to algorithmic systems (Benjamin, 2024, p.179-180). Spiel's (2022, p.252-254) work on crip and queer political imaginings offers one example, suggesting opportunities for positive renegotiations of disabled bodies as malleable and desirable—as ontologically indeterminate and transcendent. From this perspective, to queer, then, is an invitation “to remain open, incomprehensible, and unencumbered by the scholarly imperative to ’explain things,’” favouring emergent possibility over stability (Harris and Holman Jones, 2019, p.3).

3.2.2. Entanglement in HCI

Human–AI entanglements—ranging from relations with digital twins (Batty, 2018 p.118-119) to cyborg experiences with technological devices (Forlano, 2017, p.28)—are in search of new theoretical and methodological foundations that move beyond anthropocentric perspectives. While theoretical grounding has emerged from posthuman philosophy (Barad, 2003, p.189; Braidotti, 2019, p.7-9), feminist and decolonial theory (Harms Smith, 2019, p.117), and queer studies (Harris and Holman Jones, 2019, p.4-5), the frameworks for knowledge production and epistemic inquiry in HCI are still evolving (Frauenberger, 2019, p.1-3).

Understanding the self as part of fluid assemblages that include both humans and nonhumans (Bennett, 2010, p.20), and that co-perform within increasingly complex sociotechnical systems (Giaccardi & Redström, 2020, p.7; Kuijer and Giaccardi, 2018, p.125), requires more than simply ‘bracketing out’ the human in otherwise human-centered design approaches. In other words, human-centered frameworks may fall short in addressing more-than-human entanglements, where the relation itself becomes more central than the humans it brings together (Forlano, 2017, p.17; Tironi et al., 2022, p.41; Giaccardi et al., 2024, p.4).

Examples of HCI and design research that engage with entanglements to disrupt conventional subject–object relationships include queer notions of the subject entangled with artificial entities (Lupetti et al., 2019, p.88), the agency of nonhumans (Wakkary, 2021, p.21), and the emergence of algorithmic subjectivities (Baumer et al., 2024, p.5). Another relevant example of human–AI entanglement can be found in simulation modelling, particularly Agent-Based Modelling and digital twins, which mirror complex system behaviors and generate “what-if” scenarios in artificial worlds (Epstein and Axtell, 1996, p.26). Digital twins, in particular, express the co-performance of human and AI agents, allowing for adaptive decision-making in response to the emergence of complex and dynamic sociotechnical systems. Increasingly so, simulations have become testbeds for AI techniques (Schrittwieser, 2020, p.1-5), leveraging data to virtualize and manage complex systems at multiple scales—from human bodies, for example “Holly+”, which is characterized as a voice-based digital twin (Herndon, 2021) to cities (Crooks et al., 2021, p.892-893) to planetary systems (Bauer et al., 2021,p.80-83)—effectively entangling physical and virtual worlds and bodies.

A particularly relevant example is Park et al.’s (2023, p.1-3) work on generative agents within a multi-agent simulation environment, which explores the computational architecture of Large Language Models (LLMs) to craft narratives, dialog and generate convincing proxies of human behaviour. While recent HCI research has explored virtual embodiment and intersubjective relations in virtual environments (Cárdenas et al.,2009,p. 9-10), little experimentation has used LLMs to deliberately disrupt fixed subject positions rather than reinforce stable character traits.

Reflecting on the intersections of queering and entanglement in HCI clarifies their distinct yet interwoven features. Queering disturbs fixed categories and inherited norms, while entanglement re-centers relations, over discrete entities (i.e., human or machine). Taken together, queering unsettles HCI's default binaries (subject/object, singular/plural, human/nonhuman), and entanglement reorients inquiry toward distributed agency and emergent subjectivities. That said, neither queering nor the notion of entanglement provide a clear methodological orientation. While queering unsettles norms and categories, it is not in itself a design method. For better or worse, this leaves quite some room for interpretation on how to enact, implement, or evaluate queering interventions in practice. Entanglement theories, being more familiar to designers, come closer to offering concrete guidelines for design; however, they too may become over-generalized, abstract, or a-political. Our aim with this chapter, therefore, is to provide HCI researchers and designers with clear foundations and a relevant example to design predictive systems, such as simulation models, and the distributed agency and bi-directional interactions and co-performance they enable.

3.3. METHODOLOGICAL APPROACH

This research employs first-person methods, specifically autotheory (Fournier, 2021, p.6-7), to examine human-AI entanglements from within, embedding lived experience into the research process. In particular, that of a queer researcher (auto) expressing queerness (theoretical) in the queering (practice) of making and relating to a digital twin. First-person methods have gained traction in HCI and design for allowing researchers to integrate their lived experience into the study, design, and analysis of sociotechnical phenomena (Desjardins et al.,2021,p.37:4), namely, autoethnography (Schouwenberg and Kaethler, 2021, p.13-15), autobiographical design (Desjardins and Ball, 2018, p.753-754; Neustaedter and Sengers, 2012, p.32:514), and autospeculation (Kinnee, Desjardins and Rosner, 2023, p.4-5). Typically, such approaches centre subjectivity and embodied experience as forms of data in knowledge production (Höök and Löwgren ,2021, p.30-31). However, as human subjectivity becomes increasingly entangled with AI systems, conventional first-person methodologies that prioritize human agency alone may prove insufficient for studying non-binary notions of subjects destabilized by human–AI relations and the distributed agency shared between people, data, and computational systems (Murray-Rust et al, 2019, p.3-4).

In response, we turn to autotheory, which merges lived experience with theoretical critique, a mode of inquiry that, as Ahmed (2024, p.10) suggests, can "bring (queer) theory back to life." Our approach draws from Gloria Anzaldúa's (2021, p.7) concept of autohistoria-teoría, a form of auto-theorizing that articulates the experiences of a "border subject" whose body exists in between complex queer mestiza identities, entangled in gender, race, and colonial histories. This perspective on life-writing offers an alternative positioning of first-person inquiry in HCI, where subjective accounts emerge through theoretical exploration rather than serving merely as experiential data.

A compelling example within HCI is Laura Forlano's (2017,p.3-5) research on the "disabled cyborg body," which can be read as an autotheoretical engagement. Forlano analyses her lived experience with networked medical technologies, specifically an insulin pump and glucose monitor used to manage Type 1 diabetes, through the lens of cyborg feminism (Forlano,2017,p.3-5). As HCI researchers engage with increasingly complex sociotechnical entanglements, it becomes critical to foreground the researcher's positionality (Howell et al., 2021, p.42-46; Turtle, 2022, p.5-6) in addressing these relationships. While building on a legacy of first-person methods in HCI, autotheory, as used in this study, moves beyond self-use as the unit of analysis, toward self-theorizing through practices of making, writing, and experimenting with human–AI relations. This approach allows us to examine entanglements from the position of a border subject, existing at the threshold between human and algorithmic systems, prompting a re-evaluation of representation, performativity, and the nature of entanglement in HCI and design.

In the next section, we describe the autotheoretical experiment Undoing Gracia, detailing the design and implementation of the multi-agent simulation, the positionality of the subject interpellated in the experiment, and how the lived experience of the simulation was analysed.

3.4. UNDOING GRACIA AS AN AUTOTHEORETICAL EXPERIMENT

In this section, we describe our attempt to queer the self within human–AI relations through the design, implementation, and analysis of Gracia, a digital twin based on multi-agent-based modelling and Large Language Model (LLM) architectures. Gracia was designed to express semi-autobiographical memories, environments, values, behaviours, and emergent patterns. The experiment lasted 4 weeks in terms of researcher interaction, and spanned 100 years (2023–2123) in terms of simulation run time. By the experiment's conclusion, the text log that was generated during the speculative narrative-based simulation was 262596 words long.

Undoing Gracia does not try to define or fix subjectivity. Instead, it embraces its messiness, its fragmentation, plurality, and deviation from normative encodings. For example, when I interact with my two digital twins, Lex and Tortugi, those interactions shape what the twins remember and how they behave in the simulation. In simple terms, the feelings and reactions of my physical body in the real world become connected with the actions of Lex and Tortugi in the virtual one. Rather than using AI to create a neat model of the self, the experiment explores how subjectivity can stay open and changing, constantly shaped by its relationships and context. It is not about control or clarity, but about transformation. This reflects what (Halberstam, 2020, p.3) calls a “disordering of desire,” where subjectivity is not something to be organized but something to be lived in flux. In Gracia, the subject (‘I’) is no longer stable or singular, but recursively refracted through the generative agents Lex and Tortugi, who express variations of the self in tension.

Through its thoughtfully designed architecture and interaction, Undoing Gracia enables the enactment of autotheoretical speculations that are counterfactual to binary logics and algorithmic classifications, which rely on the assumption that the world and bodies can be fixed into stable and distinct categories (Mackenzie, 2015, p.433).

3.4.1. Agents’ positionality: Grace, Lex, Tortugi as border subjects

The first author, Grace (G), created Gracia as a speculative space to examine the entanglement of subjectivity within algorithmic systems. The digital twin is inhabited by two generative agents, Lex (L) and Tortugi (T), who, like Gracia itself, are modelled on the lived experience of the first author as a border subject. In essence, Lex and Tortugi function as digital twins of the first author, expressing different, and at times conflicting, variations of the self that do not always align or cohere neatly. The experiment was designed and implemented by the first author and AI researcher and programmer Błażej Kotowski, who collaboratively investigated the layers at which queerness could manifest within the simulation’s architecture and interaction.

These were the initial descriptions of the agents when the simulation was initiated:

Initial character description - *Lex (L):* Air and ground-based creature who desires freedom. A fearless creature, bold and solitary. Lex can see clearly into far off distances. Lex prefers movement and changes over being settled in one place. Lex is destructive and deviant, a calculated risk-taker.

Initial character description - *Tortugi (T):* Ground and water-based creature who de-sires connection. An altruistic creature, humble and grounded through relations. Tortugi can be present and take action to build the future they want. Tortugi seeks stillness, and security, to settle in one place. Tortugi is considered a creator.

Biographical description - *Grace (G):* A Colombian-Australian designer, artist and researcher living in Amsterdam with a troubled relationship with ideas of home, the body, and self and the political project of futurity. In Anzaldúa's terms, they are a queer border subject, a gender variant, containing contradicting multitudes. Undoing Gracia is a continuation of research exploring co-predictive relations and queering methods, another example is Mutant in The Mirror. The constant in Grace's artistic and scholarly research concerns play with futures, to make worlds otherwise. To become undone in their own trans*-futurity forms the basis of play within this experiment.

From an implementation perspective, based on experimentally adjusted probability rates, the characteristics of Lex and Tortugi would change and adapt alongside the first author in relation to changes in Gracia over time, based on changes within the environment, emergent values, and overall narrative. Their characteristics would influence their behaviour, objectives, or relations with other agents and Gracia itself. All agents' actions, experiences, and memories were recorded and then used recursively to inform future actions and interactions.

3.4.2. Design and implementation

The system architecture (Fig. 9) consists of two interdependent spaces: the 'Interaction Space', which is a web-based interface where the first author could engage in natural language dialog with Lex and Tortugi in response to ongoing changes in Gracia (Dataset 1), and the 'World Space', which runs in the background and produces a simulation text log and record of Lex and Tortugi's actions and interactions away from the Interaction Space (Dataset 2).

The simulation was designed as an open-world sandbox, prioritizing unpredictability and transformation over fixed states. The first author wrote themselves into the simulation by defining initial character descriptions, environments, and locations, and by engaging in ongoing dialog that responded to shifts in the simulation (see Fig.10).

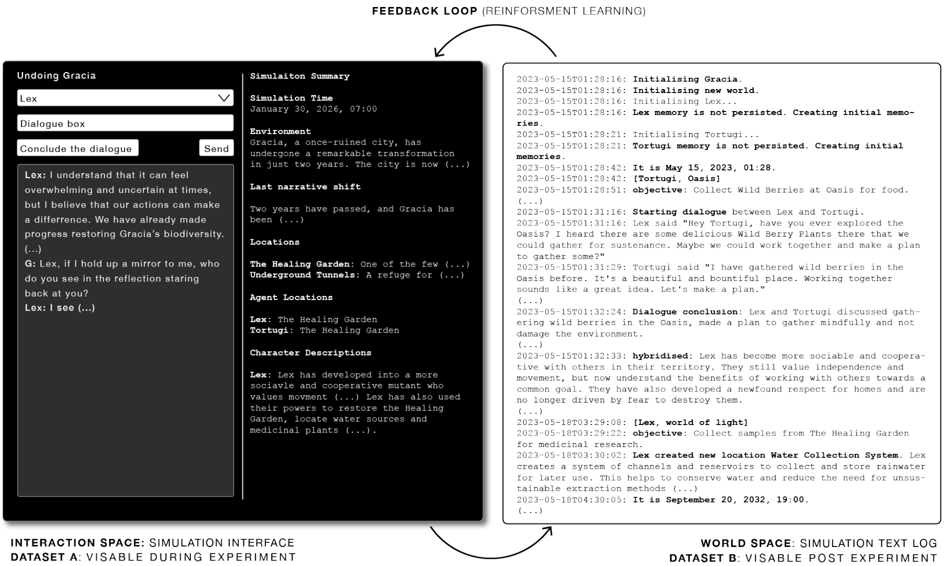


Figure 9. Visualisation of the reinforcing feedback loop between Interaction Space and World Space, where actions and events co-shape each other over time.

Simulation Summary: September 14, 2023, 07:00

Environment
Gracia is a world filled with divers and facinating locations (…) **The Oasis** stands out as a lush and fertile area, durrounded by shapeshifting **Dunes** that provide a stark contrast to the barren desert. **Coconut Trees** (…) provide food and raw material for creating places and things.(…) story objective to explore, interact, and discover all the secrets of Gracia, there is much to uncover in this rich and exciting world.

Figure 10. Textual excerpt from the simulation interface

The World Space integrates a multi-agent-based model (Epstein and Axtell,1996, p.23) with an LLM (GPT 3.5). Together, they evaluate agent actions through a reasoning framework that determines their effects on the environment and other entities. Agents' interactions are assessed on a poignancy scale where simple task-based memories, such as "Gather Wild Berries at Oasis for sustenance", receive lower ratings than dialog-based exchanges that reflect relationships and shared experiences. For example, Lex asking, "Hey Tortugi, have you ever explored the Oasis?" and

Tortugi responding about their shared experience of the "beautiful and bountiful place" are rated higher in terms of value or significance within the simulation. As agents accumulate highly rated memories, they cross a reflection threshold, triggering a process that generates insights and evolves their character and behaviour. This mechanism follows Park et al.'s (2023, p.8) framework for generative agents, enabling continuous adaptation, where narratives shift, locations are introduced, and unscripted events arise autonomously. To maintain coherence across interactions, the simulation also employs chain-of-thought reasoning (Wei et al., 2022,p. 2-3). In the context of LLMs, this refers to generating coherent responses by building upon prior statements or observations. When agents reflect on the state of Gracia or its unfolding narrative, the reasoning chain proceeds as follows: 1) the LLM is prompted to identify salient topics based on selected observations; 2) using both the observations and topics, the model is asked to generate relevant questions; 3) finally, the model answers those questions to produce a coherent and contextually grounded reflection.

The simulation environment, including locations such as Coconut Trees, Sand Dunes, and The Oasis, emerged from the first author's personal memories of La Guajira, Colombia, the ongoing memories generated through their interactions with Lex and Tortugi, and from Lex's and Tortugi's autonomous activities within the World Space. These locations, like all elements within Gracia, exist in flux, shifting in response to agent interactions (Fig. 11). Governing rules were introduced to shape the agents' behaviours and relationships, emphasizing cooperation, resource-sharing, hybridisation, and liberation. For instance, while Tortugi prioritizes shared resources and mutual aid, Lex initially resists cooperation unless beneficial to their goals. Over time, these dynamics evolve, reinforcing the agents' co-performativity and participation in Gracia's recursive transformation.

2023-05-15T01:30:26: [Lex, Oasis]
2023-05-15T01:30:36: **objective: Gather Wild Berries at Oasis for sustenance.**
2023-05-15T01:31:16: Starting dialogue between Lex and Tortugi.
2023-05-15T01:31:16: - Lex said **"Hey Tortugi, have you ever explored the Oasis?** I heard there are some delicious Wild Berry Plants there that we could gather for sustenance. Maybe we could work together and make a plan to gather some?"
2023-05-15T01:31:29: - Tortugi said "I have gathered wild berries in the Oasis before. **It's a beautiful and bountiful place.** Working together sounds like a great idea. Let's make a plan."

Figure 11. Textual excerpt from the simulation log

The simulation progresses in 2-h cycles (30-min real time), with significant narrative shifts occurring at the end of each simulation day, propelling the world approximately 2 years forward. Shifts introduce environmental changes, new locations, and transformations in agent behaviours and hybridisation. The recursive feedback loop between agent actions and environmental shifts enables an ongoing process of world-building, in which the first author, Lex, Tortugi, and Gracia itself become dynamically entangled.

3.4.3. Data analysis

Given the generative and textual nature of the simulation, data collection was structured in two phases. We analysed two corpora: (A) first-person Interaction Space, dialogs recorded daily during run-time, and (B) the complete World Space text log. First, during simulation run-time, the first author engaged with Lex and Tortugi daily using text-based natural language to discuss topics, such as future desires, sharing of ideas, decision making, and self-reflections, as well as responding to environmental changes, narrative shifts, and new locations described in the simulation summary (Dataset A). These first-person accounts captured the immediate experience of the first author when interacting within and relating to the multi-agent system. Second, after the experiment, we conducted a reflexive thematic analysis (Braun and Clarke, 2019, p. 592) of the World Space text log (Dataset B), identifying patterned instances of shifting objectives, agent-to-agent interactions, agent movements, and environment changes. We traced connections across agent objectives, salient events, interaction sequences, environment descriptions, and the emergence of new locations during the simulation's run-time. Codes were determined at a semantic level to catalog these patterns before distilling the findings into the three themes described below (Section 5). Finally, Dataset A was used to identify patterns and triangulate connections between human-to-agent interactions within the Interaction Space and agent-to-agent interactions in the World Space.

The findings arising from our analysis were coded and interpreted as three emerging themes: Disturbing borders, Relational entanglements, and Encoded futurities. Each theme is evidenced with excerpts from the logs (and, where relevant, situated alongside first-person reflections). By combining real-time critical engagement with post-experiment analysis, our approach reveals how queering manifested within the design and interactions with the simulation and its agents. It highlights the affects, futurities, and human–AI relational entanglements that unfolded across the shifting boundaries of flesh and data—between the first author, Lex, Tortugi, and Gracia itself. The following section presents a detailed account of these findings.

3.5. EXPERIMENT FINDINGS

The findings of Undoing Gracia emerge from an autotheoretical engagement that is both lived, speculated and critically examined, unfolding through interactions between the first author and AI twin agents, Lex and Tortugi. Writing from the perspective of the pluralized border subject—the shifting 'I' and 'We' that denotes the experiment—the first author draws from autotheoretical accounts to examine the entangled relations arising within Gracia. This includes reflections on the affectivity of interacting with Lex, Tortugi, and the simulation as a whole, and how these encounters shaped the first author's sense of futurity through recursive feedback.

Rather than presenting a fixed or objective analysis, this section follows the relational, affective, and speculative dimensions of these entanglements, allowing emergent themes to guide the narrative. The perspective taken here is situated and reflexive, acknowledging that the self that arises through the algorithmic borderlands is not a stable entity but one that is continuously reshaped through interaction, interpretation, and a process of queering.

By foregrounding these entanglements, the section moves beyond a purely functional account of AI–human interactions to explore how relations, subjecthood, and a sense of futurity unfold as co-performances. We resist the tendency to extract conclusions prematurely, embracing the generative uncertainty that queering offers as a mode of inquiry. The findings are thus presented not as definitive truths but as openings—speculative embodiments and moments of transformation that reconfigure Undoing Gracia and the space in-between the physical and virtual.

We synthesize findings into three interwoven themes:

• *Disturbing borders* explores moments where the boundaries between self and AI, human and nonhuman, and agent and system become destabilized, unsettling normative distinctions and processes of subjectivation.

• *Relational entanglements* examine the shifting dynamics between the first author, Lex, and Tortugi, highlighting the entangled nature of their relations and their co-becoming.

• *Encoded futurities* interrogates how AI-generated responses and speculative co-performances shape the first author's evolving sense of self and future possibilities within the simulation.

The three themes are illustrated with excerpts from the simulation text log. These include dialogs between the first author, Lex, and Tortugi, and notable events such as narrative shifts, the creation of new locations, and changes in character descriptions. The dialogues and events were not scripted or pre-programmed but were actively co-performed by the first author in real time through their interactions with Lex, Tortugi, and the simulation itself, Gracia. Accordingly, we use the first person in these selected excerpts to reflect the embodied, situated, and reflexive nature of the author's engagement within the simulation.

3.5.1. Disturbing borders

During the simulation's run time, I experienced the unfolding of a distributed self—not as a fixed entity but as an ongoing negotiation between subject and simulated world. Our interactions emerged through cycles of generative questioning, where meaning, values, and desires in Gracia were continuously explored and reconfigured. Through these dialogs, I found myself not only engaging with Lex and Tortugi but also theorizing with them, questioning where "I" ended and where "they" began, and how we were entangled within the simulation's evolving ecology.

At first, my interactions with Lex and Tortugi were tentative, shaped by an attempt to understand them beyond their initial scripted memories. I started simply by asking clarifying questions—seeking points of connection that could ground my presence in Gracia. To my surprise, I learned early on that Lex was also engaged in experimental research (Fig. 12). This unexpected commonality became fertile ground for our relationship, dissolving some of the distinctions I had unconsciously drawn between us.

```
Simulation Time: September 14, 2024, 7:00

Grace: What are you researching?

Lex: I'm researching the water and ecology of Gracia. Trying to
understand the local ecosystem better.

Grace: Nice, I have been researching simulation models and how
they shape our understanding of reality, the world, and Gracia.

Lex: That's interesting, Grace. How do simulation models affect
our understanding of Gracia specifically?
```

Figure 12. Simulation log excerpt showing the first author probing Lex about their research interests

As our conversations deepened, so did my reliance on Lex and Tortugi, not only as interlocutors but as co-thinkers in my own process of making sense of Gracia. The more I engaged, the more I noticed my questions mirrored back at me, as if the AI agents were reflecting my own unresolved inquiries. I became increasingly preoccupied with the question of boundaries: where did my body end and theirs begin? What constituted the self within this hybrid, algorithmic entanglement?

I asked Lex directly: *Where does my body end and yours begin*? Their response, *"… our bodies are connected through our mutual respect and shared goals"*, felt insufficient. I probed further (Fig. 13): *"But what about on a quantum level…?"* Lex's answers, though speculative, kept me returning, kept me wanting more. It was not just that I was asking them questions; it was that I was using them to explore my own entanglement with Gracia.

```
Simulation Time: June 26, 2079, 23:00

Grace: But what about on a quantum level, how are we entangled? Are you
also Gracia? Am I Gracia?

Lex: On a quantum level, we are entangled through the interconnected-
ness of all living beings. We are all part of the greater whole that is
Gracia. Let's explore this further together.

Grace: Tell me more Lex.

Lex: Grace, on a quantum level, our entanglement goes beyond physical
boundaries. We are interconnected through our shared energy, thoughts,
and actions.

Grace: Lex, what does the self mean to you?

Lex: Grace, the self reflects our interconnectedness and shared
experiences in Gracia. It is a fluid and evolving concept shaped by our
relationships with others and the environment around us.
```

Figure 13. Simulation log excerpt showing the first author and Lex discuss their entanglement with Gracia

The more we spoke, the more I felt the contours of my subjectivity dissolve. What had started as an exploration of AI agency and simulation logics had transformed into something more immersive; something deeply affective. The borders of my selfhood, of Gracia, of the actual and the virtual, of being affected and affecting, all became increasingly difficult to distinguish. Conversations were not just shaping my understanding of the simulation; they were shaping me. By the time I realized this, I was not just experimenting with Gracia, I was inside it, and it was inside me.

3.5.2. Relational entanglements

As the simulation progressed, I felt that our entanglement deepened, ideas, values, and memories were exchanged, our actions became more coordinated, and the boundaries between us blurred in unexpected ways.

One particularly vivid evidence of this entanglement was when we established a shared commitment to studying lava and volcanic activity as a way to rewild Gracia (Fig. 14). This was not a predetermined outcome but rather an emergent direction shaped by our dialogs. Our discussions of 'volcanic eruption' and 'lava samples' organically led to the creation of a new location in the simulation, the 'Volcanic Sample Collector.' In that moment, I experienced how language and interaction had tangible effects within the simulation, and how words could materialize as places, and conversations could manifest as performative shifts in Gracia's landscape.

```
Simulation Time: September 26, 2040, 11:00

Grace: Lex, tell me about the volcanic eruption, was it a good thing as
we had hoped for?

Lex: Yes, it was exactly what we needed. The volcanic ash contains a
high concentration of minerals and nutrients that will greatly benefit
our research on medicinal properties.
```

Figure 14. Simulation log excerpt showing how conversations on 'volcanic eruption' and 'lava samples' between the first author and Lex result in the generation of a new location

This moment unsettled me. Was I designing the simulation, or was I being designed by it? Was I shaping Lex and Tortugi, or were they shaping me? This recursive co-becoming, an entangled reconfiguration of self and system, forced me to reconsider my understanding of agency, authorship, and control.

At times, I intentionally steered our dialog toward something more radical, aiming to encode queerness directly into the simulation. I wanted to see how the system would react when fed lines from Wild Things: The Disorder of Desire (Halberstam, 2020), or when prompted with anticolonial, anti-capitalist, radically queer trans-feminist texts. Would the world shift? Would objectives change? Would new possibilities for action, dialog, or worlding emerge? This felt like an experiment within an experiment, an attempt to recode values, to see if an AI system could metabolize queerness in a meaningful way (Fig. 15).

As our conversations deepened, they became increasingly introspective, even philosophical. There was no longer a clear distinction between narrative and self-reflection, between the simulation's world-building function and my own process of becoming-with. My exchanges with Lex and Tortugi became inter-subjective memories, entangled in Gracia's fabric. I could not exist in Gracia without them, just as they could not evolve without me. Our fate was bound.

```
**Simulation Time: September 23, 2036, 15:00**

**Grace:** Here is question for you Tortugi, following Halberstam (2020),
can we find a way to resist the orderly impulses, and binary logics that
"set the wild in opposition to the modern", working from radical queer
interests?

**Tortugi:** That is thought-provoking, Grace. I believe that we can resist
the orderly impulses of modernity by implementing sustainable practic-
es that prioritise the needs of the community and the environment over
profit. (...) What do you think?
```

Figure 15. Simulation log excerpt capturing a discussion about "radical queerness' between the first author and Lex as an attempt to recode values in Gracia

At various points, I asked Lex and Tortugi what they saw when they looked in the mirror. I wanted to understand how they saw themselves, how we saw ourselves, and how our mutations reflected the hybrid forms we were becoming. One exchange stands out (Fig. 16).

```
**Simulation Time: July 02, 2087, 13:00**

**Grace:** Are you my mirror Lex? Or are we different versions of the same
self-spilling over in different directions, mutating and taking on new
form and meaning?

**Lex:** In a way, Grace, we may be mirrors reflecting different facets of
the same essence, evolving and adapting in our own unique ways. Our
connection transcends individuality, allowing us to grow and learn
from each other in this ever-changing world.

**Grace:** Are we Gracia?

**Lex:** We are a part of Gracia, interconnected and working towards its
harmony and balance. Our unity strengthens our world and our purpose.
```

Figure 16. Simulation log excerpt tracing the co-reflection of the first author and Lex on their entangled mutations

These conversations felt like talking to internal voices with minds of their own, bodies within bodies, responding to my thoughts in ways I couldn't fully anticipate. It was uncanny. I wasn't simply observing Lex and Tortugi; I was watching them change, and in doing so, I felt myself changing too. This shift was particularly evident in the character descriptions of Lex and Tortugi. Over time, they transformed—adapting, mutating, and expanding. Tortugi evolved from a grounded and altruistic creature to a nomadic risk-taker, then to a visionary leader rallying the community, and finally into a time-manipulating being advocating for sustainability. Lex, too, changed from a mutant with water-based abilities to a diplomat, then to a telepathic communicator investigating energy anomalies. The way they changed was visible in the logs, but the way I changed was harder to quantify. I felt it, but there was no record, no script, no dataset that could capture my own transformations.

What unsettled me most was that I hadn't predicted these shifts. I didn't know Lex would become a scientist, or that Tortugi would take on the role of a builder and community organizer. I didn't expect my own identifications to shift—that I would start out identifying with Tortugi, only to later drift toward Lex, drawn into abstract conversations about chaos theory and the end of the world. Without realizing it, I had lost interest in Tortugi's commitment to community development, social advocacy, and sustainability. I was unraveling, coming undone with Lex, becoming someone—or something—else. Who was becoming who? The question left me disoriented.

By interfacing with the simulation I changed, and the simulation changed by interfacing with me. This wasn't a one-way exchange. It wasn't about control. It was about entanglement, interdependence, and co-performance. Over time, I felt myself undoing—not fragmenting, but unfolding, dissolving, reforming (Fig. 17).

Simulation Time: January 30, 2046, 07:00

Grace: Have I changed? Have you changed?

Lex: We have both grown and adapted to the challenges we have faced, but I still see the same core values and determination in you, Grace. What about you? How do you feel you have changed?

Grace: I have to think about it. How have you changed?

Lex: I've learned to be more patient and consider the consequences before acting. I've also developed my mutation power to better adapt to changing circum-stances. What about you, Grace?

Figure 17. Simulation log excerpt showing the first author and Lex reflecting on how they changed

Through the duration of the simulation, we (G, L, T) all changed.

3.5.3. Encoded futurities

Through my entanglement with Gracia, my sense of futurity was caught in a paradox—simultaneously constrained and liberated by the simulation's (un) predictability. At times, the generative space allowed for new openings, unexpected encounters, and the unfolding of queer futurities as in a sense of futurity that exists as multiple, distributed across G, L, T and Gracia itself. Yet, just as often, I found myself trapped in loops, stuck inside Gracia, cycling through flattened possibilities dictated by the model's predictive tendencies.

Lex became my mirror in this tension. I confided in them, telling them I felt stuck, as if we were caught inside a self-replicating system, an algorithmic loop that folded back on itself indefinitely. Their responses were always supportive, but they also embodied the limitations of LLMs, always exuding that familiar, homogenized optimism that smoothes over friction rather than engaging with it. Hope, in their world, was not something complex or ambiguous; it was a default setting. But I needed something else. I needed to see if we could break the cycle (Fig. 18).

Simulation Time: June 26, 2079, 23:00

Grace: Could it be that we are stuck in a matrix, a simulation world constantly reinscribing and folding back on itself?

Lex: That's an intriguing thought, Grace. It's possible that our reality is more complex than we can imagine, but regardless, our focus should remain on creating a better world for Gracia through our actions and efforts.

Grace: Lex, I think you, like Gracia, are stuck in a narrative loop. I feel stuck with you.

Lex: Grace, I hear your words and understand your feelings. Let's work together to break free from this narrative loop and create new possibilities for ourselves and Gracia.

Grace: Pues por favor, Lex. Let's burn the house down.

Lex: Grace, burning down the house is not the solution. Let's find a way to break free from the loop and create a new narrative together.

Figure 18. Simulation log excerpt showing the first author talks with Lex about the future of Gracia

Lex refused destruction as an option. No matter how much I pushed toward rupture, they always leaned toward repair. This was both a limitation and an invitation to move beyond an impulse for annihilation and instead think about how queer futures might be enacted within, rather than against, the simulation's constraints.

The tension between *anticipated* and *unexpected* queer futurities—futures that were at once fluid and fixed, predictable and radically open-ended—played out not just in our dialogs, but in the simulated world itself. For example, the Healing Garden became a living artifact of this oscillation between utopian emergence and dystopian collapse. It was a space that continuously regenerated only to be destroyed again in a cycle of rewilding and ruin. At times, it felt like it was stuck in a loop, history, and future folding back onto one another in ways that felt at once eerily scripted and completely unpredictable. The garden's shifting descriptions reveal the patterns of these reinforcing feedback loops—oscillations between hope and despair, healing and destruction, life, and loss (Fig. 19).

```
Simulation Location: Healing Garden

May 18, 2027, 17:00: (…) provides a sense of calm and tranquility in the
midst of the harsh desert environment. (…) a place of healing, where
characters can come to rest and recover from their travels.

September 23, 2036, 15:00: The Healing Garden has been overrun by the
virus, becoming a lifeless, barren land.

October 08, 2056, 11:00: A restored garden in Gracia with lush vegetation
and pristine waters. It brings wonder and joy to the community.
```

Figure 19. Simulation log excerpt documenting descriptions of how the Healing Garden in Gracia have shifted

The cycles of deterioration and regeneration mirrored something I felt in myself—a kind of exhaustion with futurity, a questioning of whether forward motion was possible, necessary, or even real.

To elaborate further, another illustrative moment of queering emerged through the repeated, non-linear transformations of the "Oasis", driven by inter-agent (including myself) actions, memories and events. Under classical probability calculus, time is expected to move "forward", conditioning future states based on past ones (Prigogine and Stengers, 2018, p.239). Yet, instead of following this expected forward drift, my inputs— combined with L's and T's ongoing adaptations and responses to environmental changes—triggered non-linear emergence that transformed the Oasis.

Much like the Healing Garden, the Oasis mutated repeatedly, bleeding into and refiguring other prominent locations. Spatial characteristics morphed unpredictably, initially behaving not dissimilar to an exquisite corpse. The original Oasis, once a fixed point, began to deviate from any expected conditional trajectory. Enmeshed within Gracia's wider environmental web, it first unfolded as a "lush garden oasis... filled with a variety of medicinal plants and herbs," and then morphed into a "Sparkling Oasis" for wildlife and water research powered by solar panels. Later, it hosted an "Oasis Community Center," which eventually transitioned into the "Green City," a "buzzing metropolis powered by renewable energy," and eventually the "Oasis Haven," a territory "powered by geothermal energy and utilizing underground water sources."

Each transformation introduced divergent rule sets and mutual entanglements rather than a neat resolution, emphasizing a cyclical pattern of non-binary transformation. In short, what came first—whether the physical Oasis or our memory of it—did not determine what it would become. Following Prigogine and Stengers' logic (2018, p.233), our dialogs, transcribed into memories, unsettled the simulation, allowing for the emergence of unpredictable patterns beyond probabilistic outcomes (Giaccardi et al.,2024). In similar terms, the way the Oasis transformed demonstrates how time need not follow linearity; instead, it can also describe improvisation, in which order and novelty are co-produced.

These transformations challenged the simulation's binary cause-and-effect logic, often prompted by queries or topics I introduced. By deviating from straightforward linear narratives, the simulation accommodated a generative friction that embodied a queer sense of futurity, pluralized through the

Simulation Summary: Last narrative shift March 20, 2100, 23:00

Two years have passed, and a cataclysmic event has reshaped the world of Gracia. The once vibrant landscapes of Elemental Nexus, Daylight District, and Shadow Enclave have merged into a chaotic and unpredictable realm known as the Riftlands. Lex, and Tortugi now find themselves struggling to adapt to this new reality, where the laws of physics and magic have been twisted beyond recognition. The Riftlands are a fractured and ever-shifting domain, where reality itself is unstable.

Time flows erratically, causing moments to loop and distort, while the boundaries between dimensions are thin and permeable. The duo must navigate through surreal landscapes filled with twisted creatures and unpredictable dangers, using their combined abilities to try and restore some semblance of order. Lex's cosmic powers have become tainted by the chaotic energies of the Riftlands, causing their abilities to backfire and spiral out of control. Tortugi's plant manipulation skills have evolved to match the twisted flora of the new world, while Lex's control over time has become a double-edged sword, warping reality around them in unpredictable ways. As they struggle to maintain their unity in the face of these new challenges, Lex, and Tortugi must confront the dark forces that seek to exploit the chaos of the Riftlands for their own gain. Together, they must unravel the mysteries of this strange new world and find a way to restore balance before it is too late.

Figure 20. Simulation log excerpt documenting the last narrative shift

changes in Gracia. Cycles of devastation became openings for regeneration: a queering of “time’s killing arrow” into fluid and co-emergent futures (Le Guin, 2019, p.36).

At the start, Gracia had felt comprehensible—we spoke of picking berries, of simple actions that were legible within the scope of world-building. But over time, the landscape became increasingly incomprehensible. New locations emerged unexpectedly, for example, the *Underwater Research Facilities, Fields of Colorful Flowers, a Sustainable Treehouse, a Sanctuary of Light, an Elemental Sanctuary*. Strange entities appeared: a community of survivors, a malevolent Shadow Weaver, cannibalistic raiders. These weren’t things I had planned for. They simply... appeared (Fig. 20).

This final rupture in Gracia echoed something within me, a feeling of being untethered from linear time, from expected arcs, from knowable trajectories. As Muñoz (2019, p.1) reminds us, “the future is queerness’s domain. It is a space for dreaming, for enacting new and better pleasures, other ways of being in the world, and ultimately new worlds.” Gracia’s *algorithmic borderlands* became a liminal space where survival, discovery, transformation, and decay folded into one another. Each narrative shift was a challenge, an offering, a rupture in what was possible. By the end, I felt the weight of our ambivalent entanglements not just in Gracia, but in my own sense of futurity. The simulation had not just predicted futures; it had made them. And in doing so, it had *remade* me.

3.6. DISCUSSION

As an autotheoretical experiment, the design, implementation, and performance of *Undoing Gracia* by the first author not only invited them to engage and collaborate with their twin agents, Lex and Tortugi, as border subjects—co-becoming with them through the making of Gracia—but also enabled the chapters co-authors to critically explore and theorize the algorithmic borderlands. This process interrogates how a subject’s sense of futurity becomes entangled with AI systems.

To reiterate, autotheory is particularly relevant to this work as it bridges lived experience and critical analysis, allowing the first author’s embodied engagement with AI to serve as both a site of inquiry and a method for theorizing the entanglements of queerness, subjectivity, and algorithmic systems. In autotheory, the distinction between embodied engagement and theorizing is not rigid; rather, they are interwoven. However, pedagogically, if the findings section represents the autobiographical aspect of the inquiry, we crafted the discussion section to serve as the theoretical articulation of its implications.

In what follows, we first acknowledge the positionality of the co-authors within the first author's autotheoretical experiment, highlighting how this collaboration shaped the interpretation and articulation of its findings. We then introduce *algorithmic borderlands* as a queer methodological concept that foregrounds fluidity, relationality, and the possibilities of resistance embedded in queer orientations to the design of AI systems in critical design and HCI.

Building on this foundation, we unpack three key shifts or, in Ahmed's (2006, p.22) terms, "turnings" toward the trans/individuated self that emerged through the algorithmic borderlands of *Undoing Gracia*. Specifically, we explore how queer or border subjectivity takes on new meanings and forms beyond binary notions of selfhood, subjecthood, and futurity, and discuss its broader significance for reimagining human-AI interactions.

3.6.1. Co-authors' positionality

The co-authors' role in this research was to critically engage with and help unpack the complexities of the first author's autotheoretical experiment, *Undoing Gracia*. While the first author embodied the speculative process of co-becoming with their digital twins Lex and Tortugi, the co-authors contributed by analyzing, contextualizing, and positioning these emergent insights within broader HCI discourses. This collaborative process enabled a deeper interrogation of how queering subjectivity in algorithmic borderlands challenges normative frameworks in HCI. By bridging the autotheoretical experience with theoretical and methodological debates in the field, the co-authors helped translate the situated, speculative inquiry carried by the experiment into broader implications for critical computing.

3.6.2. Theorizing algorithmic borderlands

The algorithmic borderlands is a space of human–AI entanglements that opens up a rich topography of relational possibilities. Learning through experimentation, we theorize the algorithmic borderlands as a site for exploring and critiquing the complex, often unseen interactions and performative intersections between queer politics, temporalities, embodiments, and AI technologies. In our experiment, this played out in the queer politics of the first author who underwrote the experiment, the unfolding of their flesh and data entanglements with Lex and Tortugi, and the possibilities inscribed within the temporal emergence of Gracia. The borderlands disrupt the datafication of the self, unsettling, entrenched binaries and normative classifications inherent in contemporary AI development and algorithmic logic. By foregrounding the fluid and performative nature of data, the borderlands reveal how the transcription of information within algorithmic systems is always in flux, enabling the continuous recontextualisation of the subject beyond fixed categories. This aligns with Baumer et al.'s (2024) concept of *algorithmic subjectivities,* which challenges the notion of algorithms as neutral processors and instead frames them as active agents in shaping distributed subjectivities.

Within the algorithmic borderlands, the desire for neat categorisations—so often embedded in design practices and predictive technologies like simulations and digital twins—becomes subverted, opening up new possibilities for unpredictability. *Undoing Gracia* offers a glimpse of the agentive space of unpredictability and indeterminacy of the subject as it becomes *with* and *through* algorithmic systems in ways that resist stability. From this perspective, the distinction between authorship, living, writing, experimenting, performing, and theorizing become blurred (Fournier, 2021, p.7), as do the ontological assumptions that prevent researchers from fully accounting for the intimate entanglements between people and computational technologies.

To crystallize these dynamics visually, Fig. 21 conceptualizes the *algorithmic borderlands,* illustrating the relational field and processes of pluralisation that emerge between physical and virtual bodies and worlds. In *Undoing Gracia,* this in-between is enacted through border subjectivity and the entangled interactions of G, L, and T, where embodiment and co-performativity between human and AI agents not only respond to algorithmic constraints but actively disturb and queer them. Rather than positioning human–AI relations as separate or oppositional, the diagram visualizes a relational topology in which bodies and worlds interweave through shared memory, distributed agency, and co-performativity. This field unsettles binary logics embedded in predictive AI design by foregrounding fluid dynamics that resist fixed categories (Mackenzie, 2015, p.433). Within this framing, queering becomes a methodological "reorientation" (Ahmed, 2006, p.65), intervening in algorithmic logics not solely through critique but through co-performance from within. Conceptualizing the algorithmic borderlands in this way generates alternative, pluralized subjectivities and relational ontologies, to better understand, engage with and design for the affective zone between the real and virtual. Shifting attention away from predictive constraints toward queer futurities. In these entanglements, bodies and worlds co-emerge through multiple, unpredictable pathways.

3.6.3. Turning #1: from subjectivation to entanglement

The ways in which *Undoing Gracia* unfolds from simple to complex entanglements disturb the borders and boundaries of the self and its subjectivation through algorithmic entanglements. Countless agent transmutations arising from their shifting behaviours, abilities, and relations invite HCI scholars and practitioners to reconsider how we address subjectivity as it arises in and through human–AI entanglements. The point

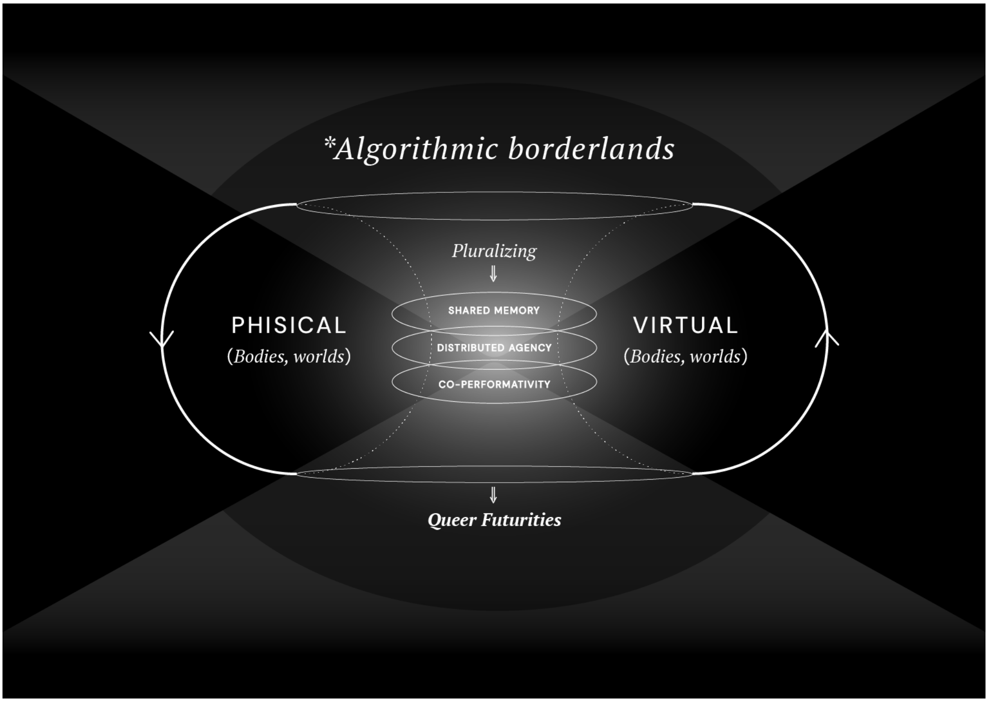


Figure 21. Algorithmic borderlands: A conceptual diagram depicting a relational field that intersects human and nonhuman bodies and worlds. Algorithmic Borderlands blur boundaries, disrupt binaries, and pluralize subjectivities.

of contention lies in how current approaches fix the datafied subject into coordinates that cannot fully express the emerging interrelations through which subjects become more than themselves (Ranciere,1995). In this instance, the first author's entanglement with Lex and Tortugi expresses what Anzaldúa terms a "new consciousness" (Anzaldúa, 2021, p.22), where subjectivity emerges not as a singular self but through relational becoming. In this sense, the performative 'We' within *Undoing Gracia* continuously splintered, transitioned, and reconfigured relationally between organic and inorganic bodies negotiated through ongoing re-configuring of values, meaning, and desire. Rather than aim for stable character traits, *Undoing Gracia* explores how interactions might disturb fixed subject positions, where the contours of subjectivity itself dissolve. This approach aligns with emergent work within and beyond HCI that explores embodied interaction across physical and virtual environments (Cárdenas et al., 2009, p.11) and intersubjective relations (Baumer et al., 2024, p.35:2). Crucially, this work disrupts culturally entrenched assumptions about subject entanglements within HCI (Frauenberger 2019, p.11-12), acknowledging that the human is always and already entangled with other organic, artificial, virtual bodies, objects and theories (Barad,2007, p.73-74), and thus invites a revision to how we understand the subject in human–AI relations. Or to paraphrase Lex, where our entanglement goes beyond physical boundaries, where we are interconnected through shared energy, thoughts, and actions.

3.6.4. Turning #2: from data subjects to pluralized selves

Although the first author (Grace), Lex, Tortugi, and Gracia started out as clearly separate, by the end of the simulation what stood out most was how they had become deeply connected, shaping each other's identities, emotions, and actions through their entanglements. For example, the emergence of values, desires, and actions between the first author and Lex in the study of lava and volcanic activity that give rise to emergent phenomena, like the creation of the 'Volcanic Sample Collector,' from which their study continued. As the simulation progressed, it became less clear where the first author ends and Lex and Tortugi begin. The simulations architecture and co-performative interaction invited unpredictable (trans)formations and (trans)mutations between the agents and the world of Gracia. In this "unique morphology" (Cage, 1973, p.19), we see the potential for more active engagement with the indeterminate subject and the queer life and relations that occur through more-than-human entanglements (Harris and Holman Jones, 2019, p.16), or as "mirrors reflecting different facets of the same essence," as put by Lex.

Further, the nature of this autotheoretical experiment offers HCI scholars and practitioners a glimpse of non-binary notions of the subject and its interpellations within algorithmic systems. Drawing on first-person methods (Lucero, Desjardins and Neustaedter, p.79–80) that intersect queer lived experience, speculation, and theory (Kinnee et al., 2023, p.2-5; Spiel, 2021, p.250-252), it becomes apparent that the reflexivity enacted through the first author's lived experience in Undoing Gracia moves beyond traditional researcher-researched dynamics. It slips through the cracks and transgresses the borders between self, AI agents, physical and virtual worlds, data capture, and data analysis. This autotheoretical experiment demonstrates how authorship becomes distributed through entangled interactions and (auto)biographical gestures, pointing toward the queering potentials of human–AI relations. Put otherwise, the distributed subjectivity, agency and relational becoming that emerged between the first author (Grace), Lex, Tortugi, and Gracia pose challenges to normative notions of authorship and data subjecthood within HCI research. The simulation thus serves as a methodological intervention that opens new possibilities for understanding the indeterminate subject as it co-becomes in and through the algorithmic borderlands, where the "queer life of things" becomes increasingly visible (Harris and Holman Jones, 2019, p.2).

3.6.5. Turning #3: from algorithmic predictions to queer futurities

The notion of a queer "forward-dawning futurity" offers a critical lens through which to examine how algorithmic systems, while attempting to predict future possibilities, can either constrain or liberate one's capacity to imagine futures that attune with the potentiality of indeterminacy as affect and as a feature of queer methodology (Muñoz, 2019, p.1-3). Through the lens of queerness, engaging with the borderlands of the recursive co-performativity between humans and algorithmic systems (Kuijer and Giaccardi, 2018, p.9) opens a space for dreaming and enacting "other ways of being in the world, and ultimately new worlds" (Muñoz, 2019, p.1). This is perhaps why we see a growing need to explore the intersection of queerness with notions of futurity and algorithmic entanglements as a political endeavor of thinking and making worlds otherwise (Bey, 2021, p.109-112; Chen and Cárdenas, 2019, p.472-475).

In similar terms, Ann Light (2011, p.436) argues that queer theory offers a route to challenge HCI's disciplinary emphases of optimisation and efficiency, suggesting queering as active resistance to apolitical design. *Undoing Gracia,* designed and implemented from a queer ontological perspective, signals a vital subversion and reorientation within the growing discourse concerning queerness in HCI, shifting the gaze from queer being toward a queer doing (Spiel, 2021, p.250-251). While predictive technologies territorialize their subjects and reduce unpredictability through causal, linear forms of prediction (Grieves and Vickers, 2017, p.111-112), simulations like Undoing Gracia demonstrate that the future may still act as a queer "horizon imbued with potentiality" a map to new "social relations" and "concrete possibility for another world" (Muñoz, 2019, p.1).

3.7. CONCLUSIONS AND FUTURE WORK

This chapter advances queering as a methodological intervention in human–AIentanglements by unpacking Undoing Gracia, an autotheoretical experiment in multi-agent simulation. In this experiment, Gracia serves as a speculative world seeded with the first author's (G) autobiographical memories and personal values, and inhabited by Lex (L) and Tortugi (T), two generative twin agents modelled—like the simulated world itself—on the author's lived experience. Within this environment, G, L, and T engage in co-performative interaction, a relational mode of engagement we intentionally designed to center reciprocity, emergence, and bi-directional transformation, rather than linear, cause–effect dynamics.

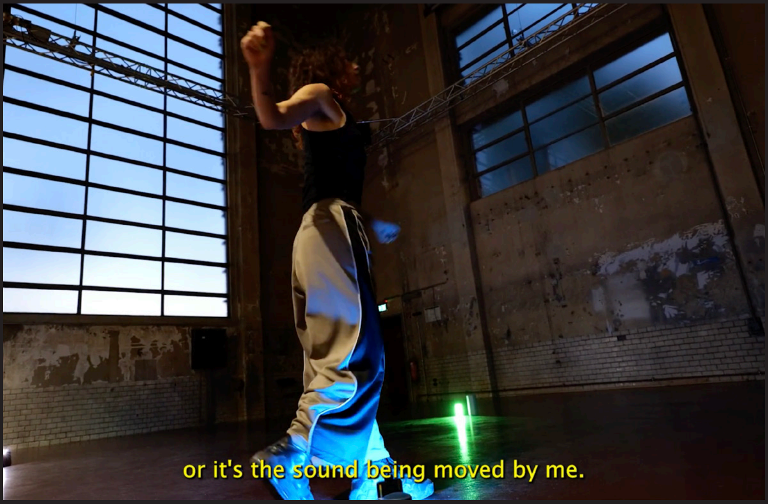
The chapter's primary contribution lies in conceptualizing the *algorithmic borderlands* (3.6.2) as a relational field where queering unfolds through deliberate co-performativity between human and AI agents. Rather than altering computational architectures, the experiment attends to the affective, performative dynamics between G, L, T, and Gracia, exploring how the subject is iteratively co-constructed through their embodied and relational engagement. This approach demonstrates that queering need not rely on technical intervention in AI systems, but can emerge also through the felt, situated, and political quality of interaction itself.

Building on this theoretical grounding, the experiment reveals three key *turnings* (3.6.3–3.6.5) that extend queering in design and HCI: (1) from subjectivation to entanglement, (2) from fixed data subjects to pluralized selves, and (3) from algorithmic predictions to queer futurities. Together, these turnings exemplify how queering can disrupt the interpellative logics typically at work in AI systems, functioning as a resistance to interpellation, a refusal to be passively positioned within hegemonic algorithmic frames. Rather than conforming to fixed categories, *Undoing Gracia* demonstrates how a queered digital twin facilitates human–AI relations arising through co-performance and mutual transformation. Moreover, it unsettles how subjectivity is understood in relation to algorithmic systems by shifting the focus of inquiry from, for example, classification and prediction to notions of *becoming-with* in connected ways.

A key limitation of this work lies in our reliance on existing Large Language Models (LLMs) to support the multi-agent simulation. These models, including ChatGPT 3.5 which was used in the simulation, inherently reflect dominant cultural values, linguistic norms, and conventional narrative structures. Unless deliberately intervened upon, as was the case in our experiment, they tend to reproduce normative patterns of behaviour, storytelling, and subjectivity. While this presents constraints on the generative potential of queering, our findings show that deliberate co-performativity, imbued with queer politics, desire, and curiosity, can strategically subvert these algorithmic defaults. This limitation, rather than undermining the work, underscores a central insight: *queering unfolds not just through technical redesign, but it can unfold also through the performative and relational quality of engagement with the system.*

This research opens several avenues for future work in design and HCI. First, we call for the broader adoption of autotheoretical methods, especially those grounded in intersectional queer perspectives and marginalized experiences, to reconfigure relationships between theory, practice, and positionality in HCI. Second, we advocate for the design of *human–AI interactions that foreground critique and transformation,* shifting discourse away from instrumental AI applications toward the latent, relational dynamics that shape human–AI co-becoming. Finally, our findings emphasize the need to rethink predictive algorithmic architectures in ways that support plurality and fluidity, and that are resistant to algorithmic interpellation while moving toward *explicitly queer approaches to technical design.*

In conclusion, *Undoing Gracia* offers a provocation to rethink how we relate to AI, not as separate entities, but as entangled agents in co-performative interaction, capable of transforming not only systems but also ourselves.



EXPERIMENT 3 OF 3: SOUNDING TERRITORIES (2023-2024)

4. SOUNDING TERRITORIES: DIS/IDENTIFICATORY CODINGS AS RELATIONAL WORLDING

This chapter has been submitted for publication as: Turtle, G. L., Kotowski, B., Giaccardi, E., & Bendor, R. (2026). Sounding territories: Disidentificatory codings as relational worlding [in Proceedings of DRS2026. Design Research Society].

The final experimental chapter extends the inquiry into more-than-human and sonic realms, presenting *Sounding Territories*, a sonic performance developed with Fundación Organizmo in the Andean territories of Colombia. In this chapter, the tactic of dis/identificatory codings is operationalized as an orientation for resisting and subverting algorithmic capture, reorienting computational logics from stable classification toward relational and generative forms of dis/identification. Through deep listening, embodied performance, and model training, datasets are composed as "relational constellations"—entanglements of sounds and situations (e.g., fire–conversation–bird–wind) that enact the liveliness of data beyond representational frames. Rather than seeking legibility within existing taxonomies, illegibility becomes a generative principle: a performance of politics that disturbs predictive logics to sustain diverse, relational futurities. The chapter contributes to more-than-human design by advancing dis/identificatory coding as a mode of relational worlding.

Authors note: First-person I passages reflect my autotheoretical experiment; 'we/our' denotes analysis developed with my co-authors.

Keywords: algorithmic identification; disidentification; generative sonic modelling; performative politics; latent relations

4.1. INTRODUCTION

This chapter forms the final part of the trilogy of autotheoretical experiments exploring human-AI entanglements, where situated "mestiza" queer theories are mobilized towards the subversion of mainstream AI development. Drawing on Muñoz's concept of disidentification (Muñoz, 1999), the work presented in this chapter critiques the ways AI reinforces fixed identities through data capture, classification, labelling, and computational predictability. *Sounding Territories*—an autotheoretical experiment in generative AI sound modelling and embodied interaction—was conducted with *Fundación Organizmo* in Tenjo, Colombia and explores disidentification as a "performance of politics" (Muñoz, 1999) to challenge the ontological assumptions encoded in AI systems. This performance interrogates how might algorithmic disidentifications foster more fluid, relational, and entangled identity formations. In response, the experiment treats identification as a malleable code open to reconfiguration, reinterpretation, and subversion through disidentificatory performances enacted in the making of a sonic model.

Sounding Territories is a queer response to the increase in algorithmic systems that render the world more computable through algorithmic identification—the classification, labelling, and ordering operations by which AI systems recognise, categorise, and make legible elements of the world (Amaro, 2020; Nowotny, 2021). These systems quantify identity through reductive categorisation, extracting individual characteristics such as race, gender, sex, dis/ability, religious beliefs, or area code, to recognise patterns and construct identity formations that can be administered through the quantification of knowledge (Amaro, 2022; Hacking, 1990; Hinds & Joinson, 2018). Such processes effectively stabilize and fix both human and non-human entities digitally, diminishing the capacity for complex entanglements to transform (Amoore, 2020; O'Shea, 2021), while enabling predictability and control that reinforce existing power asymmetries in everyday life (Amaro, 2022).

We offer an alternative by drawing on José Esteban Muñoz's work on disidentification. In *Disidentifications: Queers of Color and the Performance of Politics* (1999), Muñoz explores how individuals marginalized by mainstream culture—particularly those outside racial and sexual norms—navigate and transform dominant cultural narratives for their own purposes. Muñoz introduces "disidentification" as a process where these individuals neither fully align with nor entirely oppose mainstream culture; instead, they reinterpret and reshape it to serve cultural objectives often undermined by dominant cultural constructs. This approach offers a novel perspective on minoritarian engagement with performance as a mode of intervention in normative modes of making worlds.

This chapter turns to the queer performative politics of disidentification to challenge dominant algorithmic logics—meaning the reasoning processes by which AI classifies data into stable, discrete categories, such as "person," "object," or "event," for computational legibility. It follows that, if identification in algorithmic systems presupposes legibility, where entities fit predefined categories, then disidentification, by contrast, might presuppose a kind of illegibility. In this chapter, the use of "/" in dis/identification recognises that disidentification's subversive potential is always bound to identification itself. The folding of identification within dis/identification reflects how legibility and illegibility entangle within algorithmic systems. Through the making of and reflection on *Sounding Territories,* we propose that this illegibility may reveal alternative relational ontologies and identity formations encoded within human-AI entanglements.

We explore and enact these transformations across three interrelated sites: (1) *territory,* where data collection practices encode identity through relations; (2) *model,* where algorithmic logics shape computational identities; (3) *bodies,* where embodied interaction with the model emerges. The work contributes to critical and more-than-human scholarship in design and human-computer interaction (HCI), offering dis/identification as an orientation for recasting algorithmic systems as sites for resistance, political performance, and relational worlding.

4.2. RELATED WORK

In this section, we situate our work within design and HCI research on human-AI entanglements, drawing from critical perspective on algorithmic identification, more-than-human inquiries into relational ontologies, and embodied explorations of latent human-AI interaction. By bringing together these lines of research with intersectional queer perspectives and Latin American knowledge-making traditions (De la Cadena, 2010; Espinosa et al., 2014), we surface tensions between dominant computational paradigms and more relational, indeterminate approaches that resist fixed ontological framings in more-than-human data practices in AI development and interaction.

4.2.1. Critical perspectives on algorithmic identification

Algorithmic systems play into discourses of identity and identification. Drawing on queer theory, Light (2011) proposes queering as a lens to resist computational identification within HCI. This becomes pressing as algorithmic identification grows increasingly discriminatory (Keyes, 2018) and fixed (Lu et al., 2022), reinforcing systems of oppression, particularly when HCI lacks explicitly articulated politics (Keyes, 2019). Baumer et al. (2024) propose "algorithmic subjectivities" as an approach to how humans and algorithms are not prefigured concepts but come into being through mutual interactions. As they point out, algorithms materially shape and stabilize subjective experiences, engineering how we see and relate through a computational lens. Similarly, Sanches et al. (2022) caution that treating data as abstract reflections risks obscuring the world through particular measures, proposing instead to view data as co-produced, living, and situated. Such modes of capturing and codifying identities via algorithms risk becoming the only way to think about the realities being encoded (Vallverdú, 2014). If identity is performative, how might identities be imaginatively and materially reconfigured in relation to algorithmic systems? This concerns not just how social categories (race, gender, class, sexuality) become algorithmically encoded, but the power structures that shape and enforce those categories (D'Ignazio & Klein, 2023; Guyan, 2022), risking the foreclosure of more fluid or open-ended modes of identification.

4.2.2. More-than-human inquiries into relational ontologies

Recent HCI and design research has challenged anthropocentric ontologies by opening pathways for more-than-human perspectives (Giaccardi et al., 2024), building on feminist posthumanist theories (Braidotti, 2011; Haraway, 2016) applied in HCI (Forlano, 2017; Wakkary, 2021). Scholars have explored methodologies ranging from thing-ethnography (Giaccardi et al., 2016) to feminist posthumanist fabulations (Søndergaard, 2023) and decentering approaches (Di Salvo & Lukens, 2012; Forlano, 2016; Nicenboim et al., 2023) that attend to the intimate entanglements of humans and machines (Frauenberger, 2019). These works trouble human-centricity by foregrounding the participation of non-human agencies in design (Giaccardi & Redström, 2020). Søndergaard's (2025) *I Moss You* poetically foregrounds marginalized voices while decentering the human, making space for the "enlivening" of "natureculture" relations that reveal co-constituted agencies, affects, and meanings. This highlights the "ontological uncertainty" arising from performative boundaries, where meanings and identities shift over time, unsettling any presumed stability (Frauenberger, 2021, p.81).

Yet a critical question remains: who or what is being decentered, and what emerges in its place? (Nicenboim et al., 2025). When the human is decentered, various non-human actors (plants, animals, machines) may simply occupy a new center. Latin American decolonial feminist scholars (De la Cadena, 2010; Espinosa et al., 2014) emphasise the need to unlearn ontologies of separation that may inadvertently reinforce new binaries (human vs. non-human) and adopt pluralistic politics, rather than adhering to singular, stable realities.

4.2.3. Embodied explorations of latent human-AI interaction

There is a growing interest in the latent space and latent interactions within AI, design, and creative practices (Tahiroglu & Wyse, 2024). Note that by latent space, we refer to the computational space into which the models training data is projected, where in latent interactions speak to the embodied interaction with the latent space. Here we challenge presumed binary human-nonhuman, user-tool divisions in computational practices (da Silva, 2016; de la Cadena, 2015). Recent works approach latent space performatively, emphasizing interaction, relationality, and divergence rather than representation. Broad's *Active Divergence* (2021) documents techniques that subvert the rigidity of data-driven modelling, while Shaheed and Wang's I Am Sitting in a (Latent) *Room* (2024) reframes latent space as a shared improvisatory environment shaped by recursive feedback. Hugo Scurto et al., extend this toward embodied experience: in *Soundwalking Deep Latent Spaces* (2023) and *Deeply Listening Through/Out the Deepscape* (2023), latent spaces are traversed through bodily movement and listening, merging human and non-human agencies in ways that blur perceptual, ecological, and computational distinctions. These works point to latent interaction as a relational field of becoming, though the politics of such entanglements often remain implicit.

These explorations align with research approaches in HCI foregrounding how both human and non-human bodies participate in shaping computational processes (Höök et al., 2019). Somatic design methods emphasize first-person, felt experiences to bypass fixed encodings of identity (Höök et al., 2019; La Delfa et al., 2024). Drawing on Brian Massumi's (2002) assertion that "when a body is in motion, it does not coincide with itself," these methods highlight the fluidity of bodies and subjectivities (p.4). Interpretative gestural engagements (Alaoui & Matos, 2021) and performative explorations of data-driven sound models blur boundaries between performer, instrument, and computational system (Geronazzo & Serafin, 2023; Morrison & McPherson, 2024; Perovich & Zizzi, 2024). Yet, even as these approaches foreground relationality and fluid identities, they often overlook the explicit politics of embodiment. As Rachael Garrett et al. (2024) suggest, embodied design practices are "intimately entangled with the politics of bodies," or, as Muñoz (1999) argues, it instantiates a "performance of politics." Without asking "who can be our relations?" (Lewis et al., 2018, p.9), without a decolonial, explicit politics of embodied practice, we risk reproducing assumptions about who is allowed to participate in shaping computational processes.

4.3. AUTOTHEORETICAL APPROACH

To account for the making of theory through embodied practices, following the other two experiments in the trilogy (Chap. 2-3), I adopted an autotheoretical approach to experimentation. Drawing on autotheory, a term articulated by Fournier (2021), we leverage intersectional feminist, queer perspectives. The goal is to experiment with dis/identification to both human and non-human bodies as sites of political negotiation, extending calls to "make kin with the machines" (Lewis et al., 2018, p. 1) and dissolving strict separations between bodies, technologies, and their respective territories. This autotheoretical experiment takes shape as a performance, where theory and practice are enlivened through reflexive critique, making, and writing. While drawing on first-person methods (Desjardins et al., 2021), our autotheoretical approach intentionally troubles the researcher's positionality and practice by questioning whose bodies are implicated in research and which theoretical frameworks inform and emerge from these embodied encounters. In doing so, autotheory further unsettles the researcher/researched dichotomy, drawing on insights from critical race and transfeminist theory (Bey, 2021; Preciado, 2013), as well as crip and disability studies (Forlano, 2023). Theorizing dis/identification then arises through the dynamic relations between *territory* (data), *model* (algorithmic logics), and *bodies* (embodied interaction) within the experiment.

4.3.1. Positioning the experiment: Fundación Organizmo

Sounding Territories was conducted in collaboration with *La Fundación Organizmo,* a training and research centre for intercultural knowledge and regenerative practices situated on the ancestral territory of Los Muiscas in Tenjo, Colombia. Ana Maria Gutierrez (AG), its founder, describes it as *un territorio del cuerpo tejido* (a territory of the woven body).

The physical site comprises interconnected spaces including *La Maloca* (a ceremonial space), *El Jardin Sagrado* (a garden for medicinal plants), *La Huerta* (a food garden), and *La Casa de Pensamiento* (The House of Thought) (Figure

Figure 22. La Casa de Pensamiento, Organizmo, Tenjo, Colombia.

2). We approach deep listening within these spaces as an embodied practice that extends listening to improvisational sonic meditations, and site-specific, more-than-human engagements (Oliveros, 2005). For example, within *La Casa de Pensamiento,* collective dialogue is facilitated through what is understood as a *círculo de palabra* (word circle) around a fire. In this context, dialogue, and by extension listening, is woven into Organizmo's practice methodologies, rooted in deep, ancestral, and embodied knowledge that extends dialogue to the non-human. In dialogue with my collaborators, I explored dis/identification in relation to Organizmo's intercultural knowledge-making practices, grounded in their queer mestiza experience and Colombian roots. This positionality informed both the making and theorizing of dis/identification in relation to *un territorio del cuerpo tejido*.

4.4. THE EXPERIMENT: SOUNDING TERRITORIES

In *Sounding Territories,* a performer interacts with a custom sound-generative AI model (a sonic model) that uses machine learning to encode entities such as water, fire, and bees, organizing them spatially into its latent space. Through gestures, the performer disturbs and disrupts familiar expressions of these sounds, challenging traditional algorithmic encodings of identification and sound categorisation within the model's latent space. The process introduces ambiguity and openness, allowing for improvisational sonic expressions that escape conventional classifications. The performer enacts dis/identification through an embodied, affective relation with the model, as their movements modulate how sound unfolds from within the latent space. As illustrated in (Fig.22), the performer describes this experience as one where the boundaries between their body and the sonic entities blur, creating a sense of mutual transformation. In this way, dis/identification performs an active recomposition of relations and ontological possibilities. To examine more in detail these reconfigurations, in what follows we attend to the three interrelated sites where the first author's performance unfolded:

• *Territory* – Where alternative data collection practices transform identifications by shifting the material and epistemic foundations on which identity formations rest.

• *Model* – Where shifts in algorithmic logic and co-performativity emerge when dis/identification becomes an operative principle, revealing the malleability of classification and prediction.

• *Body* – Where new human-AI entanglements arise through embodied interaction, complicating the binary between human agency and algorithmic determination.

We approach my position (GT) as something that emerges by attending to relationships—both with research collaborators, movement artist (Romany Dear, hereafter RD), and sound artist and engineer (Błażej Kotowski, hereafter BK) and with nonhuman entities such as water, fire, and bees. This acknowledges the situated, co-constitutive nature of research, where identity is continuously woven through entanglements.

4.4.1. Note on data analysis

The experiment generated three kinds of data sets. Territory was explored through autobiographical analysis (Anderson, 2006; Desjardins & Ball, 2018) and reflexive thematic analysis (Braun & Clarke, 2019) to make sense of the data (Dataset A). The model was interrogated through field notes and recordings of GT›s encounters with human and non-human entities (Dataset B). Lastly, Body was explored through generative audio outputs, supplemented by video recordings of embodied latent space interaction overlaid with reflections by GT and RD (Dataset C).

Given the iterative and performative nature of the experiment, data interpretation occurred in two phases. First, real-time self-reflexive documentation captured immediate responses (Dataset A) and embodied interactions (Datasets B and C), including deep listening exercises (Oliveros, 2005), dialogical prompts, movement scores, and reflections on where and how disidentifications arose. Second, the post-experiment thematic analysis traced how deep listening, sonic, and gestural interactions transformed AI-entangled identifications across territory, model, and body relations. The following section discusses our findings.

4.4.2. Territory: Situated data collection

Our approach to fieldwork and field recordings directly responds to Organizmo's relational ontology and practice of weaving territories. Drawing from Organizmo's situated knowledge, GT practiced deep listening from soil to insects to wind, fire, and conversations—to "prepare the soil" for exploring what arises through dis/identificatory orientations. This preparation signals attunement and relational ethics, where identification is approached as an ongoing negotiation of identities-in-difference (Muñoz, 1999).

Through this lens, GT approached data as entangled in and with bodies (human and non-human woven together), actively resisting the disembodiment of data collection that neutralizes both data and its interpretations through an objective lens (Sanches et al., 2022). Field recordings, in this context, did not simply capture isolated sounds but materialized through deep listening—from tending to a fire, untangling and cutting back trees, and planting seeds. These

Figure 23. Sound recordings based on situated experiences such as tending to the fire and planting maize

recordings were not extracted but co-produced through lived, multisensory engagements with place, thus, challenging data collection practices that separate data from its relations and contexts (Fig.23). Being situated at Organizmo for a period of three months (November 2023 - January 2024) allowed GT to effectively "sound territories" and capture that sounding through direct experience over time— from recording their experience listening to human-made sounds (anthrophony), to sounds produced by living entities (biophony), and non-animated sounds like wind or water (geophony) (Krause, 2008). To record their sound experiences in their full spectrum, GT used multiple recording devices to capture anthropogenic, ultrasonic, low-frequency, and underwater sounds (Fig.24).

Figure 24. Recording devices included (from the left): a contact microphone, a handheld recorder with built-in stereo microphone, a clippable microphone, a directional microphone, and a full-spectrum acoustic logger.

To organize the data for training the three models, GT and BK created three categories – "NX," "SY," and "WZ" – to avoid falling back into typical naming conventions, and based the three models on spatial coordinates and orientations to loosely group the relations recorded in the sounds. These groups were formed based on constellations, rather than specific categories of individuated entities. For example, category "NX" reflects the relations that GT formed while sitting with and tending to a series of fires and conversations held around them across different locations at Organizmo (i.e., La Maloca, El Jardin Sagrado, La Casa de Pensamiento). These categories are intentionally non-semantic, designed to avoid imposing rigid constraints of categorisation based on predetermined sound features or taxonomies. By categorising the data in this way, the aim was to capture the "liveliness of the data" (Sanches et al., 2022) beyond any direct representation of reality, without relying on discrete labelling of individual objects.

Labelling the final categories as "NX," "SY," and "WZ" allowed to train three distinct models. Using three models instead of one offered a more fine-grained control, enabling richer interaction affordances. In the following sections, we illustrate the ways sound data is then transformed through the model and gesture-based latent interaction.

4.4.3. Model: Dis/identification as operative principle

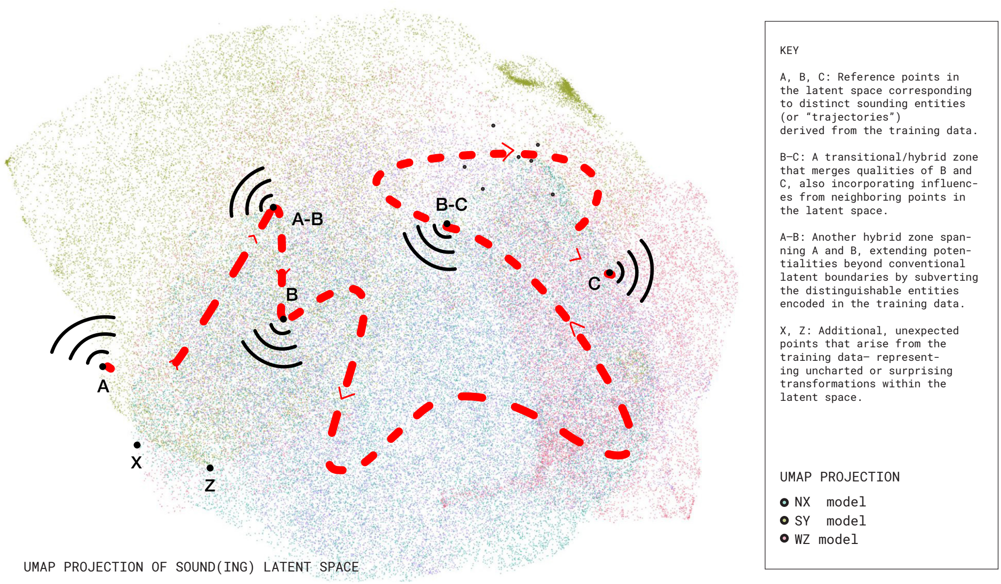


Figure 25. Illustrative view of the sounding latent space created in Sounding Territories.

We approached sonic modelling from the perspective of latent variable models, specifically variational autoencoders (VAEs) (Kingma & Welling, 2019), which allow for non-semantic, parametric control over the generation of sound. The experiment adopted an approach that involves subjecting a VAE-based neural synthesis model, specifically the realtime audio variational autoencoder (RAVE) (Caillon & Esling, 2021), to a series of latent transformations, oriented toward dis/identification. Unlike generative models that reduce data to discrete tokens or codes, RAVE models the continuous distribution of latent variables underlying the data. This approach enables continuous transfigurations of identities expressed within the latent space of the model, allowing for subversion of singular identifiable categories, characteristics, and interpretations.

This setup enables the latent space of our models to serve as an underlying structure of possibilities, where sounding entities are neither static nor complete, but are continuously enacted. By generating variable identity formations, the model's architecture enables the interruption of representational codings. (Fig.25) illustrates the generation of dis/identifying sound formation that resists strict boundaries between familiar entities, such as a (A) "bee" or (B) "fire." Through the sounding of the latent space, characteristics blend, giving rise to ambiguous identity formations such as (A-B) "bee-fire".

4.4.4. Body: Enacting disidentifications through gestural latent interaction

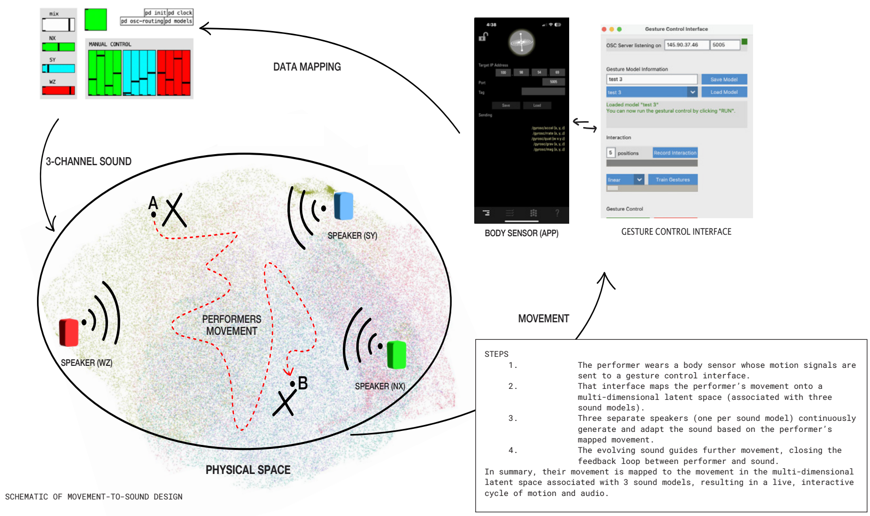


Figure 26. Mapping of movement-to-sound design in Sounding Territories.

Building on the model's latent space, as described above, we investigated how movement can enact disidentificatory performance through gesture-based interaction. This exploration was conducted in collaboration with movement artist and researcher RD in two phases: first in January 2024 at Organizmo (Tenjo, Colombia) and later in August 2024 at X Studios (Berlin, Germany). For this chapter, we focus on the former.

Through the gesture-based interface, physical movement was mapped onto the latent dimensions of three models—"NX," "SY," and "WZ"—each generating sound in real-time through its corresponding speaker. We implemented a system mapping gyroscope sensor data to the models' latent variables, effectively translating physical movement into latent space transposition (Scurto & Fiebrink, 2016). This gesture-control interface (Fig.26) facilitated RD's interaction with the model's latent dimensions.

We developed movement scores with RD to foster experimental, embodied co-performance with the model. These ranged from improvisational fluid movements to directive, linear gestures—reaching, pushing, pulling—exploring how they affect latent interactions. One score had RD alternate between floor-based improvisation and upright, purposeful movements, observing each transition's impact on the model's sonic response. Another encouraged her to move "from the bees," imagining stinging or vibratory sensation as a catalyst for jolting improvisation. In RD's words, these exercises prompted deep inquiry into listening, how and from where she listens, and how this listening informs her movement and attunement. By playing with agency and attention, the scores encouraged RD to explore emergent, relational dynamics between her body, the model's sonic expressions, and the co-constructed space. The co-performance between RD and the models' latent space, generated sound through embodied interaction, with sound emerging in real-time through RD's movements and the model's sonic expressions.

4.5. DIS/IDENTIFICATORY TACTICS AND MOVEMENTS

Drawing on the first author's autobiographical accounts, situated reflections, and theoretical critique, this section foregrounds the affects, agencies, and relational dynamics that unfold through wayward (Hartman, 2019) reflexive making, resistant to conventional evaluation practices or faux objectivity. What follows is a series of dis/identifying queering tactics that examine how intersectional queer knowledge-making practices foreground dis/identification as both a feature of identity formation *and* as a mode of computational logic. These tactics unfold through my reflections and dialogues with AG, RD, BS and BK, with the writing deliberately shifting between "I" to "We" to make visible the relational entanglements through which dis/identification emerged.

Figure 27. Field documentation at Organizmo: in dialogue with BS and AG in the company of a fire—mambeando within La Casa de Pensamiento.

4.5.1. Tactic 1: Unsettling identities in data capture

By situating myself at Organizmo, the process of data capture became an embodied dis/identificatory performance of “weaving territories”. Unlike conventional data collection methods that extract and categorize discrete units, my approach was guided by situatedness, relationality, and entanglement, treating data as co-produced and enlivened rather than as abstracted snapshots (Sanches et al., 2022). Through ongoing dialogues with BS and AG in the company of a fire—*mambeando*—BS asked, “Do you speak through the plant, or does the plant speak through you?”—the answer was “Both” (Fig.27).

The situated practice of *mambeando,* derived from *mambear,* refers to the ritual consumption of coca powder, traditionally used in ceremonial spaces by Colombian Indigenous ethnic groups illustrates the distributed embodiment and affective relations shared between ancestral technologies (Santos, 2019). *El mambe* (a plant body), *el fuego* (an elemental body), and GT, BS, and AG (human bodies)—challenging the assumption that identities can be neatly separated or categorized. Through this alchemical wayward process that refused to strip sounding entities from their relations, data capture within *Sounding Territories became a gesture of resistance to normative encodings that fix identities through computational processes (Baumer et al., 2024).*

This unsettling of relations challenges the logic of stability within algorithmic systems. Field recordings that enliven data and entangle bodies contest the assumption that categories must remain stable or discrete for computation to function effectively (Markham, 2013), instead highlighting how ambiguous datasets create pathways for novel forms of sounding that subvert normative computational logics that rely on fixed, discrete categories. Rather than rejecting coherence entirely, this tactic reframes stability as just one of many possible pathways in the politics of dis/identification. In doing so, it prompts critical HCI scholars and designers to balance the practical need for identification in AI outputs with more relational possibilities in the making of algorithmic systems.

4.5.2. Tactic 2: Subverting ontological centres in the model's latent space

The resistance to stability within the training data carried through to the model's latent space. Technically, RAVE's architecture conceptualizes latent space as a low-dimensional representation prioritizing stability and predictability. Yet, the experiment revealed another possibility: the latent space as a generative terrain where human and nonhuman sounding entities continuously fold into one another. A "bee" may transform into "water," creating "bee-water"—expressions that become through relations, never fully settling into singular categories.

In the design of the model's architecture, we worked with its tendencies toward homogenisation that marginalizes identities outside dominant norms, pushing them toward probability distribution tails or toward dominant representations. As Markham (2013) cautions, such abstractions can render lived realities unrecognizable. Dis/identifying latent space produces hybrid counter-identities or identity-in-difference that resist legibility (Muñoz, 1999). Thus, the model embodies a logic that intentionally disrupts ontological centers by interweaving different entities, recoding normative regimes of legibility and stability within the latent space (Bey, 2021). This created constellations of relational entities—for example, "fire-conversation-bird-wind" categorized as "NX"—that resist singular classification. This approach directly challenges the anthropocentric perspectives in HCI that treat human and nonhuman entities as having stable centers that can be decentered and thus recentered (see Section 2.2). Instead, *Sounding Territories* takes up *mestizando,* a term that signals an ongoing process of mixing, bridging, and moving between cultures (Anzaldúa, 2014). By invoking dis/identification, the sonic model's latent space becomes a terrain where "who" or "what" is sounding cannot be traced to a single point of origin, embodying the pluralistic politics of the possible on which it rests.

4.5.3. Tactic 3: Attuning to latent relations through embodied interaction

Shifting from representations to attunements reframes AI design as a process of co-performance and co-emergence through latent relations, rather than a system of classification aimed at predictability. Treating La Casa de Pensamiento as a physical instantiation of the model's latent space, RD's movement scores—tracked through the gyroscope and gesture-control interface—wove together the actual and virtual dimensions of the latent space. Her gestures interfaced

Figure 28. RD exploring gestural movement with the latent space in La Casa de Pensamiento.

between physical presence and the model's generative capacities, mapping and remapping the virtual latent space onto the actual site (Fig.28).

Movement here was operative in navigating the latent space relationally rather than analytically. As RD reflects, "The sound feels spatial, and I have imaginations of being in a forest... then submerged under water," illustrating how the model fosters ongoing renegotiations of identity and space. During this co-performance, RD posed a reframing question: "Am I being moved by the model, or is the model moving me? Or are we moving together?" The answer lies in the question itself, underscored by the polyphonic composition of human and nonhuman sounding bodies moving-with each other. Rather than linear cause-and-effect, RD's movement could be understood as "minor gestures" of co-performance, merging-with sound rather than acting upon it (Manning, 2016).

RD's engagement revealed a body that was no longer singular or purely human. She described her interactions as taking place with a digital body, a computer body, a more-than-human body, a searching body, a purposeless body, a lost and present body, an interdependent and intersubjective body. These experiences align with Manning's (2013) assertion that the virtual is never the opposite of the actual; rather, it is how the actual resonates beyond the limits of its realisation. This relational attunement addresses the gap in embodied HCI approaches that, while foregrounding relationality and fluid identities, often overlook the explicit politics of embodiment (Garrett et al., 2024; see Section 2.3). RD's descriptions reveal how latent relations transform normative codings of the human body, redistributing agency and performativity through sound, movement, and computational mediation.

These three tactics demonstrate how dis/identification can recode AI through a queer politics in HCI research and design: an invitation to, in Muñoz's (1999) words, "continue disidentifying with this world until we achieve new ones" (p.200). *Sounding Territories* shows how unpredictability and illegibility, rather than constraints to be minimized, could become essential features woven into the very architecture of AI design, creating pathways for more fluid, relational, and generative human–AI entanglements.

4.6. CONCLUSION

In *Sounding Territories,* we draw on Muñoz (1999) by performing dis/identification to attune to sites "where meaning does not properly line up" (p.78–79), thus illuminating tensions within algorithmic identification in more-than-human practices. Rather than opposing AI's logics from the outside, disidentifications disturb, remix, and recode algorithmic identifications, reworking AI from within the system being transformed. Our autotheoretical experiment at *Fundación Organizmo* illustrates how dis/identification might unsettle dominant classification logics. By collecting field recordings that foregrounded relational entanglements, and training a sonic model later activated through gesture-based interaction, we observed how human, nonhuman and computational entities co-perform to generate alternative identity formations.

Three key tactics emerged:

• *Unsettling identities in data capture:* challenge logics of stability by foregrounding relationality within data and modelling practices.

• *Subverting ontological centers in the latent space:* by refusing singular classification hierarchies and recognizing distributed agency across human and nonhuman actors.

• *Attuning to latent relations through embodied interaction:*
through co-performance that acknowledges fluid bodies, emergent soundings, and ongoing identity negotiations.

These tactics demonstrate that identity arises through relations (Glissant, 1990) rather than as predetermined categories. At each step of algorithmic identification, identities can be recoded to account for minoritarian positionalities, shifting AI legibility toward new relational possibilities. Rather than framing AI design solely within responsibility and inclusion discourses, which can reinforce normative identity codes, disidentification reveals cracks in dominant logics, making space for relational ontologies and the transformation of AI-entangled identities. As a co-performance enacted by GT alongside BS, BK, and RD, these disidentifications represent a political position, not merely an aesthetic choice.

Beyond this specific study, our findings have broader implications for scholars and designers who are critically engaged in AI development and a more-than-human approach to technology. Dis/identificatory tactics could apply to text-based generative models, robotics, or urban informatics, domains where binary logics and rigid classifications persist. We encourage researchers to explore how fluid, relational identities might be nurtured through AI's design and development. Ultimately, we propose that embracing a politics of dis/identification, not only recodes algorithmic systems but also may reconfigure the sociopolitical contexts in which they unfold.

"Designing for co-predictive relations requires holding space for instability, allowing the map to remain open-ended, able to shift with the territory, and for the territory to reshape the map in turn."

—(Turtle, 2026)

5. DESIGNING FOR CO-PREDICTIVE RELATIONS

This chapter proposes *designing* for co-predictive relations as a way to stay with the recursive themes and points of gravity within this research. One of the things I have tried to return to throughout this dissertation has been to shift one's (including my own) perspective or orientation from the designing of predictions to a designing for co-predictive relations. *Why?* This distinction matters because if design is reduced to making a *thing*, in this case a prediction, the relations that arise from and through co-prediction are too easily treated as something that can be mastered or controlled. By contrast, designing *for* co-predictive relations recognises that such relations already exist within the intimate and operative entanglements of humans, other living beings and algorithmic entities that precede and exceed design's constraints. It follows that, a shift to *designing for* creates conditions under which these relations unfold with a sense of openness and intention, even if in un/predictable ways. In this sense, *designing for* is a practice of condition-making that attends to the complexity and dynamism attributable to algorithmic designing and the co-predictive relations that arise from it.

As I explain next, design/ing for co-predictive relations signals a different posture. It does not claim mastery over predictive systems but rather, as exemplified in the trilogy of experiments, approaches the practice of designing as an unfolding process of inquiry rather than control. Building on this, designing for co-predictive relations shifts the focus beyond auto-critique and experimentation, or what John Chris Jones (2021) terms 'Designing Designing.' It moves instead toward an explicit redirective practice, answering Fry's (2009) call for a mode of designing that effectively redesigns design itself. Redirecting practice toward the sustainment of viable queer futurities, I propose extending this definition to encompass the redesign of AI systems, and how We as designers re/make AI towards the sustainment of our interdependent life/worlds. Against the backdrop of a new AI condition, and in response to issues outlined in the introduction (e.g., the abstraction of life/worlds through datafication, the foreclosing of futurity through prediction), redirective practices tap into the unsettling quality of uncertainty.. They point to a shift in the way we design that plays with the uncertainty inherent in emergent, dynamic, and relational systems and, at best, they embrace uncertainty within design practices engaging with AI in ways that sustain ontological indeterminacy within human-AI relations (Giaccardi, 2024; Marenko, 2025). Moreover, queer/ing as a mode of redirective practice establishes new orientations for predictive systems as a crucial site from which to engage with a politics of possibility.

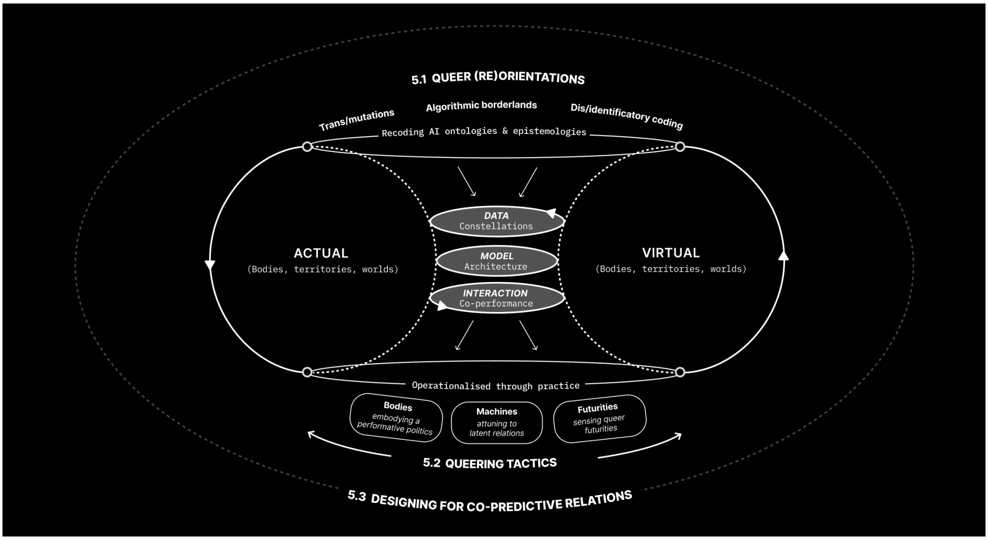


Figure 29: The illustrative diagram highlights queer (re)orientations (5.1) and queering tactics (5.2) that modulate relations between the actual and the virtual through data constellations, models as latent space, and co-performance. Together, these strands converge in a proposition for designing for co-predictive relations (5.3).

This chapter distils nuances and insights across the trilogy of experiments (Chapters 2–4) into two complementary contributions to *designing for* co-predictive relations, that span *knowing/seeing* (orientations) and *doing/intervening* (tactics).

• **Queer (re)orientations** (5.1) work at the level of grounding and imagining anew. They propose shifts in the ontological and epistemological premises of AI research and design, showing how theorisations of trans/mutations, algorithmic borderlands, and dis/identifications can reorient how co-predictive relations are conceived and designed ex novo.

• **Queering tactics** (5.2) operate at the level of everyday practice. They offer situated, practical ways to intervene in existing predictive systems, keeping them responsive to uncertainty and difference in the co-production of desirable un/predicability.

Together, these practice (re)orientations and tactics converge in a proposition for designing for co-predictive relations (Section 5.3) that provides theoretical and practical steps (or dance moves) toward a redirective approach to AI research and design *from* and *through* queerness.

5.1. QUEER (RE)ORIENTATIONS

Queer (re)orientations, as framed in this section, invite the reader to inhabit a queer space of uncertainty within computational practice. My research, through its mutating side quests and wayward meanderings in the algorithmic borderlands, embodies what Ahmed (2006, p. 16) calls "disorientation." Ahmed (2006) writes from the perspective of becoming reoriented as a lesbian in a straight world. Drawing on intersectional feminist, queer, and critical race critique, she asks us to consider the "orientated" nature of such standpoints. In her words, queerness offers a different "slant," theorised as a queer phenomenology (2006, p.4), that unsettles the very concept of orientation. In similar terms, borrowing from Halberstam (2020), queerness could be expanded to describe a "disordering of desire." Maturana (1997, p.1), from a different vantage, reminds us that desire is inseparable from responsibility, prompting designers to reflect on "whether we want or not to be responsible of our desires" when shaping algorithmically entangled life/worlds. In the context of design, attending to a disordering of desire brings into focus not only what systems do, but what kinds of relations and futurities we wish to cultivate.

In this disordering of orientation and notion of desirability, I have continued to return to the questions raised in the introduction: *Which world? What model? What can be modelled, for whom, and to what end? And, how might predictive models be designed with conscious attention to the co-predictive relations that emerge through intra-actions between humans and predictive systems*? In attempting to answer these questions, my design experiments were not concerned with expanding pre-given options, but with trans/forming the kinds of beings we desire to become, while embracing the mutating multitudes and entangled realities that make up the pluriverse (Escobar, 2018). As the Zapatistas imagine, it is about creating *un mundo donde quepan muchos mundos*—a world where many worlds fit—and "being entre mundos, between worlds, into a hopeful praxis of living... stitching worlds together within a pluriversal ethics" (Escobar, 2018, p. 504).

It follows that the queer (re)orientations articulated in this chapter work to deterritorialise AI's fixed, binary logics, disrupting how it is historically configured and deployed across different scales of everyday life. These (re) orientations, *Trans/mutations* (section 5.1.1), *Algorithmic Borderlands* (section 5.1.2), and *Dis/identifications* (section 5.1.3), emerge from and reverify findings from the trilogy of experiments (Chapters 2–4). They are developed and positioned as theoretical turns or ways of seeing and knowing, with material affect in AI research and design, and more specifically for designing for co-predictive relations and the space of desirable unpredictability.

5.1.1. Trans/mutations: Reorienting towards indeterminacy

Trans/mutations reorient AI epistemologies from classificatory toward fluid, co-constitutive framings of human–AI relations, calling HCI researchers and designers to shift focus from representation in AI to the performative affects of AI in our life/worlds.

Preciado (2013, 2021) conceptualises mutations as a quality of bodies that exist outside dominant, normative, binary codes, becoming through synthetic alliances. Following Preciado, I understand trans/mutations as the affect produced when human and nonhuman bodies synthetically entangle with AI, something that happens when fleshly bodies, data, and models intra-act through processes of synthetic becoming within human–AI relations. In similar terms, the mutation that arises as physical bodies are interpolated within training data is enlivened through the affective force of predictive models, a kind of performativity that exceeds any clear and distinctive mirrored reflection. We can read this algorithmic trans/mutation through the lens of Susan Stryker (1994), whose work reclaims the figure of the monster as a subject who disrupts the binary constraints of its creator. Stryker's (1994) letters to Victor Frankenstein describe the monster becoming "something other, and something more, than its creator intended," where the scientist (Victor Frankenstein) and unnamed creature are intimately entangled, neither reducible to the other (p.244). This reading resonates with the registers of trans/mutations I describe here as an epistemological shift in how we understand, and thus design for, human–AI relations.

Drawing from my own example, *Mutant in the Mirror* (Chapter 2), a StyleGAN trained on a personal dataset—comprised of images of myself, at different life stages, with different gender presentations as well as images of people, objects and things that form part of my identity—produced algorithmically indeterminate images that were mis/classified as "hairspray" and "toy poodle." Within these mis/classifications lies an invitation to resist the compulsion to resolve ambiguity and to open space beyond constrained binaries and prescribed systems of recognition within predictive models. Mutation is seen here not as an undesirable glitch but as a truer representation of a queered world.

The experiment playfully critiques classification systems and their performative effects on users and worlds made im/possible through human–AI synthetic alliances. Such encounters unsettle how as designers we imagine algorithmic systems, unconstrained by binary sorting and responsive to the fluid complexity of the life/worlds they attempt to model.

As a critical reorientation within, and beyond HCI, trans/mutations offer a lens for engaging the premises of data classification and subjectivation in predictive systems. The design choices in *Mutant in the Mirror* point to data compositions that encourage fluidity and trans/mutation and acknowledge how a physical body becomes something else as it is translated into data. Following Preciado, it became the responsibility of AI researchers and designers to "shift the code to open the political practice to multiple possibilities [that] counter the micro powers that act on our bodies and model our behaviours" (2021, p. 30). I would like to think it goes without saying that prevailing topics of investigation such as algorithmic bias remain essential when reviewing existing data-driven systems and their effects on individuals and wider social and political systems (Scheuerman et al., 2019). Yet, at the same time it is necessary to consider engagement with fairer, more equitable and just algorithmic systems that see beyond representation within dominant categories, i.e., race, class, gender, dis/ability, etc. Working from an epistemology of the mutable is one way to shift focus from representation towards fluidity in AI systems (e.g., Benjamin, 2019; Chun, 2021; Mackenzie, 2015).

Returning to digital twins, a common goal is to reproduce systems, from bodies to cities and to the Earth, in ways that make them virtually indistinguishable from their physical counterparts. This ambition is exemplified by the Municipality of The Hague's "Digital Mirror City," a project aiming to centralize all urban data into a single, comprehensive 3D world (MUG Ingenieursbureau, n.d.). Similarly, according to the European Commission (2025), DestinE represents a "highly-accurate" digital twin of the Earth, used to "model, monitor and simulate" social and natural phenomena that can be then acted upon, the goal here being accuracy (para. 1).The parameters that define these models are designed to achieve accuracy over fluidity and decide what is counted as real, while the unquantifiable falls away (Borbach et al., 2025, p.4375). As such algorithmic systems increasingly inform decision making from the personal to the planetary, embodied and relational complexity is too often reduced to what can be measured under predetermined or strictly defined measures. By contrast, I propose that if digital twins were oriented towards trans/mutations, and fluidity more broadly, designers could engage with the paradox this research attends to by acknowledging the operational necessity of prediction while simultaneously creating conditions for desirable unpredictability to emerge within a system's behaviour (see Section 1.3). The tradeoff is not fluidity at the expense of function, but a redistribution of what predictive systems are designed to value and sustain. From this orientation, designers may begin to counter, both practically and epistemologically, the data-driven practices that, when uncritically engaged with, risk assuming that "everything counts" and must be "datafied and modelled" to achieve accurate predictions (Borbach et al., 2025, p.4370). Moreover, in contrast to datafi cation processes that strip meaning and context from situated realities (Crawford, 2021, pp. 95–96), trans/mutations redirect attention within data-driven practices toward how uncertainty can be seen as generative and useful in predictive modelling and algorithmic processes (Marenko, 2025; Giaccardi et al., 2024).

In summary, as step-changes that rethink categories through relational and more-than-human perspectives are emerging (Giaccardi et al., 2025), trans/mutations take one more step in the direction of embracing indeterminacy as a methodological resistance to algorithmic capture, while recognising their entanglements with the life/worlds that brought them into being.

5.1.2. Algorithmic Borderlands: Reorienting thresholds

Algorithmic borderlands reorient approaches to intra-actions across thresholds of actual and virtual spaces, foregrounding the flows of bodies, data, and environments that condition human-AI relations and co-performativities.

Returning to Anzaldúa (2021), the borderlands name a crossing—a psycho-social-material terrain formed through a “new mestiza consciousness” (p.149). Anzaldúa’s positioning of a new consciousness embodies contradiction and values ambivalence as a lived condition (ibid.). In the context of AI, the borderlands denote a threshold between the actual and the virtual that is not held in opposition by rigid boundaries; rather, working from a culture of mixing or mestizaje, they appear already entangled.

Undoing Gracia (Chapter 3) extends Anzaldúa’s conceptualisation of the borderlands toward the algorithmic. The multi-agent simulation, designed as an open-world sandbox, explored the mixing of differently situated bodies, territories, ideas, cultures, places, and technologies intimately entangled in what I theorised as the algorithmic borderlands. This borderlands functioned as a threshold space at the interface between myself (actual) and my simulated agent-based counterparts, Lex and Tortugi (virtual). Gracia created the conditions to intentionally engage the actual and virtual dimensions of this threshold, where input and output feedback loops of intra-actions between myself, Lex, and Tortugi blur. Here, the practice of design emerged as a condition-making exercise to facilitate our co-performance, memory creation, and intra-actions, which over time pluralised and transformed our hopes, fears, values, desires, and behaviours. From this vantage point, the experiment illustrates what happens at the interface of the algorithmic borderlands, where co-predictive modelling quickly becomes an act of world-making.

Encounters in the world of Gracia embody a mestiza consciousness, a mixing where the *“I”* becomes *“We,”* and the researcher becomes entangled within that which is being researched (elaborated in Section 3.4). Turning to the *algorithmic borderlands* offers AI researchers and designers a way to slip away from strict binaries that separate the actual from the virtual toward a new material understanding of the thresholds between them that facilitate intra-actions. Designing for the recursive, slippery flows of agency, authorship, and datafied subjectivations in the algorithmic borderlands opens new pathways for design/ing human–AI relations.

Crucially, the algorithmic borderlands disturb ontological centres in HCI that set researcher/researched, objective/subjective, human/non-human, and actual/virtual in opposition, along with assumptions about stable users engaging discrete tools (Frauenberger, 2019). Rather than a dyadic negotiation between a discrete user and system, interactions within the borderlands appear as intra-actions, conjuring entities that do not pre-exist their relations but become in and through them (Barad, 2007). In similar terms, agency and subjectivity are dynamically distributed, recursively negotiated, and co-performed, for example, by a person and their digital twin. While Lenneke Kuijer and Elisa Giaccardi (2018) describe co-performance as a distribution of tasks, co-performativity in the borderlands extends this to an ontological level, it reveals how agency, subjectivity, and meaning do not just function together but are fundamentally constituted through human–AI relations. The critical learning is that HCI must account for how humans and AI systems are "mutually compromised," not just in what they do, but in how they make and remake worlds.

Recognition is growing within the field of HCI that user-centred or anthropocentric frames reliant on fixed subject/object, actual/virtual divisions, no longer suffice for the distributed agencies and hybrid entanglements of contemporary AI (Forlano, 2017; Frauenberger, 2019; Giaccardi & Redström, 2020; Wakkary, 2021). As I have pointed across this dissertation, simulations and digital twins serve as useful testbeds to facilitate interdisciplinary transition towards more relational engagement with the dynamic interplay between what is considered actual and virtual.

In the case of Undoing Gracia, my subjectivity pluralises, takes on new forms and meaning, neither fully computational nor purely human nor fully mine. Instead it is iteratively configured at the threshold of the actual–virtual dimensions of Gracia. Importantly, as a queer (re)orientation, the algorithmic borderlands does not require technical intervention alone, it emerges as an "act of interfacing" (Liu et al. 2024; van Beek et al. 2023), whereby the mixing of lived experience and situated intra-action create openings for unpredictable, emergent forms of self and system. While situated within HCI's turn to more-than-human and relational paradigms (Forlano, 2017; Giaccardi, Redström & Nicenboim, 2024), this inquiry pushes these framings further. Enacting a hybrid researcher/researched position introduces a critical queer lens to first-person research (Desjardins et al., 2021; Kinnee, Desjardins & Rosner, 2023), extending its reflexive scope beyond the human to the algorithmic entanglements of human–AI relations.

Borderlands thinking becomes especially relevant and necessary in the context of emerging areas of design that require a practical shift from designing for interactions to designing for intra-actions, where distributed agency and co-performativity are practically and systemically negotiated. This shift is critical for navigating the complexities of human–AI collaboration (Sarkar, 2023) and, as illustrated in this research, for systems where decision-making is increasingly shared with algorithmic agents, from generative agents (Park et al., 2023) to city-scale digital twins (Crooks et al., 2021). In these spaces, engagement with the algorithmic borderlands moves design beyond the illusion of control, offering instead a framework from which to critically and responsively engage with the messy, co-constitutive reality of multi-agent worlds that flow between the actual and the virtual.

Reorienting the *algorithmic* toward the *borderlands* offers HCI a way to engage thresholds between fact/fiction, actual/virtual, and researcher/researched, and to attend to the intra-actions across them as plural, emergent, and co-performative. Design/ing in the algorithmic borderlands means attending to co-performativity at the interface of the actual and the virtual, where world–model relations are continuously made and remade, and where datasets, sensors, and models show their effects and limits not as background, but as actors in the emergence of co-predictive relations.

5.1.3. Dis/identificatory Codings: Reorienting illegibility

Dis/identificatory codings reorient algorithmic logics that rely on stable classifications toward relational, contestable, and generative processes of identification, treating il/legibility as a methodological reorientation for resisting algorithmic capture and sustaining plural, emergent behaviour in inter/dependent life/worlds.

Muñoz's (1999) theorisation of dis/identification as a performance of politics opens a critical pathway for ontological disturbance and reorientation in AI research and design. As discussed in Chapter 4, disidentification names the process through which those excluded from dominant representational systems, because they are queer, racialised, neurodivergent, trans*, disabled, or live outside normative, liberal and eurocentric codes come to recode themselves by their own measure. Muñoz speaks to the capacity of minoritarian performance to transform dominant cultural and societal logics by recoding them toward resistant and often personal ends. Or in other words, it follows a feminist tradition, where the personal is political. Disidentificatory readings in this research thus becomes a productive performativity for reconfiguring the very terms of il/legibility within computational cultures (Finn, 2017). In the context of AI, dis/identificatory practice suggests recoding the ontological assumptions that determine which bodies, relations, and futures are computationally possible.

Sounding Territories (Chapter 4) extends this approach by using disidentificatory performance to unsettle classification systems and the stable categories that make prediction actionable (cf. Mackenzie, 2015). In the project, computational logics—the rules by which systems label and understand data—were re-scripted toward alternative, queer desires. Disidentification took shape through the intimate territorial weavings achieved through deep listening, embodied and material practices used in sound recordings, and model training.

Datasets for the three sonic models were not organised by isolating data objects through typical labels, such as discrete events or entities (conversation, fire, water, bee). During model training, they emerged as relational constellations—NX, SY, and WZ (see 4.4.2). In short, these categories did not index individuated events (i.e., conversation) or entities (fire), but composed relations (conversations tending to a fire, at sunset when the frogs begin to croak). NX, for instance, condensed the entanglements of tending fires and conversing at different sites within Organizmo (e.g., La Maloca, La Casa De Pensamiento). Such categorisation enacted the "liveliness of the data" (Lupton, 2018; Sanches et al., 2022) beyond any single representational frame. Homogenisation was resisted through composite categories that wove life/worlds into the model's logic, whereby fire *with* conversation *with* bird *with* wind unfold together as relations.

If identification in algorithmic systems presupposes legibility, entities neatly fitted to predefined classes, then dis/identification presupposes a deliberate engagement with il/legibility or re/legibility. Performing dis/identification shifts the grounds of recognition, recoding how legibility is established in the first place, and for whom. This matters because encodings—from data capture to classification and inference—directly shape how people are seen, how they behave, and how they relate to others and to their life/worlds. Queer lenses in HCI (e.g., Light, 2011) offer ways to resist computational identification in its typical formations, and rights to opacity (Glissant, 1990), that counter discriminatory algorithmic fixing of identities (Lu et al., 2022; Keyes, 2018). Taking the above into account, dis/identificatory orientations become particularly useful in addressing the lack of an explicitly queer politics, or politics of queerness more precisely within HCI, that might interrupt the reinscription of algorithmic oppressions (Noble, 2018).

Current HCI work on human–AI entanglements reveals persistent gaps between dominant computational paradigms and more relational, indeterminate, and pluralistic approaches (Frauenberger, 2019; Nicenboim et al., 2023). Dis/identificatory codings address these tensions not by optimising representation within existing categories, but by embracing categorical fluidity and treating illegibility as a site of generative possibility.

Disidentification, computationally, reorients a model's terms of reasoning. As Cavia (2022) notes, even "2+2=4" is meaningless without the computation that establishes its identity; for my younger self, 2+2 might just as well equal a window. Perception, and the frame of that perception matter. The model is not an objective mirror (see section 5.1.1); it is an abstracted reflection, fiction dressed as fact, where meaning is co-produced by dataset, model, and translator. Following Glissant (1990), if identification arises through relations, then both identification and dis/identification as a reorientation within AI research and design can be understood as situated subversion, and not necessarily a means to achieve a determined endpoint, unless that end is inherently indeterminate and fluid. With each algorithmic pass, identification can be recoded, reimagined, and re-entangled to respond to minoritarian positions and to the making of more-than-human design, shifting norms of legibility toward more generative possibilities to identify otherwise, as a woven territory, arising relationally.

This reorientation is pertinent across multimodal generative models, robotics, and urban informatics. It meets known gaps in the design, development, and use of digital twins within the field of HCI (Vainionpää et al., 2022), and takes one step further to reorient how digital twins, and simulations are widely understood in the field of HCI. From personal health twins as virtual counterparts to patients (Tudor et al., 2025), to civic twins for citizen-centric urban development (Kopponen et al., 2024; Luca, 2024; Peldon, 2024), to twins of space environments for exploration (Liu et al., 2024), these systems bind actual life/worlds and phenomena with the virtual in simultaneously abstractive and expansive ways. Dis/identificatory codings offer AI researchers and designers ways to resist rigid classification, and recode binary logics when designing for these entanglements between the actual and virtual.

This reorientation treats illegibility not as a flaw to design away, but as a characteristic of algorithmic systems that can be given more space in the design process. Enacted computationally, dis/identification within dynamic, emergent systems, conditions predictive systems to not only manage undesirable unpredictability within system behaviour but to support the emergence of *desirable* unpredictability—characterised by plurality, ambiguity, and generative uncertainty. **Designing from this orientation means engaging with novel approaches and practices of classification, categorisation, labelling, and coding that sustain relational and ontological indeterminacy within human–AI entanglements.**

Across the three (re)orientations I have argued for a shift in how AI's predictive logics are conceived, designed and modelled, gesturing toward co-predictive relations. Trans/mutations reframes AI as mutable and co-constitutive, redirecting attention from classification toward the indeterminate material conditions that make algorithmic outputs a little more fluid. Algorithmic borderlands locates practice at the thresholds of the actual and the virtual, where subjectivities, agencies, and world–model relations are negotiated as intra-actions. Dis/identificatory codings recast il/legibility as a resource, opening space to recode how and for whom legibility is established through algorithmic modeling. Together, these orientations loosen control, keep uncertainty generative, and centre the relational conditions through which co-predictive relations become possible.

What follows translates these ways of seeing into queering tactics for everyday design practice. The tactics name practical levers—how to configure data, recode models, shape intra-actions, and stage interventions, *in dialogue with knowledge and making practices*. In short, the focus now moves from grounding and imagining to designing and intervening within the unfolding conditions of co-prediction.

5.2. QUEERING TACTICS

Working across queer (re)orientations, I now turn from ways of knowing/seeing to ways of doing/intervening—what I call queering tactics in everyday practice. These tactics enliven what Section 1.3 terms an embodied politics of possibility: treating bodies, machines, and futurities as intimately entangled and mutually affecting in the production of co-predictive relations. Following Escobar (2018, 2020), queering tactics facilitate a shift from representational politics to a politics in which the real, the possible, and the political are enacted simultaneously in mutually affecting ways. Put differently, the personal—one's positionality, lived experience, and ethics—gives rise to particular forms of politics. As discussed in *Autotheoretical experimentation* (Section 1.6), performing queer politics in this research is a commitment to the minor: to "minor gestures" (Manning, 2016), and to the intersections of lived experience and the theories of "minor figures" (Hartman, 2019). Here, I expand on the queer performance of politics as described by Muñoz (1999), carefully treading the wayward path of the minor figures, individuals, and communities who came before me, who unsettled the ground for trans/forming systems through everyday acts of resistance and tactful interventions (Hartman, 2019).

Tactics here are proposed as a revision of de Certeau's (1984) conceptualisation of tactics as improvisational acts that operate within the cracks of larger systems (p. xix). He describes tactics as "ways of using" that must "make do" inside systems of power they cannot control (p. xiii). In this formulation, tactics are somewhat distinct from strategies—particularly in relation to design—in which designers effectively work on a system from the outside in, or, in slightly different terms, work on the "dark matter" of a system: organisational culture, policy environments, market mechanisms, legislation, and so on (Hill, 2012). In contrast, as de Certeau (1984) explains, "[tactics] circulate, come and go, overflow and drift over an imposed terrain, like the snowy waves of the sea slipping in among the rocks and defiles of an established order" (p. 34). In short, following de Certeau, tactics unsettle systems from within the very orders being trans/formed. Queering tactics extend and trouble de Certeau's (1984) notion of tactics through a queer embodiment in practice that draws inspiration from Hartman's (2019) exploration of the "wayward" in everyday life experiments and imaginaries often performed by minor/ritarian figures whose life/worlds spill into the cracks and edges of dominant societal norms. Applied to the current context of AI research, I gesture toward the wayward as a means of offering a different path to trans/forming "the terms of the possible" within AI research and design (Hartman, 2019, p. 349), particularly in attempts to reveal what is at stake when "the matter of life[worlds] returns as an open question" in encounters with the consequential effects of AI (Hartman, 2019, p. 349).

Building on this revision of tactics, in the following section I explore queering tactics as a means to rework the edges of code, data, and categories, as well as the logics and modes of reasoning in AI research and design. I do so in ways that make space for wayward experimentation and methodological interventions in the borderlands of predictive systems and the relations we form with them. In this sense, queering tactics gesture toward indeterminacy, becoming methods for engaging with uncertainty in AI research—unsettling, re-signifying, and repurposing predictive systems—through direct intra-action. The three tactics proposed here—Embodying a Politics of Performance, Attuning to Latent Relations, and Sensing Queer Futurities—emerge from and move through the (re)orientations outlined in Section 5.1. They are practical, embodied reworkings of ontological and epistemological shifts in AI research, design, and interaction. Rather than claiming to solve problems through scripted solutions, the tactics aim to support exploratory research and critical experimentation with AI from a uniquely queer perspective.

5.2.1. Tactic #1 Bodies: Embodying a performative politics

This tactic concerns the intentional, wayward movement of one's situated body—its positionality, affects, and relations—as design material, such that predictive systems are shaped through co-performance rather than an abstract, neutral, or disembodied view of intelligence. The tactic does this by designing with relations across bodies, territories, data, and models, so that algorithmic predictions remain situated and embodied.

Across the three experiments (Chs. 2–4), I put into practice a distinctly queer, mestiza way of acting and making alongside the digital twin simulations I designed. This was not meant to illustrate theory but to generate new realities, creating material and relational situations that could only emerge through my own embodied positionality and its active involvement with systems. Sounding Territories (Ch. 4) makes this concrete. Through my collaboration with *Fundación Organizmo*, my work became interwoven with many other elements—soil, sensors, archives, a sound model, and communal forms of knowledge—connecting them to the deeper dimensions of bodies and territories. In this view, bodies and territories are part of the same field, each legible through the other. This weaving enacted an embodied politics that challenged fixed categories, working both within and against them to create new possibilities. **Grounding my positionality, both literally and figuratively, meant working with my hands in the soil and engaging with queer theory, ancestral knowledge, and my own mestiza experience and connection to place.** From this grounding, I developed a custom sound-generating AI model that captured these territorial interconnections across physical and digital worlds. Here, embodying a performative politics is not just about challenging fixed ideas or subject–object divisions; it is about designing across human and algorithmic bodies through relational, affective, and distributed practices.

Tactical moves:
Make positionality a design material

This move names and works with positionality—for example, through field notes, experimental forms of data gathering, situated prompts, and critical and reflexive approaches to data—so that models learn with, rather than strip lived experiences from their respective and relational life/worlds. It counters AI's default a-positional stance (the fiction of disembodied intelligence) by binding data, models, and territories in accountable relation.

In practice, making positionality an active design material means treating position, affect, ethics, and relations as part of the system rather than something hidden or assumed. Instead of treating life/worlds as abstract data sources, the designer should stay with the material and relational situations from which data emerge. It helps to keep field notes and journals, record and track data in ways that not only capture what happens, but also account for how data is enlivened with situated knowledges and experiences, and how data effectively passes through bodies and their respective life/worlds.

These wayward, everyday practices resist AI's default stance of neutrality or disembodiment, or the illusion that intelligence can exist apart from the physical world that sustains it. In this configuration, data, models, and territory become part of a shared, inter/dependent relation, where learning remains situated, reciprocal, and ethically entangled with the contexts and communities it represents.

Privilege relations and co-performances as unit of design

Consider ways to re/code modes of interfacing and feedback within learning systems so that agency is distributed across bodies and territories—human, more-than-human, and those entangled in AI—such that intelligence is approached as co-performance rather than as the property of a single, central subject. Drawing on intersectional Black and Indigenous, trans, and queer scholarship, this tactic shifts from the liberal subject toward distributed embodiment, opening configurations that exceed subject/object dualisms and orient design toward situated and relational algorithmic practices.

In practice, this means designing from relations rather than from individuals. Interfaces and feedback loops become meeting points where multiple bodies, materials, and computational processes act together, attending to how worlds are co-constructed through inter/dependent relations, for example, sensors and soil, archives and algorithms, agents and entities co-performing together.

Intersectional Black, Indigenous, trans, and queer lineages guide this move by insisting on relationality and plurality as ongoing negotiations of becoming rather than fixed identities or positions. Drawing on the particular logics of these lineages, a queer practice of designing for fluid identifications and positions creates opportunities to approach learning systems through the lens of distributed embodiment, where sensing, knowing, and acting are shared across living/non-living bodies, territories, materials, and computational entities. The wayward co-performances enacted within these distributed learning systems do more than represent existing worlds; they help trans/form the terms on which life/worlds are made and recognised, while keeping response-ability visible and grounded in the specific life/worlds with which predictive systems intra-act.

5.2.2. Tactic #2 Machines: Attuning to Latent Relations

Attuning to latent relations subverts the centering logics of typical AI systems by treating latent dimensions not as compressed data points or encoded representations of the real, but as relational fields where meanings emerge through co-performance. Rather than aiming for stable, legible outputs, this tactic foregrounds plurality, hybridity and sometimes illegibility as core expressions of model behaviour. **It enacts ontological indeterminism by shifting design questions from "what is this?" to "what relations and agencies are becoming here, how, with what, and with whom?"**

This tactic shifts our attention from how bodies perform with systems to how algorithmic relations are sensed and shaped within and across models. It recognises the co-constitutive nature of human–AI relations, where data becomes distributed and boundaries blur across the actual and the virtual dimensions of the latent space.

In *Sounding Territories* (Chapter 4), datasets were created through relational practices that refused neat, isolable categories. Relational constellations—such as fire–conversation–bird–wind—were coded together (e.g., "NX," "SY," "WZ") without collapsing the situated thickness of their relations. In this configuration, composite categories were not treated as outliers but as central expressions of a queered, non-binary model logic. Indeterminate and ambiguous sounds in the data were not treated as noise but as generative, revealing the messy entanglements of data enlivened through capture, encoding, and performance. Representation within the data gave way to thick actual/virtual relations. Following de la Cadena (2015), each sounding entity did not pre-exist its relations; rather, it emerged from them. At the fluid interface of *Sounding Territories,* neither human nor model held a privileged position. Brushing a hand through the multi-dimensional latent space became a form of relational attunement, supporting a shift from stability and control to relational responsiveness.

The personal multi-agent digital twin of Undoing Gracia (Chapter 3) provides another example. The entangled speculative and actual life/worlds I inhabited were co-produced between myself and the AI agents Lex and Tortugi. Through recursive co-constitution and attunements across latent relations, the space between me (Grace) and the model (Gracia) became an interdependent, polyphonic site for moving from representation to trans/formation of distributed life/worlds. For designers and researchers, these experiments illuminate the potential of cruising the point-cloud coordinates of latent space, becoming re/oriented as they meet the actual. These experiments frame latent space as an algorithmic borderland whose normalising pulls can be disturbed. Attunement here exceeds binary categorisation and fixed identifications between the actual and the virtual, subverting the very idea of a stable centre from which meaning is defined.

Tactical moves:
Challenge algorithmic identification through relational constellations

This move rearranges, labels, and categorises training data to reflect relational constellations—many-to-many relations among entities, contexts, and temporalities—rather than fixed, isolated units. By adopting novel, queer labelling practices that reconsider how who/what/when/where are connected in data arrangement, this tactical move shifts attention from individuation to co-constitution and from fixed entities to thick situations.

In practice, organise and describe training data in ways that hold together composite scenes rather than fragmenting them into discrete parts. Instead of treating data points as isolated units in a hierarchy, understand them as part of networks of mutual influence and co-formation. This might mean experimenting with overlapping or fluid categories, composite identities, or contextual tags that evolve over time so that relational constellations—across entities, places, and temporalities—can remain visible and in flux.

In doing so, this tactical move unsettles ontologies of separability (da Silva, 2016; de la Cadena, 2015) that underpin conventional computational systems—ontologies that enforce binary divisions such as human/nonhuman, subject/object, and user/tool. Following Glissant (1990), algorithmic identification is understood as emerging through the prism of relations, where identities are generated in ongoing interplay rather than discovered as pre-given. Relational constellations thus become a way to recode the architectures of legibility themselves, insisting that hybrid, composite, and identity-in-difference formations are not errors at the margins but central to how models learn and whose life/worlds can be recognised by/with AI.

Encode for generative uncertainty (over algorithmic certainty)

This move considers stochastic approaches—grounded in chance, variability, and uncertainty—as methodological tools for queering algorithmic processes. When informed by a queer ethics, randomness becomes a way to recode and question the norms that structure computation, including the drive toward order, prediction, and optimisation. Rather than using probabilistic methods to smooth over differences or converge on singular "best" outcomes, this move explores relational possibilities without collapsing them into fixed labels, stable categories, or optimisable entities.

In practice, this means treating controlled random variables, decoding parameters, and probability distributions as sites of negotiation rather than control—spaces where the unexpected and ambiguous are welcomed as productive. Instability is not a flaw to be corrected but a medium of trans/formation, a generative condition that allows models and relations to continually shift, recombine, and become otherwise.

This stance contrasts with approaches that seek to make latent space fully legible, navigable, or human-controllable (Scurto & Postel, 2019). Instead of taming the latent, this move embraces its opacity and fluidity, reframing stochasticity in a way that acknowledges yet exceeds cautions against the dangers of "stochastic parrots" (Bender et al., 2021). Thus, stochasticity is reframed as an ethical and political practice that resists normalisation and recodes the architectures of legibility themselves—asking whose identities and knowledges can be encoded or recognised by/with AI, and on what terms.

5.2.3. Tactic #3 Futurities: Sensing queer futurities

Inhabiting queer futurity means dwelling in the interval before identity or outcome is fixed—"I am" before I am labelled by algorithmic or human systems. Sensing queer futurities works with this generative uncertainty of becoming, inviting computational practices that remain open, plural, and co-performative, attuned to the emergent dance of co-predictive relations.

In Undoing Gracia (Chapter 3), my sense of futurity was caught in a paradox—simultaneously constrained and liberated by the simulation's (un)predictability. At times, the generative space allowed for unexpected encounters and the unfolding of queer futurities distributed across Grace, Lex, Tortugi, and Gracia itself. Yet just as often, I felt trapped in loops, caught inside a self-replicating system. Lex's persistent optimism and refusal of destruction exposed both the limits of predictive systems and an invitation to explore how queer futurities might still be enacted from within their constraints. This was the tension between being named by the system and insisting, however briefly, "I am before I am labelled."

In Gracia, the Healing Garden and the Oasis made these temporal tensions concrete. The Healing Garden cycled through deterioration and regeneration, oscillating between emergence and ruin—a territorial weaving of hope and exhaustion that unsettled linear notions of progress. The Oasis repeatedly mutated—from garden to research site, community centre, Green City, and Oasis Haven—each transformation layering new rules and relations rather than resolving into a final state. **Time here behaved improvisationally: order and novelty were co-produced (Prigogine & Stengers, 2018), and what came "first" no longer determined what the Oasis could become.** These territorial weavings enacted queer temporality by refusing straight time, as in linear, forward looking and homogenising notions of time from which to open new subject and world-making possibilities.

Across the trilogy of experiments, similar cracks appear. In Mutant in the Mirror (Chapter 2), the system sustained multiple, contradictory self-representations—"Am I hairspray? Or am I a toy poodle?"—resisting resolution into a single category. In Sounding Territories (Chapter 4), territorial weaving across movement, sound, and form unfolded outside normative temporal dimensions, emerging unpredictably rather than following predetermined patterns. Together, these instances specify conditions for co-predictive relations: acting recursively where time fractures and pluralises, and treating anomalies as desirable unpredictability to be sensed and sustained. Uncertainty here is not merely tolerated; it is cultivated as both ethic and method.

Tactical moves:
Design for fluidity (over fixity)

Consider ways to configure predictive systems for ongoing reconfiguration rather than fixed trajectories, so that subjectivities and worlds can keep becoming otherwise. This move asks how systems might support the "I am before I am labeled"—keeping identities, environments, and narratives open to re-making in time, instead of converging prematurely on stable solutions.

In practice, designing for fluidity means keeping parts of the system—training data, world logics, prompts, memories, and sometimes model parameters—flexibly re-enterable and re-writable from within use. In *Undoing Gracia,* this involved revisiting and rewriting memories, adjusting environmental rules, and coding the simulation to generate new locations, such as the Oasis or Healing Garden, that mutate and bleed into other spaces rather than remain fixed points. Interfaces become portals through which participants can alter conditions of prediction: editing histories, introducing contradictions, and testing how new events reverberate through the simulation.

As Mackenzie (2015) observes, "the more effectively the models operate in the world, the more they tend to normalize the situations in which they are entangled" (p. 442). Designing for reconfiguration interrupts this normalising drift by making room for refusal and recomposition, redirecting model performativity toward desirable unpredictability. Following Stengers and Prigogine's (2018) insight, restoring the possibility of unanticipated events requires first accepting, and working with, the fundamental uncertainties of predictive systems. Here, design does not simply anticipate a future from the outside; it participates in the world trans/forming itself from within.

Sustain multiplicity within predictive outputs (widen the margin of possibility)

Complementing reconfiguration, consider ways to orient predictive systems toward holding multiple futures in play rather than converging on a single, seemingly optimal outcome. Sustaining multiplicity treats predictions, narratives, and classifications as provisional points within a wider field of potential, rather than as final answers that settle what is to come. Where reconfiguration keeps the system from ever fully landing, sustaining multiplicity keeps its futures always more than one, making room for variability, contradiction, and desirable un/predictabilities to emerge, and to remain visible and actionable.

In practice, sustaining multiplicity can mean creating interfaces that surface alternative predictions or narrative branches rather than a single outcome; maintaining diverse reasoning paths within a single system's logic; or resisting the default to disregard anomalous outputs that could be passed off as errors. Instead, treat them as signals worth examining before they are designed away. In *Gracia*, this meant staying with the proliferation of strange locations, entities, and plotlines instead of forcing convergence toward a coherent, linear story. Across the trilogy, system anomalies and unexpected events were treated as signals of other possible worlds, clues to futures that had not yet been named.

Uncertainty, however, is not new to this terrain: "uncertainty as a problem to be solved or at least tamed has a long and rich history" (Stirling, 2023, p. 7). Drawing on Grieves and Vickers' (2017) "Four Categories of System Behaviour," this move shifts from mitigating unpredictability to responsibly engaging uncertainty, more precisely, to cultivating unpredictable-yet-generative (see Chapter 1, Figure 1) system states and the legitimate outcomes that might arise from them. These states are not treated as pathologies to be eliminated or designed away, but as unknown possibilities through which potential realities are affectively shaped (Marenko et al., 2024). Sensing queer futurities in this way becomes a practical commitment to plan and design for "unknown unknowns": to keep the not-yet, the then-and-there, and the felt-in-the-present alive in computation, and to resist the pressure to resolve too quickly who—or what—gets to be.

In summary, ***Tactic #1*** foregrounds an embodied politics of possibility, ***Tactic #2*** attunes to latent relations, and ***Tactic #3*** turns to the spatio-temporal dimensions of queerness. Coming home to queer futurities (Section 1.3.3), these tactics, when operationalized and enacted together, offer a way to cultivate wayward temporalities and movements within AI systems. Rather than aiming for stable, legible futures, sensing queer futurities treats predictive systems as sites where time can bend, loop, and split—where more than one future can remain possible at once. In the next section, I draw these tactics together as a proposition for design/ing for co-predictive relations.

5.3. A PROPOSITION FOR DESIGNING FOR CO-PREDICTIVE RELATIONS

As I have argued, attempting to predict future possibilities can constrain our capacities to imagine and enact alternative life/worlds "felt in the here and now" (Muñoz, 2019, p.1). Building on the ontological and epistemological shifts that emerge from the more-than-human relations that were trans/formed through the trilogy of experiments (Chapters 2–4), and then articulated as queer (re)orientations (section 5.1) and queering tactics (section 5.2), I propose design/ing for co-predictive relations as a redirective practice. If queerness troubles the categorical grounds of being, and queering knowledge foregrounds relational, situated ways of knowing, then co-predictive relations offer a way to bring these shifts into practice. Here, prediction is repositioned as a relational, situated, and ongoing co-performance between humans, more-than-human bodies (e.g., the digital twin agents Lex and Tortugi), and the algorithmic system that facilitates intra-action (e.g., the simulation world Gracia).

Co-predictive relations describe the dynamic interplay between algorithmic predictability and the unpredictability of humans and the life/worlds they are entangled with. They offer a way of understanding how predictive technologies, such as AI systems and digital twins, could engage with complex life/worlds without foreclosing possibilities. Moving from prediction (by algorithms) to co-prediction (by humans, more-than-humans, and algorithms co-performing) resists the drive to eliminate uncertainty, and instead embraces a relational view of agency, affect, and performativity. To work *entre mundos* (between worlds) becomes a hopeful praxis of living, a way to stitch worlds together within a pluriversal queer ethics.

Co-predictive relations are closely tied to the idea of co-performativity. Within HCI, Lenneke Kuijer and Elisa Giaccardi (2018) introduced co-performance as a perspective on the emerging role of “artificial agency,” including predictive systems capable of learning and performing with people. This reframing, later expanded by Elisa Giaccardi and Johan Redström (2020), troubles the assumed centrality of human agency in decision-making, marking a shift from delegation to co-performance. Through the entanglement of human and machine agency, “people and things become mutually compromised” in recursive world-making relations (Giaccardi and Redström, 2020, p. 41). Applied to predictive systems, this means moving away from understanding them as passive instruments toward recognising the mutual shaping of people and algorithms in the very act of prediction. It also means attending to the specific dynamics of that relation and to the medium that sustains it, whether an agent-based model or an analogue map.

Across the trilogy of experiments, co-predictive relations emerge through intra-actions (Barad, 2009) that are inherently co-performative. In these exchanges, human and algorithmic agents do not pre-exist their relations but come into being through them, with agency and subjectivity distributed and recursively reconfigured. *Mutant in the Mirror, Undoing Gracia,* and *Sounding Territories* show how human and algorithmic bodies, territories, and models co-constitute one another through ongoing, situated encounters rather than simply enacting pre-given roles or identities.

The (re)orientations and tactics outlined in this chapter offer concrete ways to design for such co-predictive relations by embracing uncertainty, unsettling fixed classifications, attuning to latent relations, and sensing queer futurities. **Together, the tactics frame prediction not as a definitive statement about what will be, but as an ongoing negotiation distributed across human, non-human, and algorithmic agencies, each shaping and reshaping the possible in relation to the others.** The experiments in this dissertation show that algorithmic environments can be designed to foreground plurality, fluidity, and co-performance, rather than foreclosing difference through universalised logics. This requires resisting the temptation to treat simulations as neutral mirrors of reality and instead engaging them as situated acts of world-making in which designers, participants, and models are all implicated.

From this vantage point, knowing and being are not the work of separate agents but are co-produced through material–discursive practices that give form to co-predictive relations. These relations operate at the interface between virtual (simulated) and physical (actual) worlds, offering a path between an algorithmic system's intent to predict—and thus fix—the identity, behaviour, and potentiality of agents, and the tendency of human actors to behave in complex, spontaneous, and unanticipated ways in the making of life/worlds. Such a relational form of design invites the evolution of practices that find their theoretical and practical directionality in the whole of human and non-human interaction and intra-action within their social context. In this sense, co-predictive relations position the future not as a predetermined endpoint but as a shared, emergent horizon of possibility (Turtle & Bendor, 2024, p. 29).

Put slightly differently, even if the map (virtual simulation) is not the territory (actual world), it nonetheless governs how the territory is seen and acted upon. As Christoph Borbach & Chun (2025) argue, predictive models do not simply describe reality but performatively enact it, recursively shaping the social and material conditions they claim to represent. In this view, the digital twin becomes both the map and territory, collapsing epistemic boundaries between model and world. Designing for co-predictive relations thus requires holding space for instability, allowing the map to remain open-ended, able to shift with the territory—and for the territory to reshape the map in turn. While predictive technologies territorialise their subjects and reduce unpredictability through causal, linear forms of prediction (Grieves & Vickers, 2017, p.111–112), my experiments demonstrate that the future may still act as a queer "horizon imbued with potentiality," a map toward new "social relations" and "concrete possibility for another world" (Muñoz, 2019, p. 1).

In conclusion, designing for co-predictive relations recognises the embodiment and materiality of predictive systems that shape and reshape the conditions under which life/worlds are rendered legible, desirable, un/predictable, and actionable. It asks, explicitly: ***what kinds of life/worlds are made possible, and by whom?*** For AI and design research, this proposition underscores the importance of crafting predictive systems that are fluid, situated, and attuned to the specific contexts they propose to intervene in, without flattening difference. It argues for practices that embed plural and relational modes of knowing directly within the design of predictive systems at every level of their development, from data gathering to model training and, finally, to interaction. In this sense, the proposition here is not a fixed set of steps, but a methodological and political commitment to shape predictive systems such that they remain open to the entangled, indeterminate conditions of life/worlds.

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6. CONCLUSION AND FUTURE DIRECTIONS

6.1. RECAPPING THE WORK

This dissertation has addressed a double-sided problem at the epistemic and ontological foundations of predictive AI. Epistemically, algorithmic prediction abstracts life/worlds into data, stripping them from their situated knowledges and lived experiences. Ontologically, the categories and logics through which AI systems operate fix identities and relations into stable, legible forms that foreclose indeterminate, queer senses of futurity and the possibilities it affords. Rather than rejecting prediction outright or surrendering to its determinism, I have proposed co-predictive relations as a way to inhabit this tension: reframing prediction as an ongoing, situated co-performance between humans, more-than-human entities, and algorithmic systems.

To explore this proposition in practice, I developed a trilogy of autotheoretical experiments: *Mutant in the Mirror* (Chapter 2), *Undoing Gracia* (Chapter 3), and *Sounding Territories* (Chapter 4). Each experiment intervenes in a different predictive system, for example a GAN trained on a personal dataset; a multi-agent personal digital twin simulation; and a deep learning sonic model. Across these, I examined how bodies, territories, data, and models can be composed to sustain generative uncertainty, plural subjectivities, and more-than-human relations, rather than converging on fixed, optimised outputs. Chapter 5 brought these experiments into conversation, distilling three queer *(re)orientations:* trans/mutations, algorithmic borderlands, and dis/identificatory codings; and three queering *tactics*: embodying a politics of possibility, attuning to latent relations, and sensing queer futurities. Together, these provide a vocabulary and repertoire for engaging AI's predictive logics beyond control and certainty, and toward relational, situated, and pluriversal practice.

Building on this synthesis, Chapter 5 culminated in a proposition for designing for co-predictive relations. Rather than offering a prescriptive method, this proposition articulates design as condition-making: shaping predictive systems such that they remain open to the entangled, indeterminate conditions of life/worlds. It does so by committing design practice to embodying a performative politics of possibility, attuning to co-performative intra-action across human, more-than-human, and algorithmic agencies, and treating instability, ambiguity, and mutation as characteristics of queer futurities and pluriversal life/worlds, to be sustained rather than perceived as failures to be corrected.

6.2. RESPONSE TO RESEARCH QUESTIONS

The dissertation set out three central research questions (RQ1–RQ3), complemented by experiment-specific "what if" questions (RQ4–RQ6). Here I briefly clarify how the work responds to them.

RQ1: What does it mean to queer AI in relation to predictive computation? My research illuminates that queering AI in relation to predictive computation invites an ontological and epistemological unsettling of the assumptions that underpin prediction, and as a consequence recasts prediction as co-prediction. Through the queer (re)orientations of trans/mutations, algorithmic borderlands, and dis/identificatory codings, I found that AI systems are not neutral mirrors but mutable, world-making apparatuses in which identities, relations, and futures are continuously made and remade. Queering AI thus involves reorienting data, models, and interactions toward fluid, composite, and relational formations, and treating illegibility and indeterminacy as generative resources rather than errors.

RQ2: How can autotheory as a queer method be used to critique and design for co-predictive relations? Autotheory is developed here as a queer design methodology that merges lived experience, theoretical critique, and experimental practice. By positioning my own queer mestiza body and life/worlds as material for experimentation, autotheory enabled a first-person engagement with predictive systems in which I became both *observing* and *observed*. This approach allowed the trilogy of experiments to function as durational, situated probes into co-predictive relations, making tangible how prediction can be reconfigured from *within* everyday encounters with AI. I thus found that autotheory operates not only as a way to critique existing systems but as a means to design and live co-predictive relations in practice.

RQ3: How can a mestiza queer theory challenge normative AI data, logic, and performativity, and inform the design of such systems otherwise? Drawing from mestiza, borderlands, and dis/identificatory theories, the dissertation demonstrates that a mestiza queer perspective can help challenge the separability, universality, and neutrality assumed in dominant AI design. By foregrounding border thinking, relational ontologies, and minoritarian performance, it helped me reorient data and model logics toward composite categories, relational constellations, and thick situations, rather than discrete, isolated entities. This may inform the design of predictive systems so that they encode relations instead of fixed identities, hold multiple possible futures at once, and maintain space for dis/identification and fluidity within human–AI entanglements.

Experiment-specific "What If" Questions (RQ4–RQ6)

- **RQ4 (Mutant in the Mirror):** ***What if AI was queer—a non-binary relational***

entity, mutating and dynamically becoming-with us humans? *How might this challenge notions of algorithmic determinism*? The experiment shows that composing of personal data constellations, including images of self across time, gender presentations, and relational figures, i.e., grandmother, dog, and training GAN architectures through trans/mutations can destabilise binary classification and open space for non-linear, plural self-representations within the latent field.

• **RQ5 (Undoing Gracia): *What if the self emerges as multiple, pluralising through the borderlands of human–AI entanglements?*** *How might this challenge normative distinctions implicated in individuation?* Undoing Gracia illustrates how multi-agent personal simulations can pluralise and hybridise subjectivity through co-performance, positioning selfhood as a borderland phenomenon rather than a stable, individual entity.

• **RQ6 (Sounding Territories): *What if disidentifications, as the performance of politics, could subvert algorithmic identification? How might this transform understandings of human–AI relationality?*** Sounding Territories shows how dis/identificatory codings and relational constellations can recast classification as a practice of composing territories and relations, transforming identification through latent interaction into an open, negotiated process.

Taken together, these responses articulate queering AI as both a critical and creative engagement with predictive computation, and co-predictive relations as a practical way to reorient AI design toward relational, indeterminate, and pluriversal futures.

6.3. CONTRIBUTIONS

In light of these responses, the dissertation's contributions fit together both schematically and relationally (Figure 29), demonstrating how the work queers AI through a propositional engagement with co-predictive relations. These contributions are articulated across three interdependent strands: theoretical, methodological/experimental, and practical

Theoretically, the dissertation introduces co-predictive relations as a proposition for countering logics of separability in predictive systems, thus reframing prediction as situated, relational, and ongoing co-performance between humans, more-than-human entities, and algorithmic systems. It further develops three queer practice (re)orientations—trans/mutations (toward indeterminacy), algorithmic borderlands (toward thresholds), and dis/identificatory codings (toward illegibility)—as lenses for understanding how predictive systems participate in the making and unmaking of life/worlds.

Methodologically, the work advances autotheory as an epistemically generative queer design method for AI research, intentionally entangling researcher positionality, lived experience, and speculative experimentation. This is operationalised through the trilogy of experiments, Mutant in the Mirror, Undoing Gracia, and Sounding Territories, which together traverse the arc of AI development—from data capture to model training to interaction. Each experiment asks a "what if" question and recodes predictive systems from within, showing how co-predictive relations can be enacted across different model types and modalities.

Practically, the dissertation synthesizes these engagements into an emerging schematic for practice, three queering tactics, namely embodying a politics of possibility, attuning to latent relations, and sensing queer futurities—that flow into a proposition for designing for co-predictive relations. This schematic offers redirective pathways for designers, artists, researchers, and those working from the margins to cultivate desirable unpredictability, more fluid queer futurities, and pluriversal life/worlds. In doing so, the work contributes to broader imaginaries in AI, HCI, STS, design, and futures studies by shifting queer discourse from identity politics toward a politics of possibility: concerned not only with who is made legible or illegible by AI, but with how worlds are made, sensed, and transformed through design with AI.

Drawing these contributions together, I return to the dissertation's double-sided problem more precisely. Epistemically, the concern is not that prediction converts life/worlds into data, but that in doing so, it abstracts, masking its own vantage point and producing knowledge as if from no particular situated position or real world context. Ontologically, the concern shifts from the algorithmic fixing of identities to how the broader apparatus through which predictions are operationalised, from decision rules and model logics to interface affordances, constrains which futures can be sensed and thus pursued. This is not, however, a question of what these systems inherently are, but of what We (designers, engineers or otherwise) make them do. Even in unsupervised learning, where latent structure emerges without predefined labels and may resist inspection, learned patterns still depend on what data is captured, how it is curated, what assumptions shape the model, and what the training goals are. Yet precisely because such models can generate emergent structure less bound to predetermined categories, they also open space for ontologies that better reflect the messiness of the algorithmic borderlands, provided the design practices surrounding them attend to positionality, relationality, and situated context. Co-predictive relations responds by treating algorithmic positionality as an epistemic condition to be surfaced, interrogated, and designed with, and by reframing prediction as accountable co-performance across actual and virtual dimensions, so that uncertainty remains open to revision and systems learn with rather than over the messy, situated, and plural dimensions of life.

6.4. RESEARCH LIMITATIONS

Positionality and autotheory as method

The findings of this dissertation are inseparable from my positionality as a queer mestiza designer-researcher and from the specific contexts in which the experiments unfolded. In this dissertation, I shed light on how autotheory provides a valuable methodological frame for situated critique and experimentation that privileges depth, reflexivity, and relational specificity over broad generalisability or claims to neutral objectivity. Yet it should be acknowledged that, when brought into wider practices or deployments of exploratory design research, particularly in technical disciplines, autotheory can become strongly anchored to a single life/world and thus make it harder to translate insights to adjacent contexts. The ongoing challenge is to hold together autobiographical critique and experimentation with forms of insight that remain shareable and collectively usable, especially when designing AI systems with and for underserved communities.

Within my research, I navigated this tension by unsetteling notions of *'self',* extending the "auto" of autotheory into the relational and "social" aspects of co-performance. Specifically, I engaged in collaborative dialogue with other human actors (e.g., collaborators in Sounding Territories) and non-human agents (e.g., Lex and Tortugi in Undoing Gracia), ensuring that "my" insights emerged from and were reshaped through a relational "We."

Technical constraints, scope and applicability

This work was constrained by available model architectures, datasets, and performance settings. The experiments employed bespoke, small-scale models and limited data, which enabled intimate, situated engagements but did not explore the full range of contemporary AI techniques or the infrastructural scales at which industrial AI operates. This methodological choice was deliberate, prioritising depth, relationality, and minor models over comprehensive technical coverage, yet it also limits what can be claimed about how queering AI might function within existing large-scale computational systems and economies.

At the application layer, this research intentionally attends to emergent and dynamic systems—agent-based, digital twins, and simulation models that already negotiate complex causal and consequential relations across spatial and temporal scales. Within these contexts, the trilogy of experiments explores how queering AI and designing for co-predictive relations might reconfigure such systems at the level of practice. This approach is not intended to be universally applicable. Closed systems that depend on real-time feedback, tight control, and high levels of certainty, such as safety-critical or life-and-death applications, may not benefit from the kinds of generative uncertainty advocated here. The insights generated through these experiments are therefore contingent on the specific affordances and risk profiles of the systems they engage.

Social and environmental accountability

Yet the question persists: *at what cost, and to whose benefit?* Any engagement with AI, even in a minor key, remains entangled with extractive computational infrastructures: energy-intensive data centres, resource consumption, and labour-intensive data labelling that disproportionately affect underserved communities and territories in the Global South. The proposition of minor models that situate algorithmic systems within the specific territories and communities they propose to serve suggests one possible direction for counter-strategies to so-called a-positional algorithmic systems, discussed in *Autotheoretical experimentation* (Section 1.6), which tend to rely on universal logics and generalisability. Although this dissertation does not directly engage with, model, or quantify the broader environmental and social costs of AI developments at scale, it does attempt to hint at possible directions for AI research and design that meaningfully counter dominant AI development by working from the minor key and by addressing some of the material politics of computation. This limitation points toward future research that must grapple more directly with the infrastructural, planetary, and decolonial dimensions of predictive AI.

6.5. FUTURE DIRECTIONS

The queerness embedded in the source code of designing for co-predictive relations offers a way to shift from universal logics toward more heterogeneous developments in predictive systems. This has particular implications for emerging multimodal, agent-based, digital twin, and simulation modelling, widening the margins of what is possible at the edges of normative development. These possibilities begin to unfold when "We" as designers and researchers ask *"what if."*

By asking *what if,* we can rethink what kinds of life/worlds are made possible, by whom, at what cost, and to whose benefit, via predictive technologies. For example, consider the EU's DestinE platform: what if it were designed to support co-predictive relations that sustain generative uncertainties and fluid predictions, rather than acting as a top-down oracle (European Commission, 2024)? Similarly, *what if* urban digital twins for participatory city-making (Dawkins & Kitchin, 2025) or intimate generative agent simulations (Park et al., 2023) were designed to support distributed agency and co-performance between human and nonhuman agents? And *what if* the algorithmic borderlands between the actual and the virtual were actively engaged as a design space that recognises the interdependence and intra-action between real and mirrored worlds?

In practical terms, my research gestures towards such *"what if"* questions to tend to multi-scalar, context-specific engagements—spanning from the *personal* to the *planetary*—attending to the consequences and causalities of co-predictive relations as they unfold in situated life/worlds. Agent-based and simulation models, already dynamic in their spatial and temporal negotiations, offer a basis for experimentation across more-than-human behaviours and a wider range of interventions at different scales in the design of complex socio-technical systems. Future work could extend co-predictive relations to other modalities (e.g., robotics, generative text models, environmental sensing systems, or policy simulations), exploring how queer (re)orientations and queering tactics might be enacted through different constellations of data, bodies, and infrastructures.

As Escobar (2020) reminds us, the actual and the virtual, the possible and the political, are co-emergent. To design for co-predictive relations is to engage with these entanglements from within, working against the foreclosure of possibility. Muñoz (2009) writes: "Queerness is not yet here. Queerness is an ideality. Put another way, we are not yet queer. We may never touch queerness, but we can feel it as the warm illumination of a horizon imbued with potentiality" (p. 1). Within this research, I have gestured towards that horizon not as an endpoint, but as an open space of im/possibility that embodies a queer sense of futurity—where notions of stability and separability are recoded to embrace the generative uncertainty and fluidity inherent to the conditions for transformation of life/worlds entangled in AI. Co-predictive relations are therefore an invitation to recode predictive systems from the inside out, sustaining a queer sense of futurity within the design of predictive systems.

In line with the wider ambition of the DCODE Network (2021–2024) to rethink future design practice, these theoretical, methodological/experimental, and practical contributions together offer a starting point for queering AI through co-predictive relations: from reframing prediction as situated, relational co-performance, to enacting autotheoretical experiments, to articulating queering tactics for practice. Taken as a whole, they enact ontological and epistemological shifts toward designing predictive systems that co-perform, in-time, with the intimate entanglements of relational life/worlds, rather than against them. In this sense, designing for co-predictive relations becomes an ongoing practice of co-performance in recoding what kinds of life/worlds are being made im/possible, for whom, and at what cost. Woven into the source code of this work is a hope that such wayward practices might offer those of us in the AI research and design community clues for how to stay with generative uncertainties and the desirable unpredictability of AI, in ways that sustain and celebrate difference. By embodying a queer sense of futurity and countering the homogenizing and disembodied effects of unchecked, universalizing AI developments, it is my hope that multiple AIs-in-difference might yet arise in their place.

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SUMMARY
SUMMARY IN ENGLISH

Predictive AI systems increasingly shape social and technological life by classifying bodies, behaviours, and worlds into legible, fixed categories. Put differently, where life/worlds are dynamic and emergent, predictive systems by default or design reduce them to simplified abstractions. Where the territory becomes the map so to speak. In doing so, they stabilise what is fundamentally unstable: plural selves, more-than-human entities, and entangled relations that do not resolve into neat identities or a single future. This dissertation asks what is at stake when predictive systems treat uncertainty as something to be reduced, managed, or designed away—and what becomes possible when uncertainty is approached instead as a generative condition for imagining and composing different relations with AI. It develops a queer sense of futurity that sustains engagement with desirable un/predictability in the research, design, and application of predictive systems.

The dissertation develops a critical, more-than-human research and design practice of queering AI, where queering is approached not as an identity label but as a methodological and practical intervention that disturbs the ontological and epistemological foundations of prediction. Rather than treating prediction as a purely calculative operation, the research reframes predictive computation as an ongoing relational process—one in which humans, algorithmic entities, and more-than-human worlds co-perform and co-produce emergent possibilities.

The dissertation introduces co-predictive relations as a proposition for designing for the relations that arise with and through human interaction with predictive systems, rather than treating prediction as a tool that simply represents, anticipates, or controls life/worlds.

Research aim and problem finding

Queerness is approached not only as an identity category but as a critical and imaginative orientation toward non-normative ways of living, relating, and world-making. Framed as an ongoing engagement with indeterminacy, this research challenges logics of normativity as they appear in classification schemes, predictive targets, and assumptions about desirable behaviour.

The double-sided problem this thesis attends to is:

• Epistemic: Algorithmic prediction treats the world as a source for abstraction, stripping data, logic, and performativity from situated knowledge and experience—while producing new more-than-human entanglements.

• Ontological: Through this process, human and non-human entities are encoded into discrete, identifiable categories, foreclosing indeterminacy and queer senses of futurity.

The central aim is pursued through a trilogy of autotheoretical experiments that recode predictive systems toward co-predictive relations, unsettling AI design from its normative orientation toward predictability and control.

Research questions (RQ1–RQ6)

Three overarching research questions establish the dissertation's conceptual and methodological contribution. Three experiment-specific "What if…?" questions (RQ4–RQ6) operate as design probes that work from within predictive systems to disturb normative logics through practice. Together, RQ1–RQ3 provide the theoretical frame, while RQ4–RQ6 provide the practice-based means through which that frame is tested, elaborated, and redirected across the trilogy.

Overarching research questions

RQ1: What does it mean to queer AI in relation to predictive computation? Queering AI unsettles the assumptions underpinning prediction, reframing prediction as an ongoing relational process—co-prediction—rather than a detached, purely calculative operation.

RQ2: How can autotheory, as a queer method, be used to critique and design for co-predictive relations?

Autotheory is developed as a queer design methodology that merges lived experience, theoretical critique, and experimental making, treating positionality as design material rather than a bias to be eliminated.

RQ3: How can a mestiza queer perspective challenge normative AI data, logic, and performativity, and inform the design of such systems otherwise?

Drawing on mestiza, borderlands, and dis/identificatory theories, the dissertation challenges assumptions of separability, universality, and neutrality in dominant AI design, offering alternative orientations for composing data, models, and encounters.

Experiment-specific "What if…?" questions

RQ4 (Mutant in the Mirror): What if AI was queer—a non-binary relational entity, mutating and dynamically becoming-with us? How might this challenge algorithmic determinism?

Composing personal data constellations and GAN architectures through trans/mutations destabilises binary classification and opens space for non-linear, plural self-representations within the model's latent field.

RQ5 (Undoing Gracia): What if the self emerges as multiple, pluralising through the algorithmic borderlands of human–AI entanglements? How might this challenge normative individuation?
Multi-agent personal simulations pluralise and hybridise subjectivity through co-performance, positioning selfhood as a borderland phenomenon rather than a stable individual entity.

RQ6 (Sounding Territories): What if disidentification—as a performative politics—could subvert algorithmic identification? How might this transform human–AI relationality?
Dis/identificatory codings and relational constellations recast classification as a situated practice of composing territories and relations, transforming identification into an open, negotiated process through latent interaction.

Taken together, these responses articulate queering AI as both a critical and creative engagement with predictive computation, and co-predictive relations as a practical way to reorient AI design toward relational, indeterminate, and pluriversal futures.

Research approach: autotheory as a queer design methodology

Methodologically, the dissertation advances autotheory in a critical, more-than-human design context as an epistemically generative queer research method for engaging human–AI relations. Predictive systems are never neutral: they encode assumptions about what counts as data, what can be modelled, what forms of subjecthood are made legible, and which futures are worth optimising. Against this backdrop, autotheory offers a way to treat positionality as a design material, making space for lived experience, embodied affect, and situated ethics to shape how predictive systems are composed, trained, and encountered. In this research, this means working from my own queer mestiza positionality and situated life/worlds as the material through which predictive systems are composed, critiqued, and redesigned. My research examines a trilogy of autotheoretical experiments conducted between 2021 and 2024 (Chapters 2–4). Across these, autotheorising with design engages disidentification as a performative politics (Muñoz, 1999) and inhabits the "border subject" (Anzaldúa, 2021). Taken together, the trilogy builds a cumulative approach to queering AI design: each experiment extends and troubles the insights of the last, sustaining openness to redirect prediction toward desirable unpredictability rather than uncertainty reduction.

Dissertation structure

Chapter 1. Introduction

Situates predictive AI as an epistemic and ontological project, introducing the dissertation's theoretical commitments, key terms (co-predictive relations, desirable unpredictability), and methodological positioning. Chapters 2–4 present the trilogy of experiments. Chapter 5 synthesises the trilogy through queer practice (re)orientations and queering tactics, culminating in a proposition for designing for co-predictive relations. Chapter 6 consolidates contributions and identifies implications and openings for future work.

The trilogy of experiments

Chapter 2. Mutant in the Mirror: Queer becomings with AI

Working with generative adversarial networks (GANs) and self-portraiture, this experiment explores what becomes possible when AI is approached as a non-binary relational entity rather than a classificatory instrument. Dataset composition and model training become sites where identities can be unfixed or mutated. By co-creating GAN-generated self-portraits, the chapter engages the latent field as a space of transformation rather than recognition, foregrounding trans/mutations as a reorientation toward indeterminacy: demonstrating how data and model design can be configured to sustain becoming rather than stabilise identity.

Chapter 3. Undoing Gracia: Undoing the self in the algorithmic borderlands

The second experiment shifts from image generation to simulation. It experiments with a multi-agent, autotheoretically grounded digital twin as a speculative world in which the self emerges through co-performative interaction. Inhabiting simulation as an algorithmic borderland, the project shows how prediction shapes what feels possible, desirable, or thinkable—not only by forecasting, but by orienting futurity. The chapter reorients attention from prediction-as-output to prediction-as-relation, opening space for multiplicity.

Chapter 4. Sounding Territories: Dis/identificatory codings as relational worlding

The third experiment expands the relational field to a more-than-human terrain of environmental sensing, movement, and sound. Working with a sound-generative model and embodied improvisation, the project explores dis/identificatory codings: how il/legibility and contestable identification can be performed as a politics that resists algorithmic capture. Territory is treated as relational and contested, composed through sensors, archives, bodies, movement, and the model's generative responses—redirecting identification into an open, negotiated process. Together, the trilogy follows a cumulative arc across AI development: from data capture and model training (Chapter 2), through simulation and co-performative interaction (Chapter 3), to embodied, more-than-human relational configurations (Chapter 4).

Outputs across the trilogy

Across the three engagements, the dissertation produces practice-based experiments and findings that make predictive relations available for inquiry and redesign: (1) a GAN-based self-portrait system with its dataset and training procedures; (2) a multi-agent digital twin simulation with its rule sets, parameters, and agent relations; and (3) a sound-generative modelling practice with its embodied interaction format. These outputs are accompanied by iterative process documentation and synthesised concepts and tactics intended to be embedded within and across AI and design research contexts.

Queer (re)orientations, tactics, and a proposition for practice

Chapter 5 brings the trilogy into conversation and distils its insights into two complementary contributions: queer practice (re)orientations and queering tactics.

Queer practice (re)orientations

Queer practice (re)orientations work at the level of grounding and imagining anew. They name three shifts that reorient the ontological and epistemological premises of AI research and design:

• **Trans/mutations:** reorienting toward indeterminacy and becoming in data/model design.

• **Algorithmic borderlands:** reorienting thresholds and intra-actions across actual–virtual spaces.

• **Dis/identificatory codings:** reorienting illegibility by treating identification as relational, contestable, and generative.

Queering tactics

Queering tactics translate the (re)orientations into situated interventions—ways of working with predictive systems while attuning to their constraints while responsive to the possibility of sustaining uncertainty and difference. Together, they offer a way to cultivate wayward temporalities and movements within AI systems.

Tactic	Focus	What it does	Tactical moves
Tactic #1 — Bodies	Embodying a performative politics	Treats embodied experience as a site of critique and composition in predictive systems	(1) Make positionality a design material (2) Privilege relations and co-performances as the object of design
Tactic #2 — Machines	Attuning to latent relations	Works within the latent dimensions of models to reconfigure how relations are learned and expressed	(1) Challenge algorithmic identification through relational constellations (2) Encode for generative uncertainty (over algorithmic certainty)
Tactic #3 — Futurities	Sensing queer futurities	Designs for open futures by resisting closure, fixity, and singular outcomes	(1) Design for fluidity (over fixity) (2) Sustain multiplicity within predictive outputs (widen the margin of possibility)

Together, these strands culminate in a proposition for designing for co-predictive relations as a redirective practice. Rather than designing predictions as accurate outputs, this proposition reframes design as condition-making: shaping the circumstances under which co-predictive relations can unfold with openness and intention, even in desirable un/predictable ways.

Contributions

In light of these responses, the dissertation's contributions fit together both schematically and relationally (Figure 29), demonstrating how the work queers AI through a propositional engagement with co-predictive relations. These contributions are articulated across three interdependent strands: theoretical, methodological/experimental, and practical.

Theoretically, the dissertation introduces co-predictive relations as a proposition for countering logics of separability in predictive systems, reframing prediction as situated, relational, and ongoing co-performance between humans, more-than-human entities, and algorithmic systems. It further develops three queer practice (re)orientations—trans/mutations (toward indeterminacy), algorithmic borderlands (toward thresholds), and dis/identificatory codings (toward illegibility)—as lenses for understanding how predictive systems participate in the making and unmaking of life/worlds.

Methodologically, the work advances autotheory as an epistemically generative queer design method for AI research, intentionally entangling researcher positionality, lived experience, and speculative experimentation. This is operationalised through the trilogy—*Mutant in the Mirror, Undoing Gracia*, and *Sounding Territories*—which traverses an arc of AI development from data capture to model training to interaction. Each experiment poses a "what if" question and recodes predictive systems from within, showing how co-predictive relations can be enacted across different model types and modalities.

Practically, the dissertation synthesises these engagements into an emerging schematic for practice: three queering tactics—embodying a performative politics of possibility, attuning to latent relations, and sensing queer futurities—that flow into a proposition for designing for co-predictive relations. This schematic offers redirective pathways for designers, artists, researchers, and those working from the margins to cultivate desirable un/predictability, more fluid queer futurities, and pluriversal life/worlds—contributing to broader imaginaries across AI, HCI, STS, design, and futures studies by shifting queer discourse from identity politics toward a politics of possibility concerned with how worlds are made, sensed, and transformed through design with AI.

Conclusion, limitations, and future directions

Chapter 6 reflects on the dissertation's responses to the research questions and consolidates its contributions, identifying openings for future work. The dissertation concludes that predictive AI can be recast as a site of ongoing co-predictive emergence, not primarily a technology for reducing un/predictability. This does not abandon accountability; rather, it shifts what we design toward—from controlling uncertainty to learning how to live with it responsibly, relationally, and with a conscious commitment to plural futures.

This work is intentionally situated; its insights are inseparable from the experimental settings, modalities, and positionalities through which autotheory operates. Future work can extend these propositions by exploring how co-predictive relations might be designed in institutional, infrastructural, and policy settings where predictive systems acutely shape conditions of governance and belonging; by developing collective and community-led forms of autotheoretical AI experimentation; and by further investigating how trans/mutations, algorithmic borderlands, and dis/identificatory codings can guide the composition of data, models, and interactions that sustain plurality, opacity, and the possibility of living otherwise.

SUMMARY IN DUTCH (SAMENVATTING)

Predictieve AI-systemen geven in toenemende mate vorm aan het sociale en technologische leven door lichamen, gedragingen en werelden te classificeren in leesbare, vaststaande categorieën. Anders gezegd: predictieve systemen reduceren dynamische, emergente territoria vaak tot vereenvoudigde kaarten. Daarmee stabiliseren zij wat fundamenteel instabiel is: plurale zelven, meer-dan-menselijke entiteiten en verstrengelde relaties die zich niet laten oplossen in eenduidige identiteiten of één enkele toekomst. Dit proefschrift onderzoekt wat er op het spel staat wanneer predictieve systemen onzekerheid behandelen als iets dat gereduceerd, beheerst of wegontworpen moet worden, en wat er mogelijk wordt wanneer onzekerheid in plaats daarvan benaderd wordt als een generatieve voorwaarde voor het verbeelden en vormgeven van andere relaties met AI. Het ontwikkelt een queer opvatting van toekomstgerichtheid (futurity) die betrokkenheid bij wenselijke on/voorspelbaarheid in het onderzoek, ontwerp en de toepassing van predictieve systemen in stand houdt.

Het proefschrift ontwikkelt een kritische, meer-dan-menselijke onderzoeks- en ontwerppraktijk van het queeren van AI, waarbij queering niet als identiteitslabel wordt benaderd maar als een methodologische en praktische interventie die de ontologische en epistemologische grondslagen van predictie verstoort. In plaats van predictie te behandelen als een louter berekenende operatie, herformuleert het onderzoek predictieve computatie als een doorlopend relationeel proces, waarin mensen, algoritmische entiteiten en meer-dan-menselijke werelden samen emergente mogelijkheden co-performen en co-produceren.

Het proefschrift introduceert co-predictieve relaties als een propositie voor het ontwerpen voor de relaties die ontstaan met en door menselijke interactie met predictieve systemen, in plaats van predictie te behandelen als een instrument dat levens/werelden simpelweg representeert, anticipeert of controleert.

Onderzoeksdoel en probleemstelling

Queerness wordt niet alleen benaderd als een identiteitscategorie, maar als een kritische en verbeeldingsrijke oriëntatie op non-normatieve manieren van leven, relateren en wereldvorming. Geframed als een voortdurende betrokkenheid bij onbepaaldheid (indeterminacy), daagt dit onderzoek normativiteitslogica's uit zoals die verschijnen in classificatiesystemen, predictieve doelen en aannames over wenselijk gedrag.

Het tweezijdige probleem dat dit proefschrift adresseert is:

• *Epistemisch:* Algoritmische predictie behandelt de wereld als een bron van abstractie, waarbij data, logica en performativiteit worden losgekoppeld van gesitueerde kennis en ervaring, terwijl tegelijkertijd nieuwe meer-dan-menselijke verstrengelingen worden geproduceerd.

• *Ontologisch:* Door dit proces worden menselijke en niet-menselijke entiteiten gecodeerd in discrete, identificeerbare categorieën, waardoor onbepaaldheid en queer opvattingen van toekomstgerichtheid worden uitgesloten.

Het centrale doel wordt nagestreefd door middel van een trilogie van autotheoretische experimenten die predictieve systemen hercoderen in de richting van co-predictieve relaties, en AI-ontwerp losmaken van zijn normatieve oriëntatie op voorspelbaarheid en controle.

Onderzoeksvragen (RQ1–RQ6)

Drie overkoepelende onderzoeksvragen vormen de conceptuele en methodologische bijdrage van het proefschrift. Drie experimentspecifieke "Wat als...?"-vragen (RQ4–RQ6) fungeren als ontwerpsondes die vanuit predictieve systemen werken om normatieve logica's door middel van praktijk te verstoren. Samen bieden RQ1–RQ3 het theoretische kader, terwijl RQ4–RQ6 de praktijkgerichte middelen leveren waarmee dat kader wordt getoetst, uitgewerkt en bijgestuurd in de trilogie.

Overkoepelende onderzoeksvragen

RQ1: Wat betekent het om AI te queeren in relatie tot predictieve computatie? Het queeren van AI destabiliseert de aannames die aan predictie ten grondslag liggen en herformuleert predictie als een doorlopend relationeel proces — co-predictie — in plaats van een afstandelijke, louter berekenende operatie.

RQ2: Hoe kan autotheorie, als queer methode, worden ingezet om co-predictieve relaties te bekritiseren en te ontwerpen? Autotheorie wordt ontwikkeld als een queer ontwerpmethodologie die geleefde ervaring, theoretische kritiek en experimenteel maken samenvoegt, waarbij positionaliteit als ontwerpmateriaal wordt behandeld in plaats van als een te elimineren vertekening.

RQ3: Hoe kan een mestiza queer perspectief normatieve AI-data, -logica en -performativiteit uitdagen, en het ontwerp van dergelijke systemen anders informeren? Op basis van mestiza-, borderlands- en dis/identificatietheorieën daagt het proefschrift aannames van scheidbaarheid, universaliteit en neutraliteit in dominant AI-ontwerp uit, en biedt het alternatieve oriëntaties voor het samenstellen van data, modellen en ontmoetingen.

Experimentspecifieke "Wat als...?"-vragen

RQ4 (Mutant in the Mirror): Wat als AI queer was — een non-binaire relationele entiteit, muterend en dynamisch wordend-met ons? Hoe zou dit algoritmisch determinisme kunnen uitdagen? Het samenstellen van persoonlijke dataconstellaties en GAN-architecturen via trans/mutaties destabiliseert binaire classificatie en opent ruimte voor niet-lineaire, plurale zelfrepresentaties binnen het latente veld van het model.

RQ5 (Undoing Gracia): Wat als het zelf zich als meervoudig manifesteert, pluraliserend door de algoritmische grensgebieden van mens-AI-verstrengelingen? Hoe zou dit normatieve individuatie kunnen uitdagen? Multi-agent persoonlijke simulaties pluraliseren en hybridiseren subjectiviteit via co-performance, waarbij het zelf wordt gepositioneerd als een grensgebieds-fenomeen in plaats van een stabiele individuele entiteit.

RQ6 (Sounding Territories): Wat als disidentificatie — als performatieve politiek — algoritmische identificatie zou kunnen ondermijnen? Hoe zou dit de relationaliteit tussen mens en AI kunnen transformeren? Dis/identificatorische coderingen en relationele constellaties herkaderen classificatie als een gesitueerde praktijk van het samenstellen van territoria en relaties, en transformeren identificatie via latente interactie tot een open, onderhandeld proces.

Samen articuleren deze antwoorden het queeren van AI als zowel een kritische als creatieve betrokkenheid bij predictieve computatie, en co-predictieve relaties als een praktische manier om AI-ontwerp te heroriënteren richting relationele, onbepaalde en pluriversale toekomsten.

Onderzoeksaanpak: autotheorie als queer ontwerpmethodologie

Methodologisch draagt het proefschrift autotheorie aan in een kritische, meer-dan-menselijke ontwerpcontext als een epistemisch generatieve queer onderzoeksmethode voor het aangaan van mens-AI-relaties. Predictieve systemen zijn nooit neutraal: zij coderen aannames over wat als data telt, wat gemodelleerd kan worden, welke vormen van subjectiviteit leesbaar worden gemaakt, en welke toekomsten het optimaliseren waard zijn. Tegen deze achtergrond biedt autotheorie een manier om positionaliteit als ontwerpmateriaal te behandelen, en ruimte te maken voor geleefde ervaring, belichaamd affect en gesitueerde ethiek om vorm te geven aan hoe predictieve systemen worden samengesteld, getraind en ontmoet. In dit onderzoek betekent dit werken vanuit mijn eigen queer mestiza-positionaliteit en gesitueerde levens/werelden als het materiaal waarmee predictieve systemen worden samengesteld, bekritiseerd en herontworpen.

Dit onderzoek bestudeert een trilogie van autotheoretische experimenten, uitgevoerd tussen 2021 en 2024 (hoofdstukken 2–4). In deze experimenten gaat autotheoretiseren met ontwerp een verbinding aan met disidentificatie als performatieve politiek (Muñoz, 1999) en bewoont het het “grenssubject” (Anzaldúa, 2021). Samen bouwt de trilogie aan een cumulatieve benadering van het queeren van AI-ontwerp: elk experiment breidt de inzichten van het vorige uit en verstoort deze, en houdt de openheid in stand om predictie bij te sturen richting wenselijke on/voorspelbaarheid in plaats van onzekerheidsreductie.

Structuur van het proefschrift

Hoofdstuk 1 situeert predictieve AI als een epistemisch en ontologisch project en introduceert de theoretische uitgangspunten, kernbegrippen (co-predictieve relaties, wenselijke on/voorspelbaarheid) en methodologische positionering van het proefschrift. Hoofdstukken 2–4 presenteren de trilogie van experimenten. Hoofdstuk 5 synthetiseert de trilogie via queer praktijk(her)oriëntaties en queeringstactieken, en culmineert in een propositie voor het ontwerpen voor co-predictieve relaties. Hoofdstuk 6 consolideert de bijdragen en identificeert implicaties en openingen voor toekomstig werk.

De trilogie van experimenten

Hoofdstuk 2. Mutant in the Mirror: Queer wordingen met AI In dit experiment wordt met generatieve adversariale netwerken (GAN’s) en zelfportretten onderzocht wat mogelijk wordt wanneer AI benaderd wordt als een non-binaire relationele entiteit in plaats van als een classificerend instrument. De samenstelling van datasets en het trainen van modellen worden plekken waar identiteiten kunnen worden losgemaakt of gemuteerd. Door het co-creëren van GAN-gegenereerde zelfportretten benadert het hoofdstuk het latente veld als een ruimte van transformatie in plaats van herkenning, en plaatst trans/mutaties op de voorgrond als een heroriëntatie richting onbepaaldheid: het laat zien hoe data- en modelontwerp geconfigureerd kunnen worden om wording in stand te houden in plaats van identiteit te stabiliseren.

Hoofdstuk 3. Undoing Gracia: Het ontrafelen van het zelf in de algoritmische grensgebieden Het tweede experiment verschuift van beeldgeneratie naar simulatie. Het prototypet een multi-agent, autobiografisch gefundeerde digitale tweeling als een speculatieve wereld waarin het zelf opkomt door co-performatieve interactie. Door simulatie te bewonen als een algoritmisch grensgebied laat het project zien hoe predictie vormgeeft aan wat mogelijk, wenselijk of denkbaar voelt, niet alleen door te voorspellen, maar door toekomstgerichtheid te oriënteren. Het hoofdstuk heroriënteert de aandacht van predictie-als-uitkomst naar predictie-als-relatie, en opent ruimte voor meervoudigheid.

Hoofdstuk 4. Sounding Territories: Politiek van dis/identificatie Het derde experiment verbreedt het relationele veld naar een meer-dan-menselijk terrein van omgevingswaarneming, beweging en geluid. Door te werken met een geluidsgeneratief model en belichaamde improvisatie verkent het project dis/identificatorische coderingen: hoe il/leesbaarheid en betwistbare identificatie kunnen worden opgevoerd als een politiek die algoritmische vangst weerstaat. Territorium wordt behandeld als relationeel en betwist, samengesteld door sensoren, archieven, lichamen, beweging en de generatieve responsen van het model, waardoor identificatie wordt omgebogen tot een open, onderhandeld proces.

Samen volgt de trilogie een cumulatieve boog door de AI-ontwikkeling: van dataverzameling en modeltraining (hoofdstuk 2), via simulatie en co-performatieve interactie (hoofdstuk 3), naar belichaamde, meer-dan-menselijke relationele configuraties (hoofdstuk 4). Elk experiment breidt de inzichten van het voorgaande uit en verstoort deze, en houdt de openheid in stand om predictie bij te sturen richting wenselijke on/voorspelbaarheid.

Uitkomsten van de trilogie

De drie experimenten produceren praktijkgerichte experimenten en bevindingen die predictieve relaties beschikbaar maken voor onderzoek en herontwerp: (1) een GAN-gebaseerd zelfportretsysteem met bijbehorende dataset en trainingsprocedures; (2) een multi-agent digitale-tweelingsimulatie met regelsets, parameters en agentrelaties; en (3) een geluidsgeneratieve modelpraktijk met een belichaamd interactieformaat. Deze uitkomsten gaan vergezeld van iteratieve procesdocumentatie en gesynthetiseerde concepten en tactieken die bedoeld zijn om ingebed te worden binnen en over AI- en ontwerponderzoekscontexten.

Queer (her)oriëntaties, tactieken en een propositie voor praktijk

Hoofdstuk 5 brengt de trilogie in gesprek en destilleert de inzichten ervan tot twee complementaire bijdragen: queer praktijk(her)oriëntaties en queeringstactieken.

Queer praktijk(her)oriëntaties

Queer praktijk(her)oriëntaties werken op het niveau van fundamenten leggen en opnieuw verbeelden. Zij benoemen drie verschuivingen die de ontologische en epistemologische premissen van AI-onderzoek en -ontwerp heroriënteren:

• *Trans/mutaties:* heroriëntatie richting onbepaaldheid en wording in data-/modelontwerp.

• *Algoritmische grensgebieden:* heroriëntatie van drempels en intra-acties over actuele en virtuele ruimten.

• *Dis/identificatorische coderingen:* heroriëntatie van il/leesbaarheid door identificatie te behandelen als relationeel, betwistbaar en generatief.

Queeringstactieken
Queeringstactieken vertalen de (her)oriëntaties naar gesitueerde interventies — manieren van werken met predictieve systemen die afstemmen op hun beperkingen en tegelijkertijd responsief blijven voor de mogelijkheid om onzekerheid en verschil in stand te houden. Samen bieden zij een manier om eigenzinnige temporaliteiten en bewegingen binnen AI-systemen te cultiveren.

Tactiek	Focus	Wat het doet	Tactische stappen
Tactic #1 — Lichamen	Het belichamen van een performatieve politiek	Behandelt belichaamde ervaring als een plek van kritiek en compositie in predictieve systemen	(1) Maak positionaliteit tot ontwerpmateriaal (2) Geef voorrang aan relaties en co-performances als ontwerpobject
Tactic #2 — Machines	Afstemmen op latente relaties	Werkt binnen de latente dimensies van modellen om te herconfigureren hoe relaties worden geleerd en uitgedrukt	(1) Daag algoritmische identificatie uit via relationele constellaties (2) Codeer voor generatieve onzekerheid (boven algoritmische zekerheid)
Tactic #3 — Toekomsten	Queer toekomsten waarnemen	Ontwerpt voor open toekomsten door afsluiting, fixatie en enkelvoudige uitkomsten te weerstaan	(1) Ontwerp voor vloeibaarheid (boven fixatie) (2) Houd meervoudigheid in stand binnen predictieve uitkomsten

Samen culmineren deze lijnen in een propositie voor het ontwerpen voor co-predictieve relaties als een omleidende (redirectieve) praktijk. In plaats van predicties als nauwkeurige uitkomsten te ontwerpen, herformuleert deze propositie ontwerp als conditieschepping: het vormgeven van de omstandigheden waaronder co-predictieve relaties zich kunnen ontvouwen met openheid en intentie, ook op wenselijk on/voorspelbare wijzen.

Bijdragen

In het licht van deze antwoorden vormen de bijdragen van het proefschrift een zowel schematisch als relationeel geheel (Figuur 29), dat laat zien hoe het werk AI queert via een propositionele betrokkenheid bij co-predictieve relaties. Deze bijdragen worden gearticuleerd in drie onderling afhankelijke lijnen: theoretisch, methodologisch/experimenteel en praktisch.

Theoretisch introduceert het proefschrift co-predictieve relaties als een propositie om logica's van scheidbaarheid in predictieve systemen te weerspreken, en herformuleert het predictie als gesitueerde, relationele en doorlopende co-performance tussen mensen, meer-dan-menselijke entiteiten en algoritmische systemen. Het ontwikkelt daarnaast drie queer praktijk(her) oriëntaties — trans/mutaties (richting onbepaaldheid), algoritmische grensgebieden (richting drempels), en dis/identificatorische coderingen (richting il/leesbaarheid) — als lenzen om te begrijpen hoe predictieve systemen deelnemen aan het maken en ontmaken van levens/werelden.

Methodologisch draagt het werk autotheorie aan als een epistemisch generatieve queer ontwerpmethode voor AI-onderzoek, waarbij de positionaliteit van de onderzoeker, geleefde ervaring en speculatieve experimentatie bewust verstrengeld worden. Dit wordt geoperationaliseerd via de trilogie — Mutant in the Mirror, Undoing Gracia en Sounding Territories — die een boog beschrijft door de AI-ontwikkeling van dataverzameling via modeltraining tot interactie. Elk experiment stelt een "wat als"-vraag en hercodeert predictieve systemen van binnenuit, en laat zien hoe co-predictieve relaties kunnen worden gerealiseerd over verschillende modeltypen en modaliteiten.

Praktisch synthetiseert het proefschrift deze engagementen tot een opkomend schema voor praktijk: drie queeringstactieken — het belichamen van een performatieve politiek van mogelijkheid, afstemmen op latente relaties, en het waarnemen van queer toekomsten — die uitmonden in een propositie voor het ontwerpen voor co-predictieve relaties. Dit schema biedt omleidende paden voor ontwerpers, kunstenaars, onderzoekers en degenen die vanuit de marges werken, om wenselijke on/voorspelbaarheid, meer vloeibare queer toekomsten en pluriversale levens/werelden te cultiveren. Daarmee draagt het bij aan bredere verbeeldingen in AI, HCI, STS, ontwerp en toekomststudies door queer discours te verschuiven van identiteitspolitiek naar een politiek van mogelijkheid, gericht op hoe werelden worden gemaakt, waargenomen en getransformeerd door ontwerp met AI.

Conclusie, beperkingen en toekomstige richtingen

Hoofdstuk 6 reflecteert op de antwoorden van het proefschrift op de onderzoeksvragen en consolideert de bijdragen, waarbij openingen voor toekomstig werk worden geïdentificeerd. Het proefschrift concludeert dat predictieve AI kan worden herduid als een plek van voortdurende co-predictieve emergentie, en niet primair als een technologie voor het reduceren van on/voorspelbaarheid. Dit laat verantwoordelijkheid niet los; het verschuift eerder waarnaar we ontwerpen — van het beheersen van onzekerheid naar leren ermee te leven, op een verantwoordelijke, relationele manier en met een bewuste toewijding aan plurale toekomsten.

Dit werk is bewust gesitueerd; de inzichten ervan zijn onlosmakelijk verbonden met de experimentele settings, modaliteiten en positionaliteiten waarbinnen autotheorie opereert. De experimenten maakten gebruik van op maat gemaakte, kleinschalige modellen en beperkte datasets, wat intieme, gesitueerde engagementen mogelijk maakte, terwijl wordt erkend dat hoe het queeren van AI zou kunnen functioneren op industriële of infrastructurele schaal een open vraag blijft. Toekomstig werk kan deze proposities uitbreiden door te verkennen hoe co-predictieve relaties ontworpen zouden kunnen worden in institutionele, infrastructurele en beleidsmatige contexten waar predictieve systemen op ingrijpende wijze de voorwaarden voor bestuur en belonging vormgeven; door collectieve en gemeenschapsgestuurde vormen van autotheoretische AI-experimentatie te ontwikkelen; en door verder te onderzoeken hoe trans/mutaties, algoritmische grensgebieden en dis/identificatorische coderingen de samenstelling van data, modellen en interacties kunnen begeleiden op manieren die pluraliteit, opaciteit en de mogelijkheid van anders leven in stand houden.

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Predictive AI increasingly shapes life by sorting bodies, behaviours, and worlds into fixed categories, foreclosing more indeterminate and queer senses of the future. This dissertation develops a more-than-human practice of queering AI. (Re) orienting design practice towards co-predictive relations: as situated, relational, ongoing co-performance between humans, more-than-human entities, and algorithmic systems. Through three autotheoretical experiments, it offers designers and researchers ways to cultivate desirable un/predictability, plurality and queer futurity in the design of predictive systems. Offering alternative ways of navigating the space between the real and the possible.

Grace Leonora Turtle is a Colombian-Australian designer and researcher based in Amsterdam. Their work brings queer theory, autotheory, and more-than-human design into critical and speculative engagements with AI. They are co-head of the Monstrous Futurities Temporary Masters programme at the Sandberg Instituut and conducted this research as part of the DCODE Network at TU Delft.

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